# Chapter 5

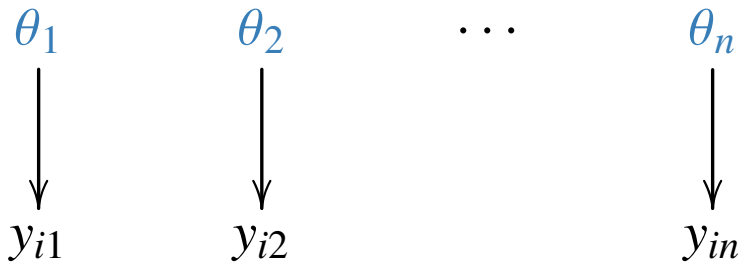
- 5.1 Lead-in to hierarchical models
- 5.2 Exchangeability (useful concept)
- 5.3 Bayesian analysis of hierarchical models (we use Stan/brms for computation)
- 5.4 Hierarchical normal model (we use Stan/brms for computation)
- 5.5 Example: parallel experiments in eight schools (useful discussion on benefits of hierarchical model)
- 5.6 Meta-analysis
- 5.7 Weakly informative priors for hierarchical variance parameters

# Hierarchical model

- In simple model: posterior for the parameters
- In hierarchical model: posterior for the prior parameters

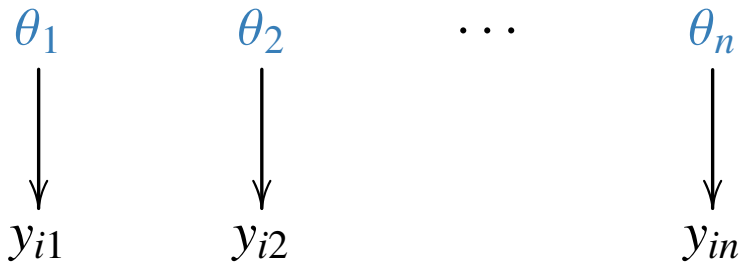
# Hierarchical model

- Example: CVD treatment effectiveness
  - in hospital $j$ the survival probability is $\theta_j$
  - observations $y_{ij}$ tell whether patient $i$ survived in hospital $j$

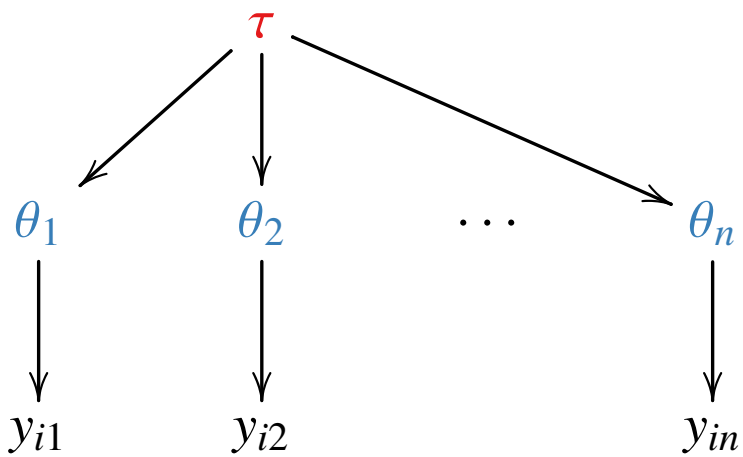


# Hierarchical model

- Example: CVD treatment effectiveness
  - in hospital $j$ the survival probability is $\theta_j$
  - observations $y_{ij}$ tell whether patient $i$ survived in hospital $j$




  - sensible to assume that $\theta_j$ are similar




  - natural to think that $\theta_j$ have common population distribution
  - $\theta_j$ is not directly observed and the population distribution is unknown

# Hierarchical model: terms

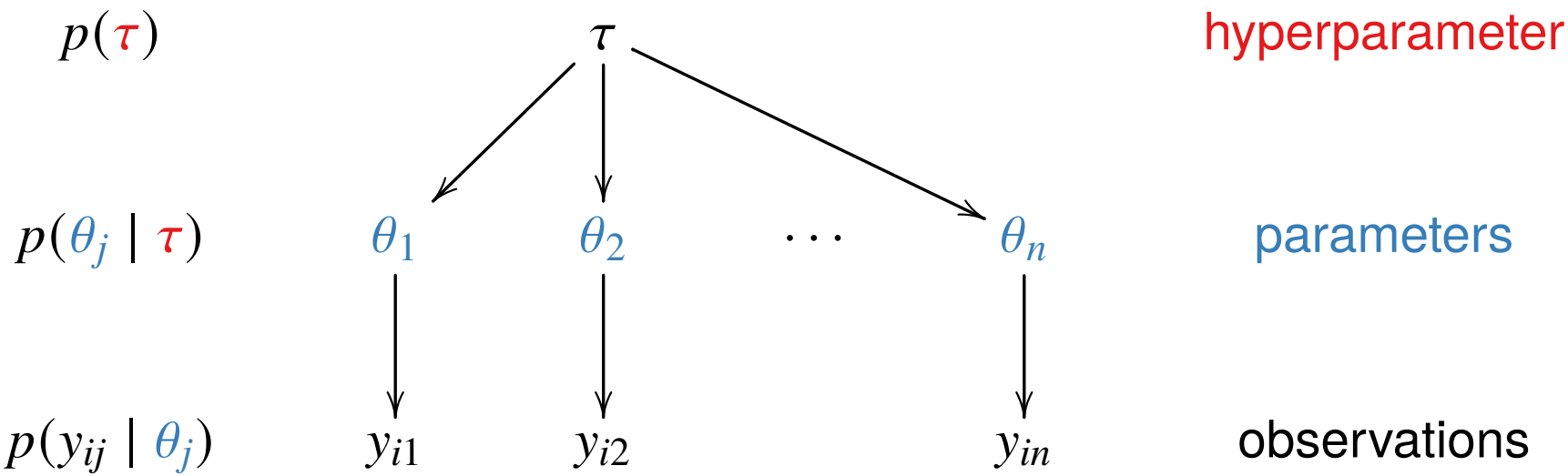
Level 1: observations given parameters $p(y_{ij} \mid \theta_j)$



# Hierarchical model: terms

Level 1: observations given parameters $p(y_{ij} \mid \theta_j)$

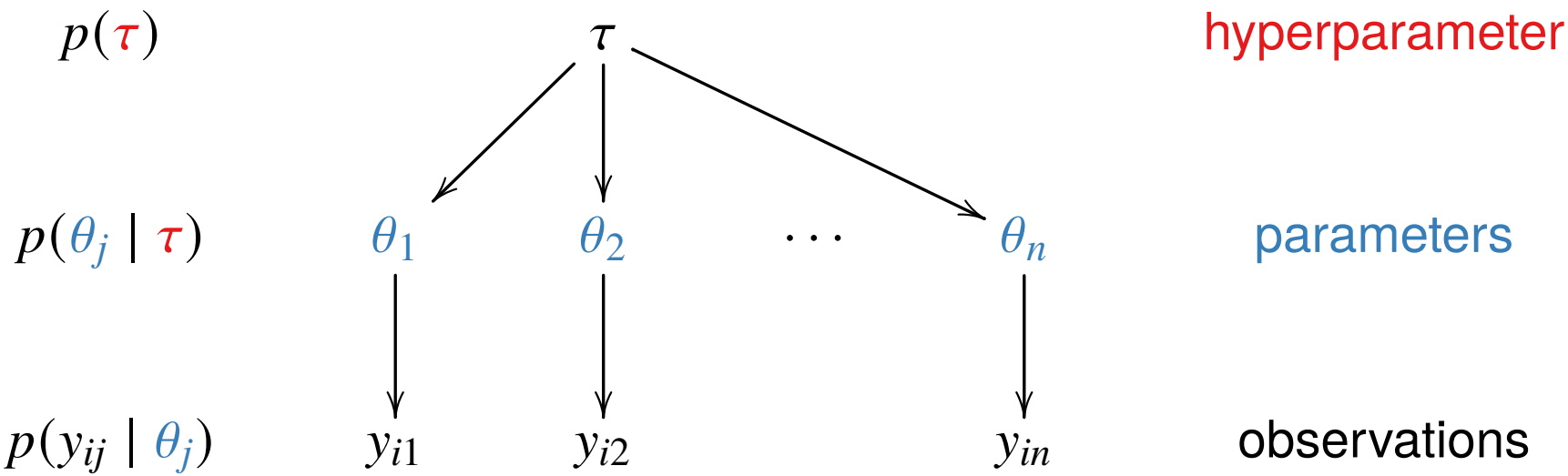
Level 2: parameters given hyperparameters $p(\theta_j \mid \tau)$



# Hierarchical model: terms

Level 1: observations given parameters $p(y_{ij} \mid \theta_j)$

Level 2: parameters given hyperparameters $p(\theta_j \mid \tau)$

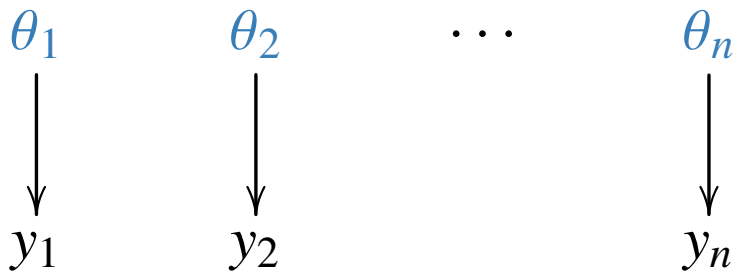



Joint posterior

$$\begin{aligned} p(\theta, \tau \mid y) &\propto p(y \mid \theta, \tau) p(\theta, \tau) \\ &\propto p(y \mid \theta) p(\theta \mid \tau) p(\tau) \end{aligned}$$

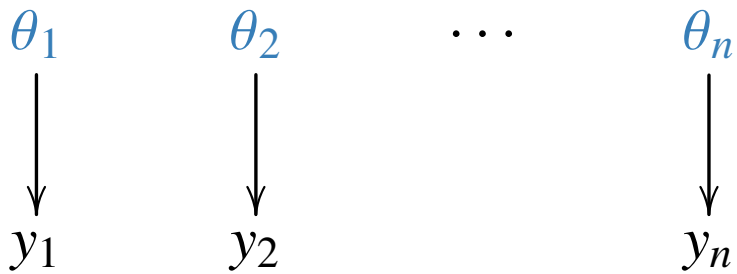
# Compare

- "Separate model" (model with separate/independent effects)

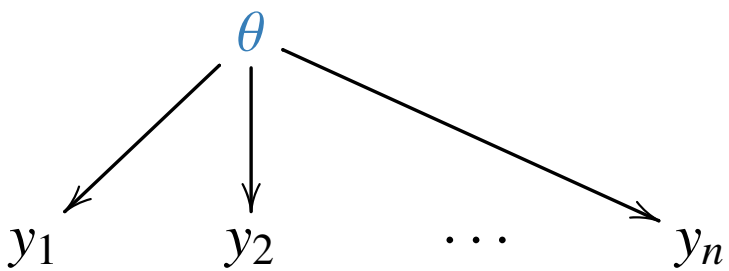


# Compare

- "Separate model" (model with separate/independent effects)

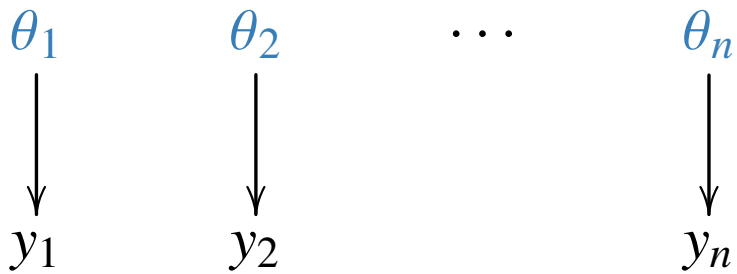



- "Joint model" (model with a common effect / pooled model)

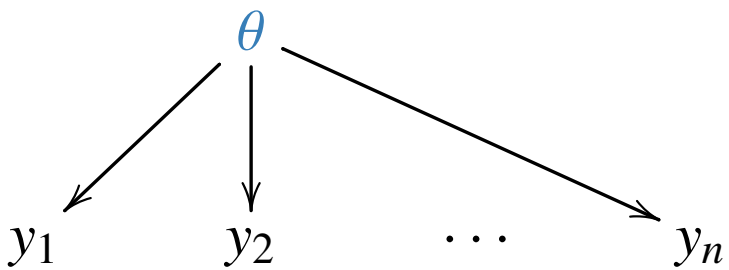


# Compare

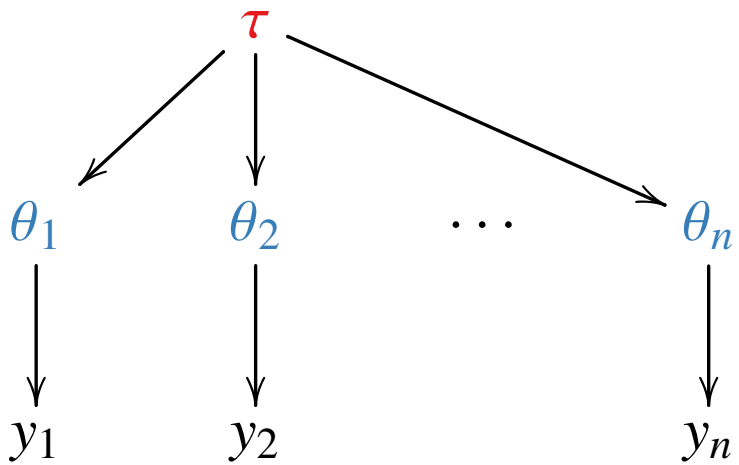
- "Separate model" (model with separate/independent effects)




- "Joint model" (model with a common effect / pooled model)




- Hierarchical model



# Hierarchical binomial model: rats

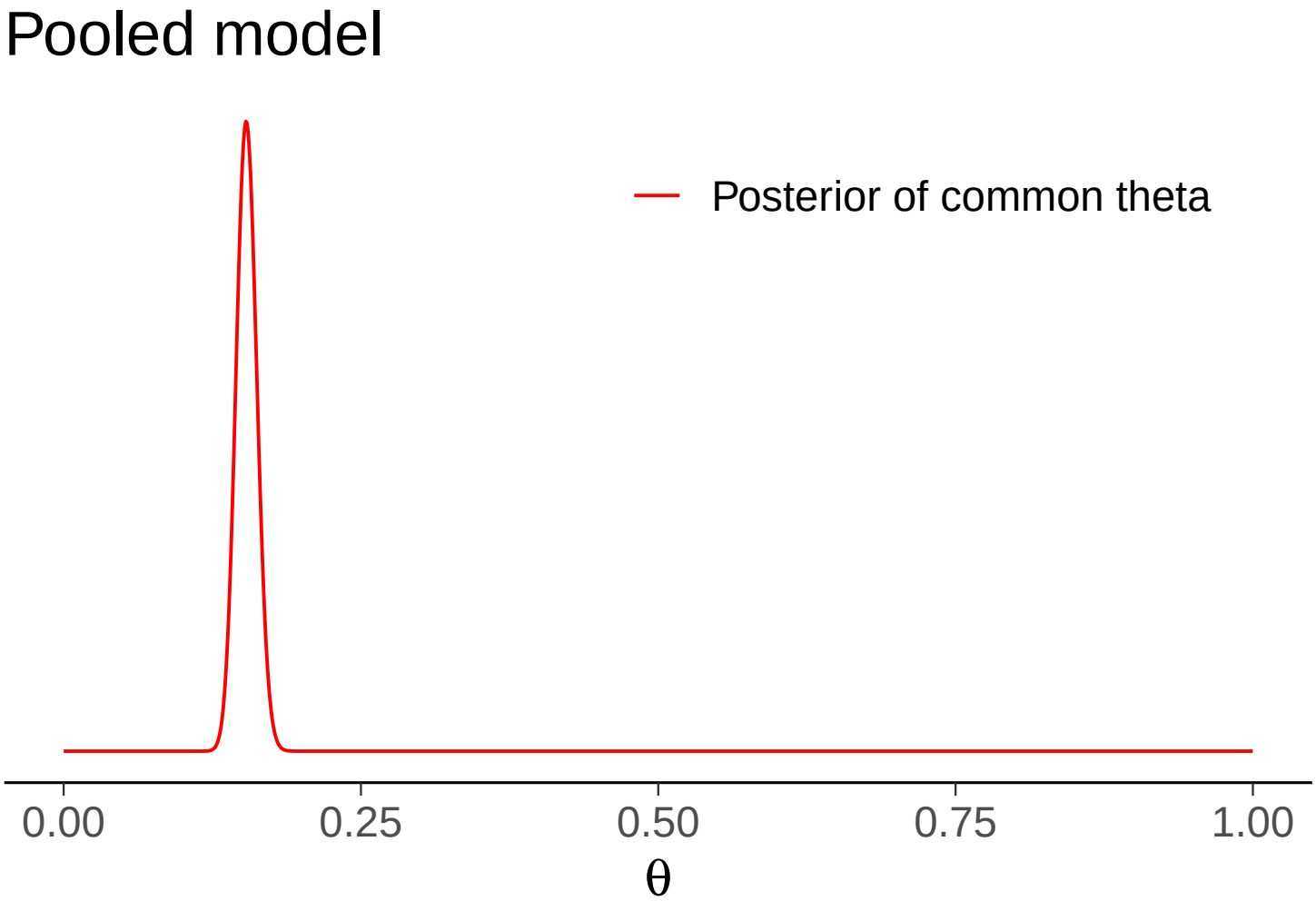
- Medicine testing
- Type F344 female rats in control group given placebo
  - count how many get endometrial stromal polyps
  - familiar binomial model example

# Hierarchical binomial model: rats

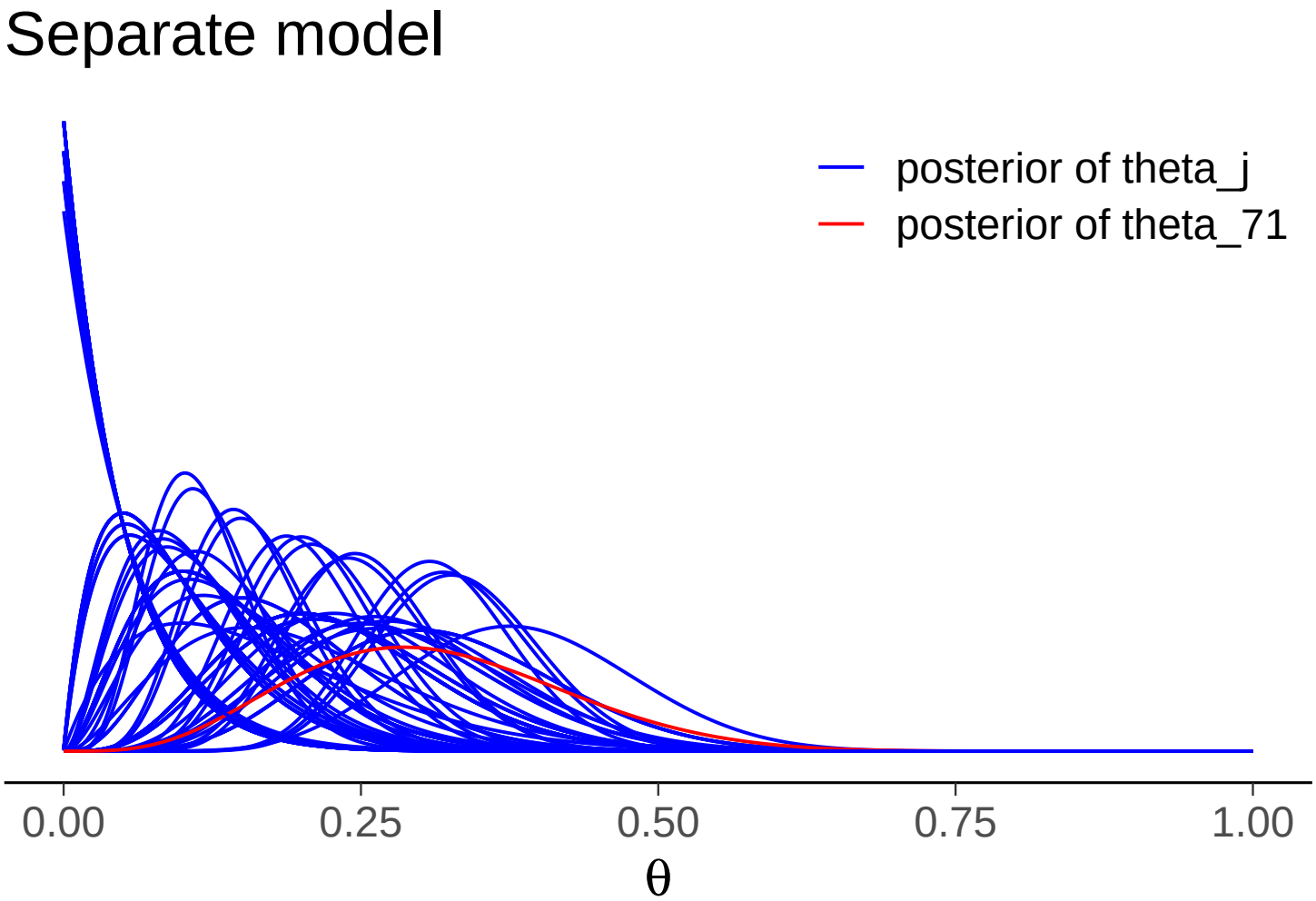
- Medicine testing
- Type F344 female rats in control group given placebo
  - count how many get endometrial stromal polyps
  - familiar binomial model example
- Experiment has been repeated 71 times

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/19 | 0/19 | 0/19 |
| 0/19 | 0/18 | 0/18 | 0/17 | 1/20 | 1/20 | 1/20 | 1/20 | 1/19 | 1/19 |
| 1/18 | 1/18 | 2/25 | 2/24 | 2/23 | 2/20 | 2/20 | 2/20 | 2/20 | 2/20 |
| 2/20 | 1/10 | 5/49 | 2/19 | 5/46 | 3/27 | 2/17 | 7/49 | 7/47 | 3/20 |
| 3/20 | 2/13 | 9/48 | 10/50 | 4/20 | 4/20 | 4/20 | 4/20 | 4/20 | 4/20 |
| 4/20 | 10/48 | 4/19 | 4/19 | 4/19 | 5/22 | 11/46 | 12/49 | 5/20 | 5/20 |
| 6/23 | 5/19 | 6/22 | 6/20 | 6/20 | 6/20 | 16/52 | 15/46 | 15/47 | 9/24 |
| 4/14 | | | | | | | | | |

# Hierarchical binomial model: rats



# Hierarchical binomial model: rats

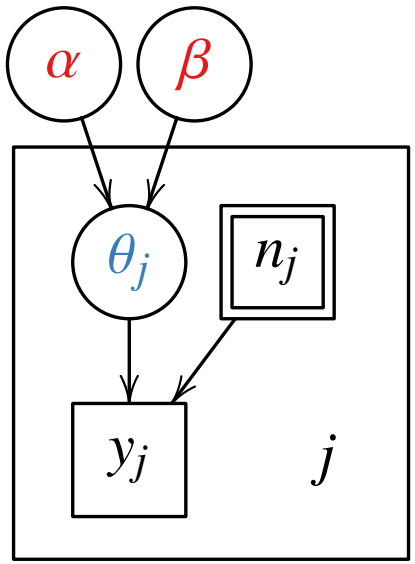


# Hierarchical binomial model: rats

- Hierarchical binomial model for rats
  prior parameters $\alpha$ and $\beta$ are unknown

$$\theta_j \mid \alpha, \beta \sim \text{Beta}(\theta_j \mid \alpha, \beta)$$

$$y_j \mid n_j, \theta_j \sim \text{Bin}(y_j \mid n_j, \theta_j)$$



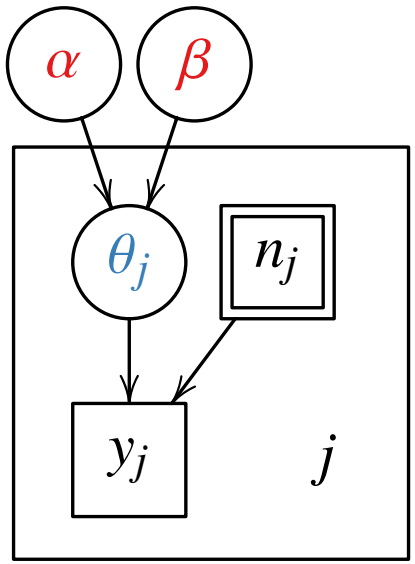

- Joint posterior $p(\theta_1, \ldots, \theta_J, \alpha, \beta \mid y)$
  - multiple parameters

# Hierarchical binomial model: rats

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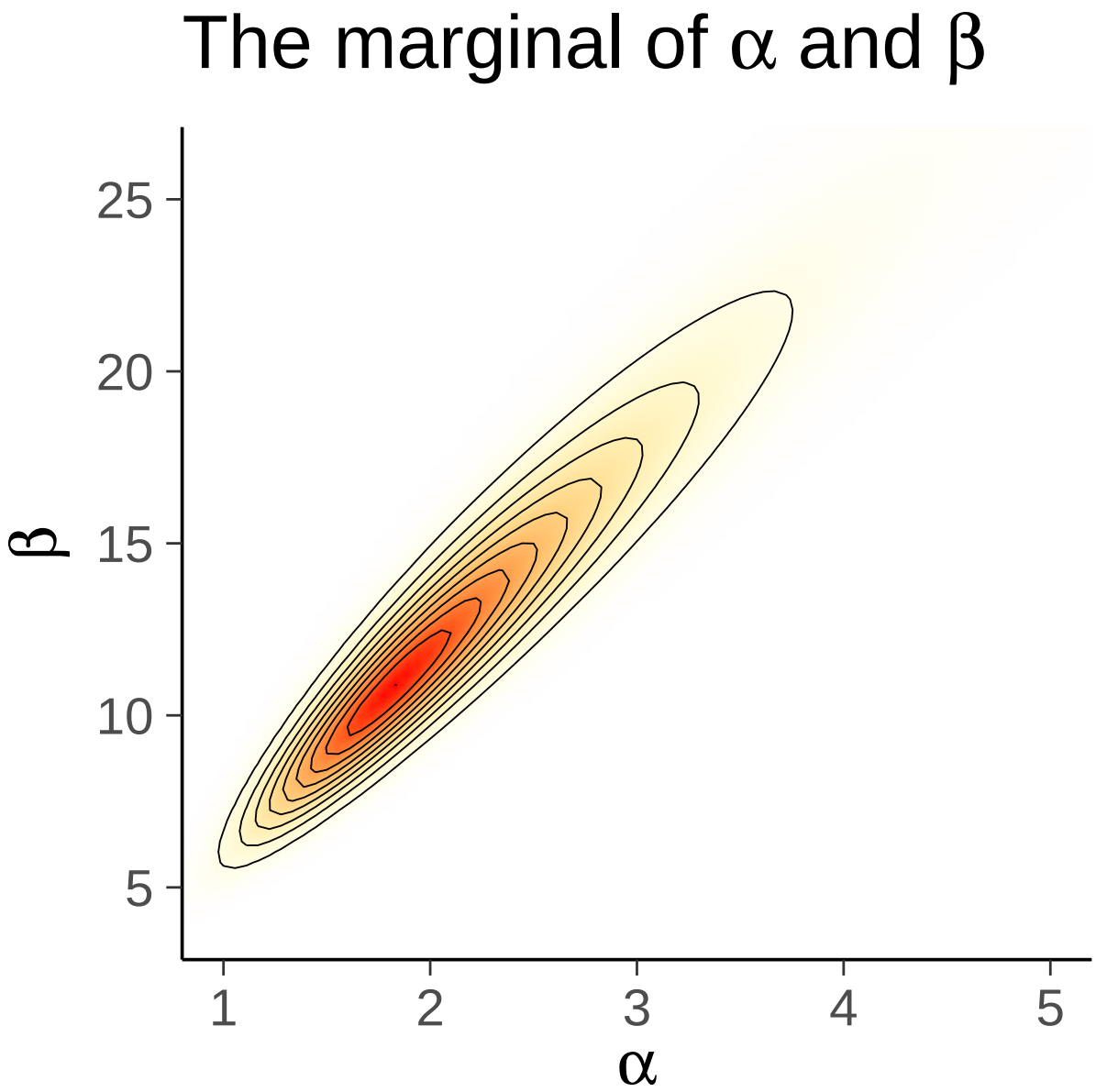



- Joint posterior $p(\theta_1, \ldots, \theta_J, \alpha, \beta \mid y)$
  - multiple parameters
  - factorize $\prod_{j=1}^{J} p(\theta_j \mid \alpha, \beta, y) p(\alpha, \beta \mid y)$

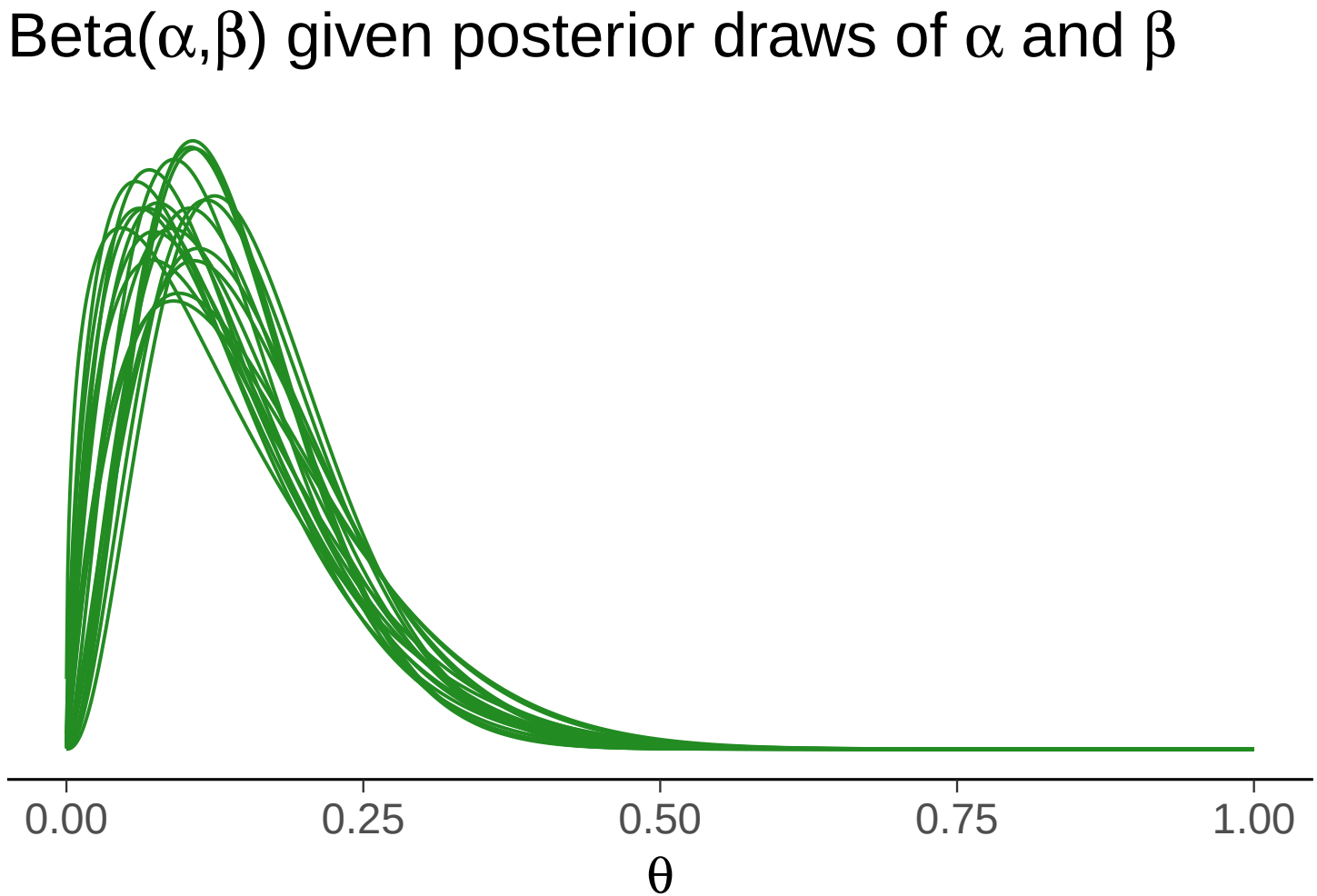
# Hierarchical binomial model: rats

- Population prior $\mathrm{Beta}(\theta_j \mid \alpha, \beta)$
- Hyperprior $p(\alpha, \beta)$?
  - $\alpha, \beta$ both affect the location and scale
  - BDA3 has $p(\alpha, \beta) \propto (\alpha + \beta)^{-5/2}$
    - diffuse prior for location and scale (BDA3 p. 110)
- demo5_1

# Hierarchical binomial model: rats

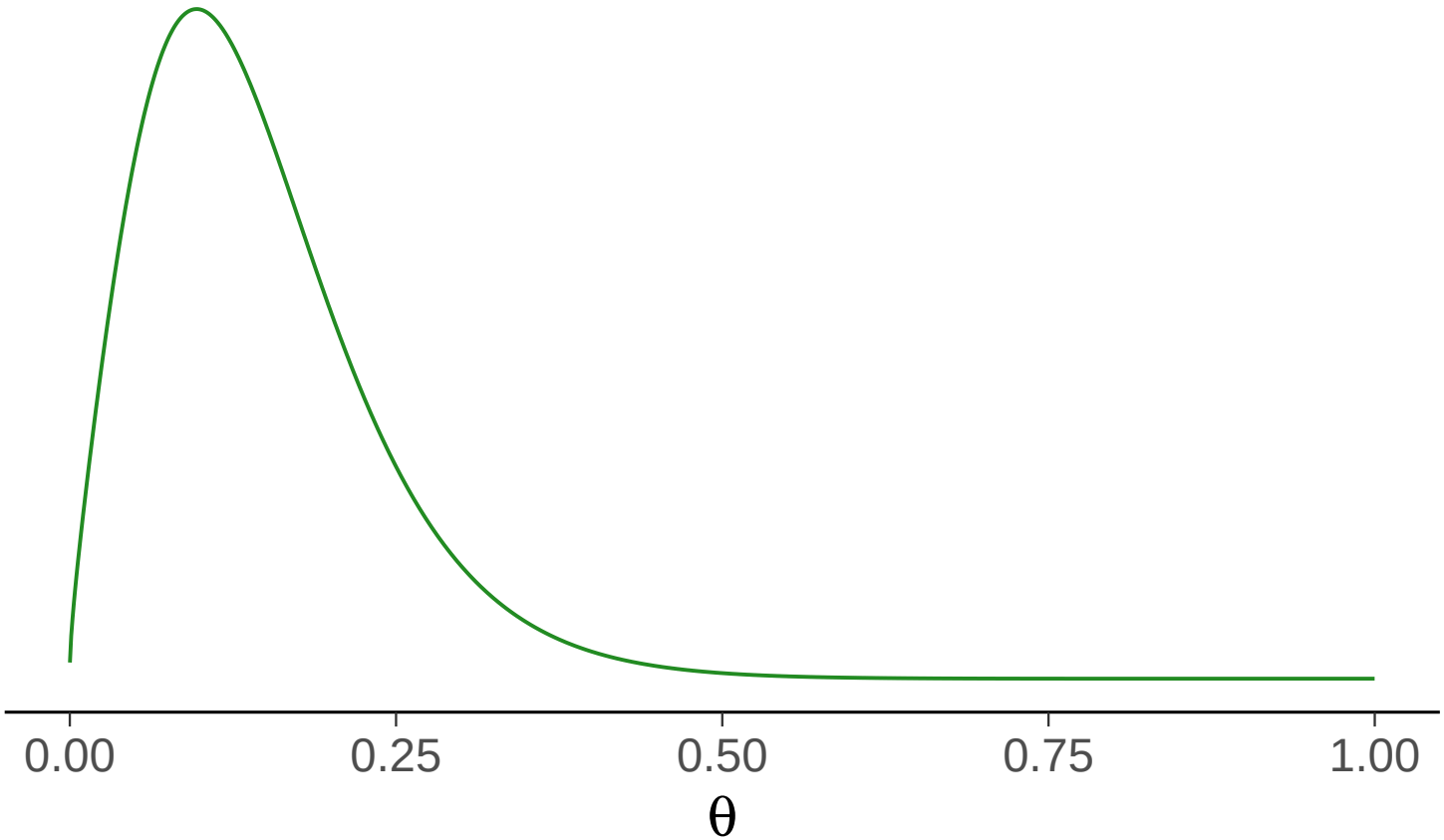


# Hierarchical binomial model: rats



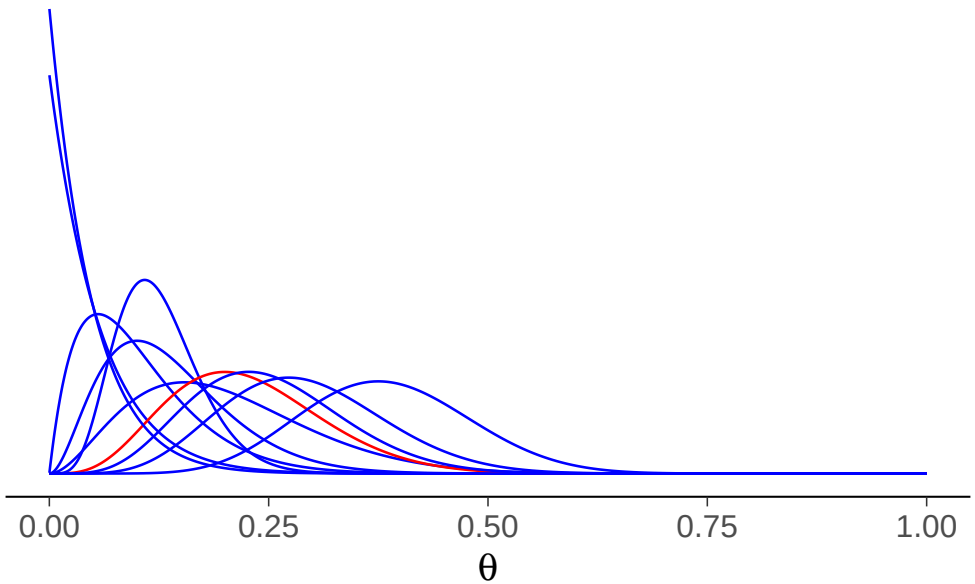
# Hierarchical binomial model: rats

Population distribution (prior) for $\theta_j$



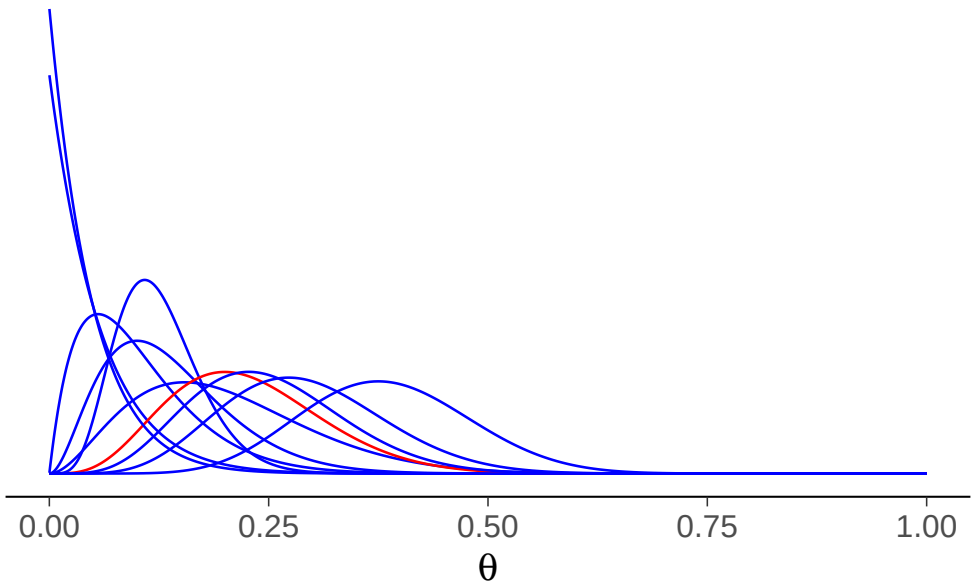
# Hierarchical binomial model: rats

Separate model

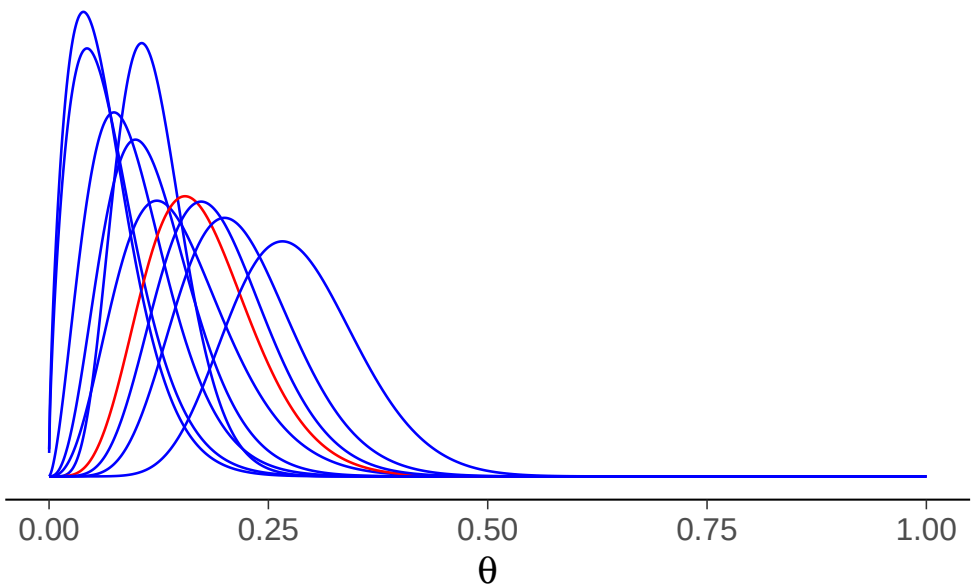


# Hierarchical binomial model: rats

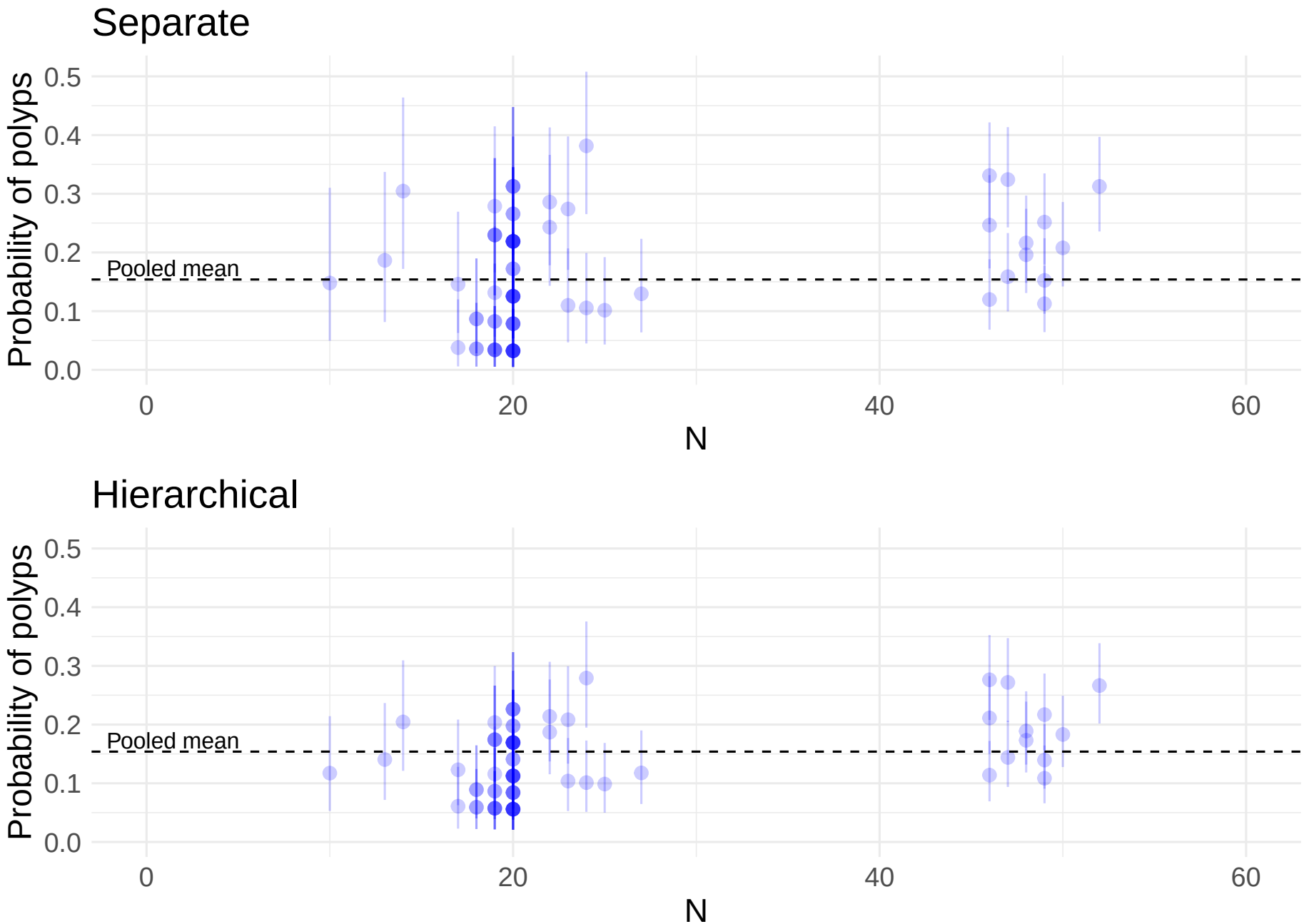
Separate model




Hierarchical model



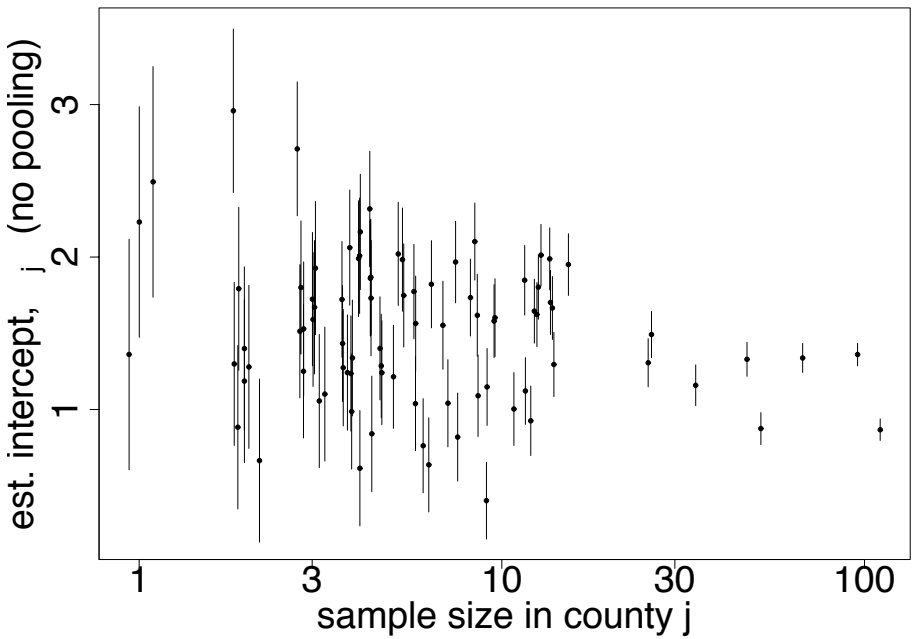
# Hierarchical model and group size: Rats



# Hierarchical model and group size: Radon

919 home radon levels in 85 counties in Minnesota:

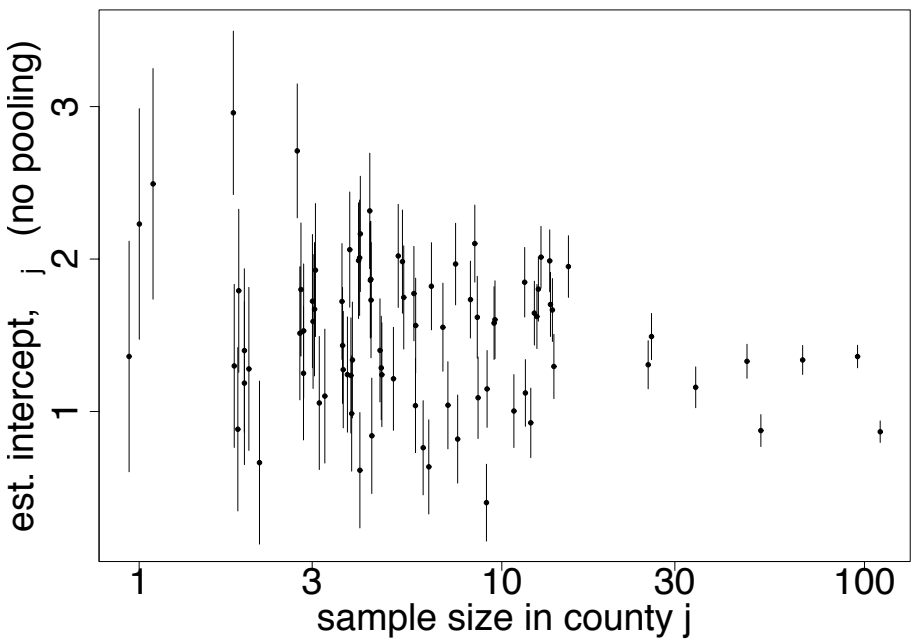
Separate



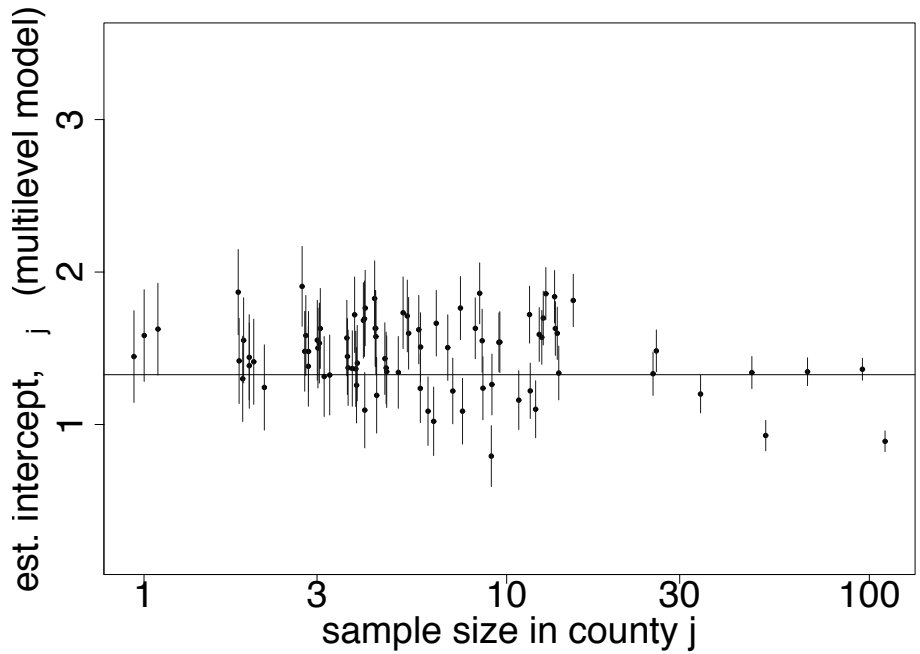
# Hierarchical model and group size: Radon

919 home radon levels in 85 counties in Minnesota:

Separate




Hierarchical



# Stan code for hierarchical model

Diet effect on chicken weights (at age 12 days)

```
data {
  int<lower=1> N_observations;
  int<lower=1> N_diets;

  vector[N_observations] weight;
  vector[N_observations] diet_idx;
}
```

# Diet effect on chicken weights (at age 12 days)

Hierarchical model with

- unknown mean weight per diet
- unknown prior mean and sd
- unknown residual sigma

```
parameters {
  vector[N_diets] mu_diet;
  real mu_0;
  real<lower=0> sd;
  real<lower=0> sigma;
}
```

# Stan hierarchical model

`diet_idx` is a vector with each element indicating the group

```
model {
  // Priors
  for (diet in 1:N_diets) {
    mu_diet[diet] ~ normal(mu_0, sd);
  }
  mu_0 ~ normal(0, 1);
  sd ~ normal(0, 1)
  sigma ~ normal(0, 1);

  // Observation model
  for (obs in 1:N_observations) {
    weight[obs] ~ normal(mu_diet[diet_idx[obs]], sigma);
  }
}
```

# Stan hierarchical model

mu_0 and sd are the population mean and sd

```
model {
  // Priors
  for (diet in 1:N_diets) {
    mu_diet[diet] ~ normal(mu_0, sd);
  }
  mu_0 ~ normal(0, 1);
  sd ~ normal(0, 1)
  sigma ~ normal(0, 1);

  // Observation model
  for (obs in 1:N_observations) {
    weight[obs] ~ normal(mu_diet[diet_idx[obs]], sigma);
  }
}
```

# Stan hierarchical model

sd is constrained to be positive and thus the prior is half-normal

```
model {
  // Priors
  for (diet in 1:N_diets) {
    mu_diet[diet] ~ normal(mu_0, sd);
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  // Observation model
  for (obs in 1:N_observations) {
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  }
}
```

# Stan hierarchical model

`sigma` is constrained to be positive and thus the prior is half-normal

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model {
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  }
  mu_0 ~ normal(0, 1);
  sd ~ normal(0, 1)
  sigma ~ normal(0, 1);

  // Observation model
  for (obs in 1:N_observations) {
    weight[obs] ~ normal(mu_diet[diet_idx[obs]], sigma);
  }
}
```

# Stan without loops

## Vectorized statements

```
model {
  // Priors
  mu_diet ~ normal(mu_0, sd);
  mu_0 ~ normal(0, 1);
  sd ~ normal(0, 1)
  sigma ~ normal(0, 1);

  // Observation model
  weight ~ normal(mu_diet[diet_idx], sigma);
}
```

# Stan vs brms

## Stan

```
model {
  // Priors
  mu_diet ~ normal(mu_0, sd);
  mu_0 ~ normal(0, 1);
  sd ~ normal(0, 1)
  sigma ~ normal(0, 1);

  // Observation model
  weight ~ normal(mu_diet[diet_idx], sigma);
}
```

---

## brms formula

```
brm(weight ~ 1 + (1 | Diet),
```

# Stan vs brms

## Stan

```
model {
  // Priors
  mu_diet ~ normal(mu_0, sd);
  mu_0 ~ normal(0, 1);
  sd ~ normal(0, 1)
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  weight ~ normal(mu_diet[diet_idx], sigma);
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# Stan vs brms

## Stan

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model {
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  mu_diet ~ normal(mu_0, sd);
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  weight ~ normal(mu_diet[diet_idx], sigma);
}
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# Stan vs brms

## Stan

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  sd ~ normal(0, 1)
  sigma ~ normal(0, 1);

  // Observation model
  weight ~ normal(mu_diet[diet_idx], sigma);
}
```

---

## brms formula

```
brm(weight ~ 1 + (1 | Diet), data=Chick12,
```

# Stan vs brms

## Stan

```
model {
  // Priors
  mu_diet ~ normal(mu_0, sd);
  mu_0 ~ normal(0, 1);
  sd ~ normal(0, 1)
  sigma ~ normal(0, 1);

  // Observation model
  weight ~ normal(mu_diet[diet_idx], sigma);
}
```

---

## brms formula

```
brm(weight ~ 1 + (1 | Diet), data=Chick12,
    prior=c(prior(normal(0,1), class="Intercept"), # p(mu_0)
            prior(normal(0,1), class="sd"),        # p(tau)
            prior(normal(0,1), class="sigma")))    # p(sigma)
```

# Stan vs brms

## Stan

```
model {
  // Priors
  mu_diet ~ normal(mu_0, sd);
  mu_0 ~ normal(0, 1);
  sd ~ normal(0, 1)
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  // Observation model
  weight ~ normal(mu_diet[diet_idx], sigma);
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## Stan

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model {
  // Priors
  mu_diet ~ normal(mu_0, sd);
  mu_0 ~ normal(0, 1);
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  // Observation model
  weight ~ normal(mu_diet[diet_idx], sigma);
}
```

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## brms formula

```
brm(weight ~ 1 + (1 | Diet), data=Chick12,
    prior=c(prior(normal(0,1), class="Intercept"), # p(mu_0)
            prior(normal(0,1), class="sd"),        # p(tau)
            prior(normal(0,1), class="sigma")))    # p(sigma)
```

# brms generated Stan code

```
// generated with brms 2.22.1
data {
  int<lower=1> N;  // total number of observations
  vector[N] Y;  // response variable
  // data for group-level effects of ID 1
  int<lower=1> N_1;  // number of grouping levels
  int<lower=1> M_1;  // number of coefficients per level
  array[N] int<lower=1> J_1;  // grouping indicator per observation
  // group-level predictor values
  vector[N] Z_1_1;
  int prior_only;  // should the likelihood be ignored?
}
```

# brms generated Stan code

```
parameters {
  real Intercept;  // temporary intercept for centered predictors
  real<lower=0> sigma;  // dispersion parameter
  vector<lower=0>[M_1] sd_1;  // group-level standard deviations
  array[M_1] vector[N_1] z_1;  // standardized group-level effects
}
transformed parameters {
  vector[N_1] r_1_1;  // actual group-level effects
  real lprior = 0;  // prior contributions to the log posterior
  r_1_1 = (sd_1[1] * (z_1[1]));
  lprior += normal_lpdf(Intercept | 0, 1);
  lprior += normal_lpdf(sigma | 0, 1)
    - 1 * normal_lccdf(0 | 0, 1);
  lprior += normal_lpdf(sd_1 | 0, 1)
    - 1 * normal_lccdf(0 | 0, 1);
}
```

# brms generated Stan code

```
parameters {
  real Intercept;  // temporary intercept for centered predictors
  real<lower=0> sigma;  // dispersion parameter
  vector<lower=0>[M_1] sd_1;  // group-level standard deviations
  array[M_1] vector[N_1] z_1;  // standardized group-level effects
}
transformed parameters {
  vector[N_1] r_1_1;  // actual group-level effects
  real lprior = 0;  // prior contributions to the log posterior
  r_1_1 = (sd_1[1] * (z_1[1]));
  lprior += normal_lpdf(Intercept | 0, 1);
  lprior += normal_lpdf(sigma | 0, 1)
    - 1 * normal_lccdf(0 | 0, 1);
  lprior += normal_lpdf(sd_1 | 0, 1)
    - 1 * normal_lccdf(0 | 0, 1);
}
```

# brms generated Stan code

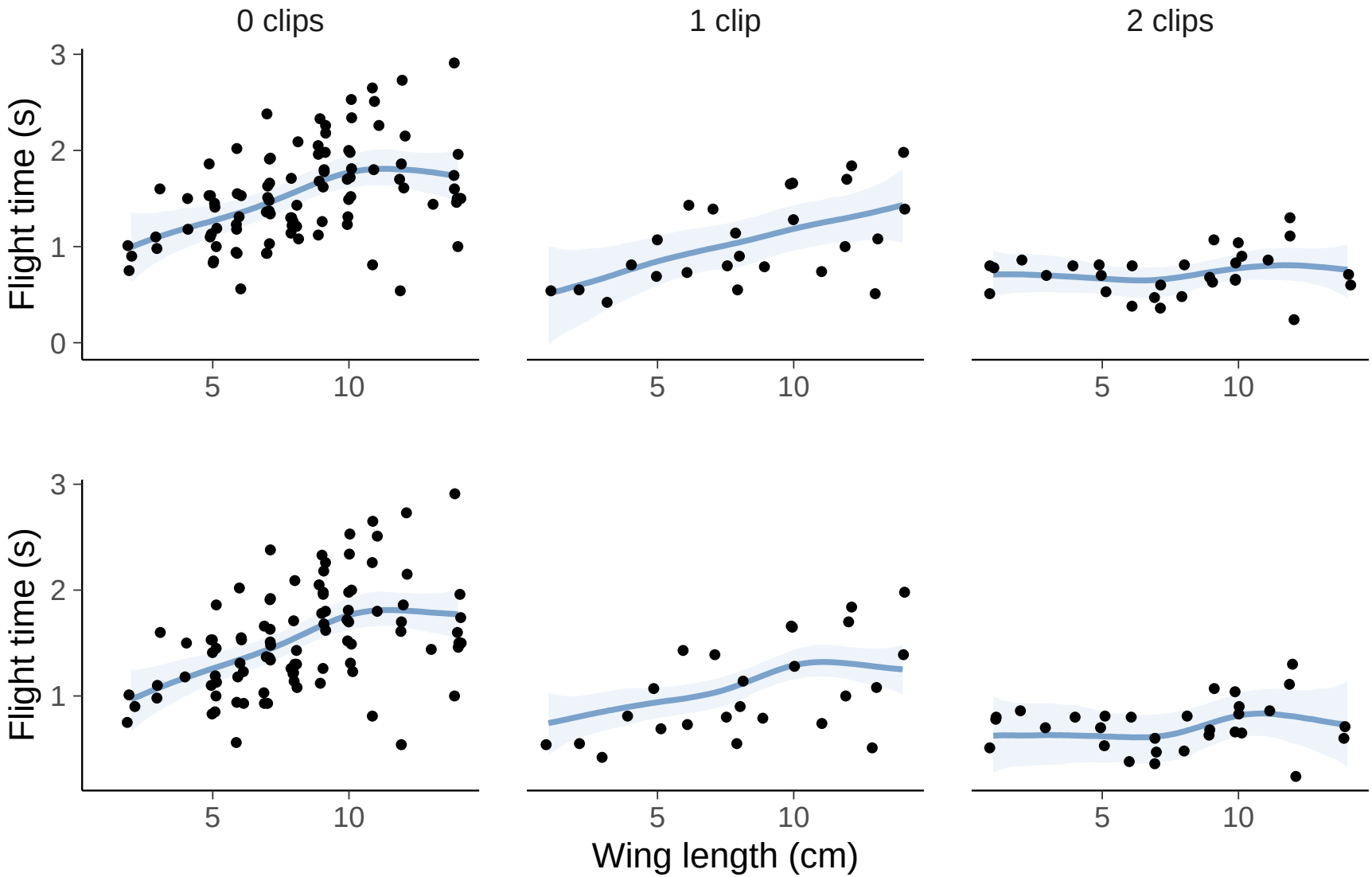
```
model {
  // likelihood including constants
  if (!prior_only) {
    // initialize linear predictor term
    vector[N] mu = rep_vector(0.0, N);
    mu += Intercept;
    for (n in 1:N) {
      // add more terms to the linear predictor
      mu[n] += r_1_1[J_1[n]] * Z_1_1[n];
    }
    target += normal_lpdf(Y | mu, sigma);
  }
  // priors including constants
  target += lprior;
  target += std_normal_lpdf(z_1[1]);
}
generated quantities {
  // actual population-level intercept
  real b_Intercept = Intercept;
}
```

# brms generated Stan code

```
model {
  // likelihood including constants
  if (!prior_only) {
    // initialize linear predictor term
    vector[N] mu = rep_vector(0.0, N);
    mu += Intercept;
    for (n in 1:N) {
      // add more terms to the linear predictor
      mu[n] += r_1_1[J_1[n]] * Z_1_1[n];
    }
    target += normal_lpdf(Y | mu, sigma);
  }
  // priors including constants
  target += lprior;
  target += std_normal_lpdf(z_1[1]);
}
generated quantities {
  // actual population-level intercept
  real b_Intercept = Intercept;
}
```

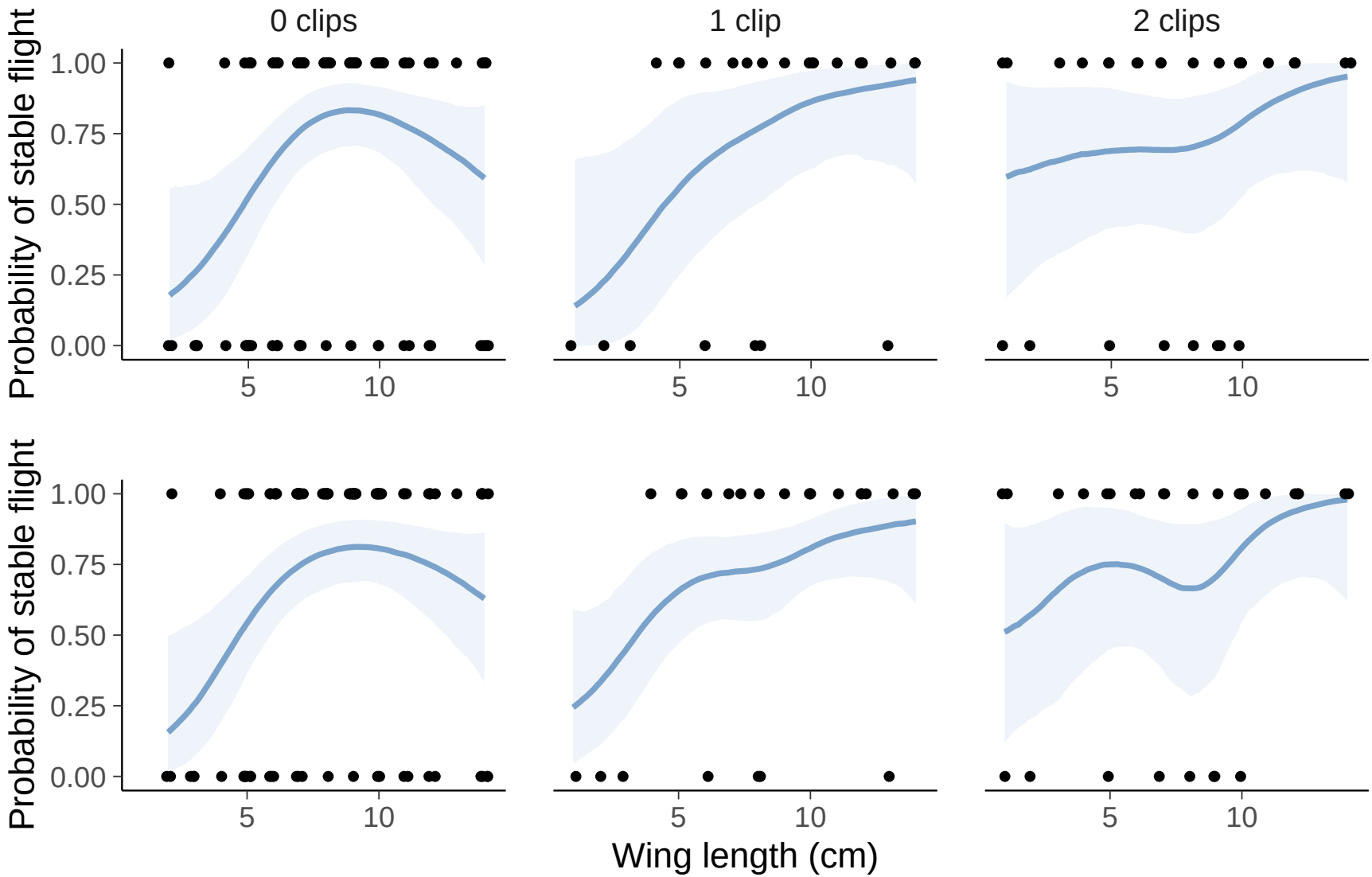
# Paper helicopters: flight time

## Separate model vs. hierarchical model



# Paper helicopters: stability

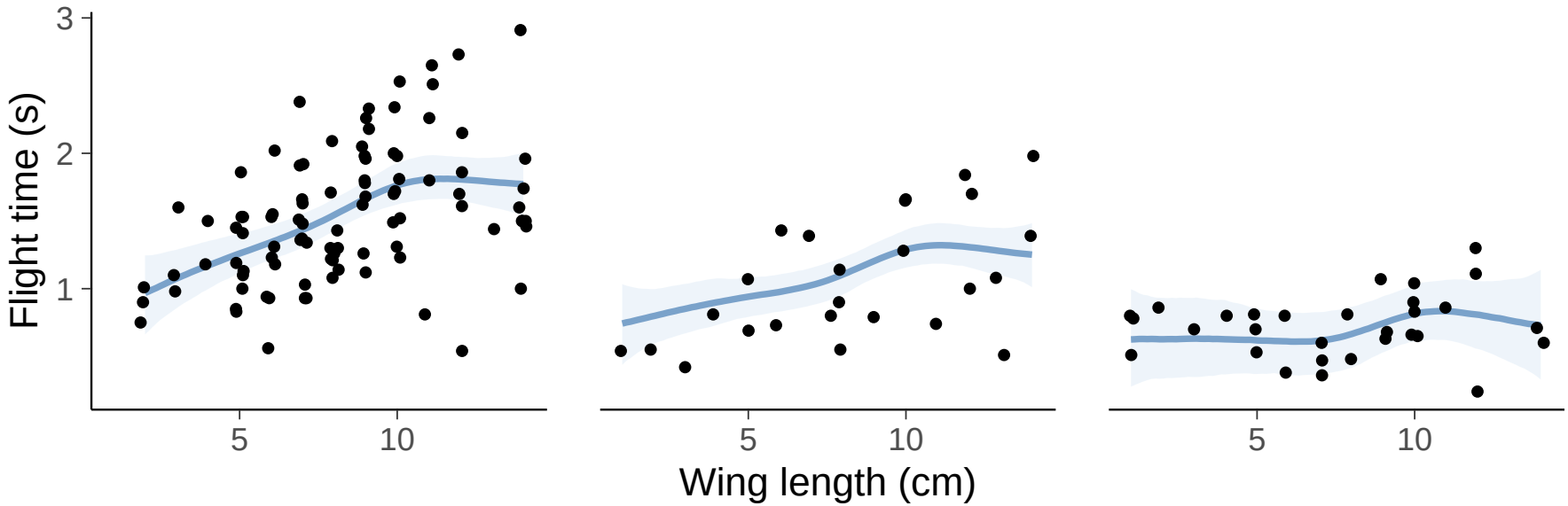
Separate model vs. hierarchical model



# Paper helicopters: brms

## Flight time

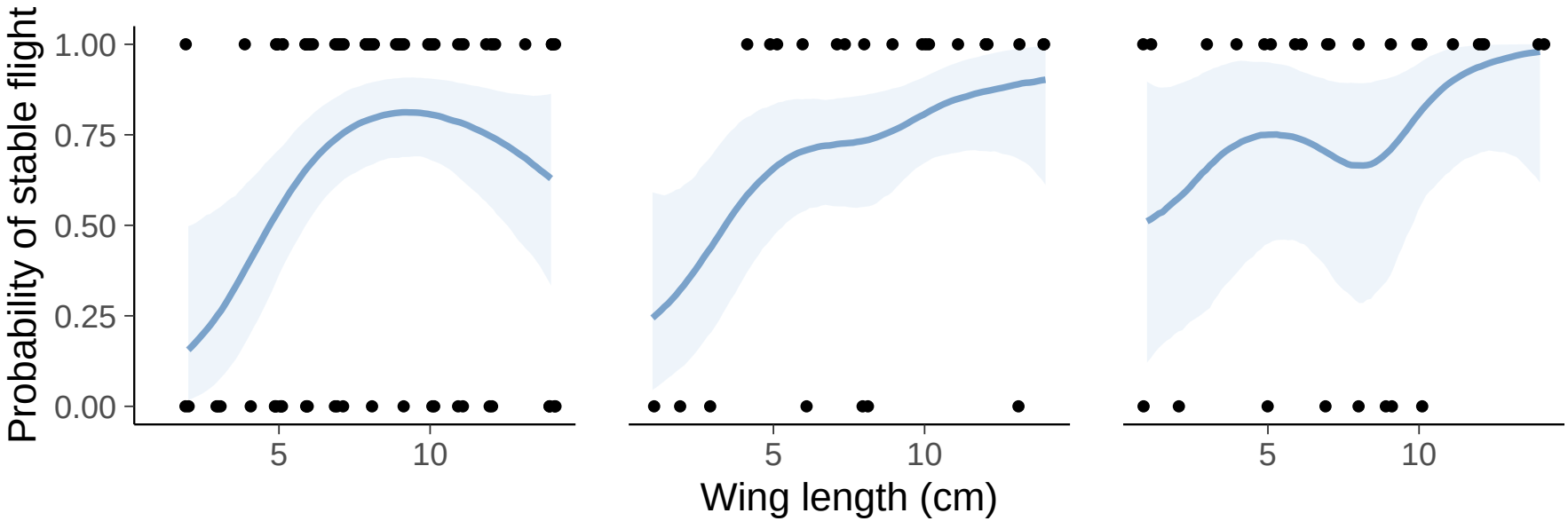
```
flight_time ~ s(wing_length) + s(wing_length, by = nclips)
```



# Paper helicopters: brms

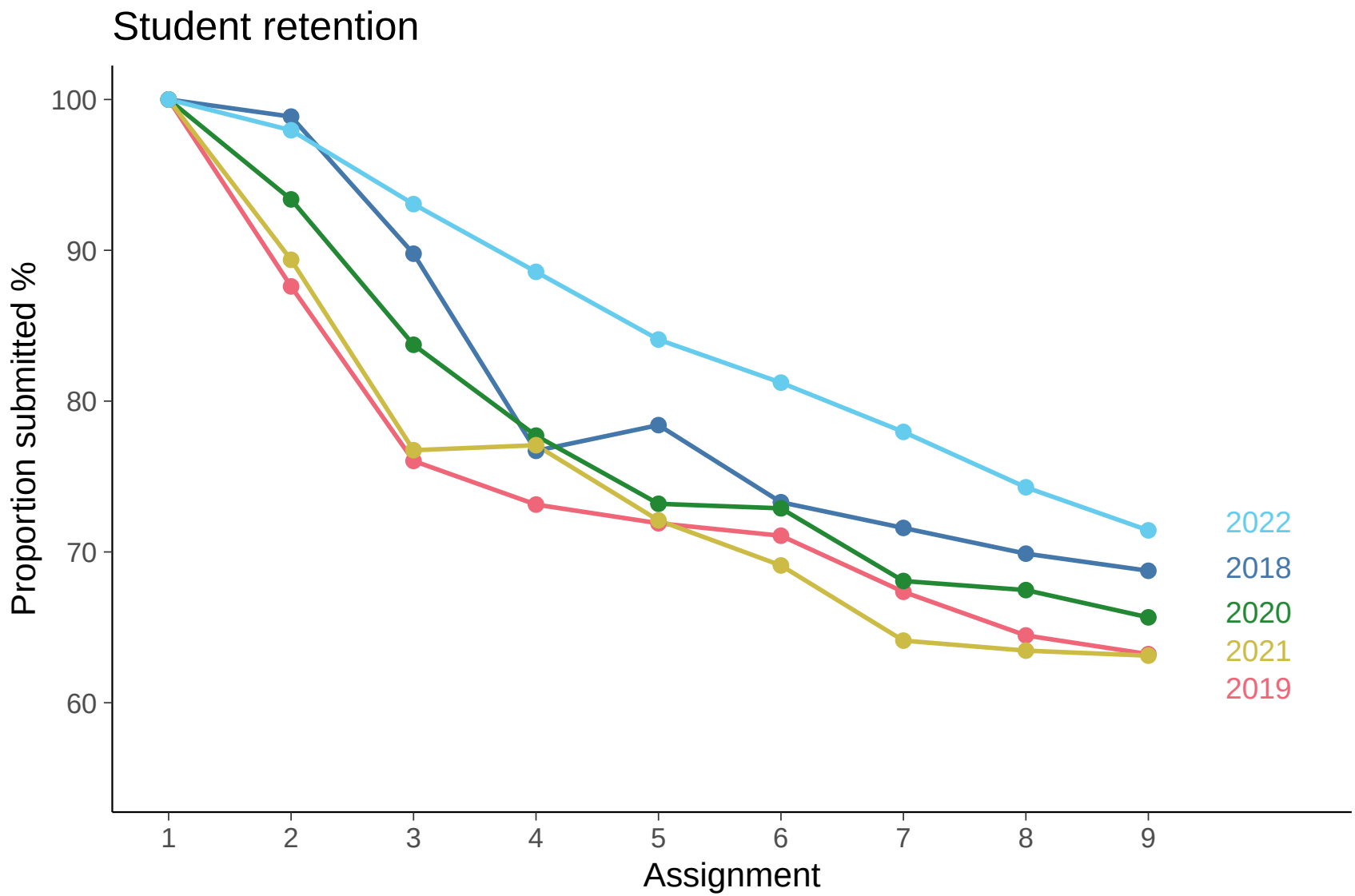
## Stability

```
stable_flight ~ s(wing_length) + s(wing_length, by = nclips),
family = bernoulli()
```

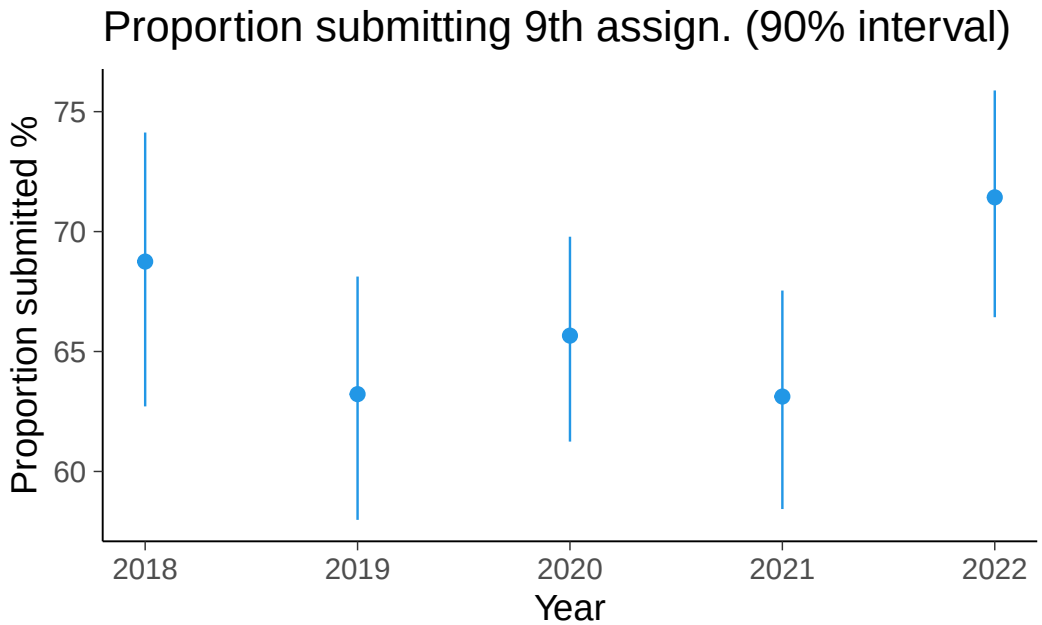


# Student retention

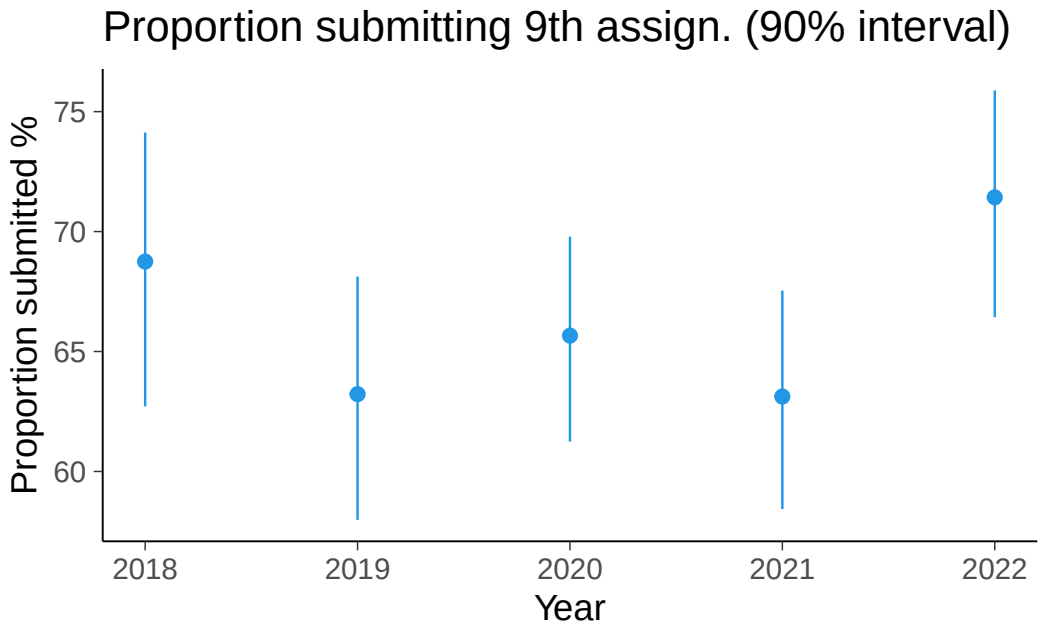
Was year 2022 better than earlier year?

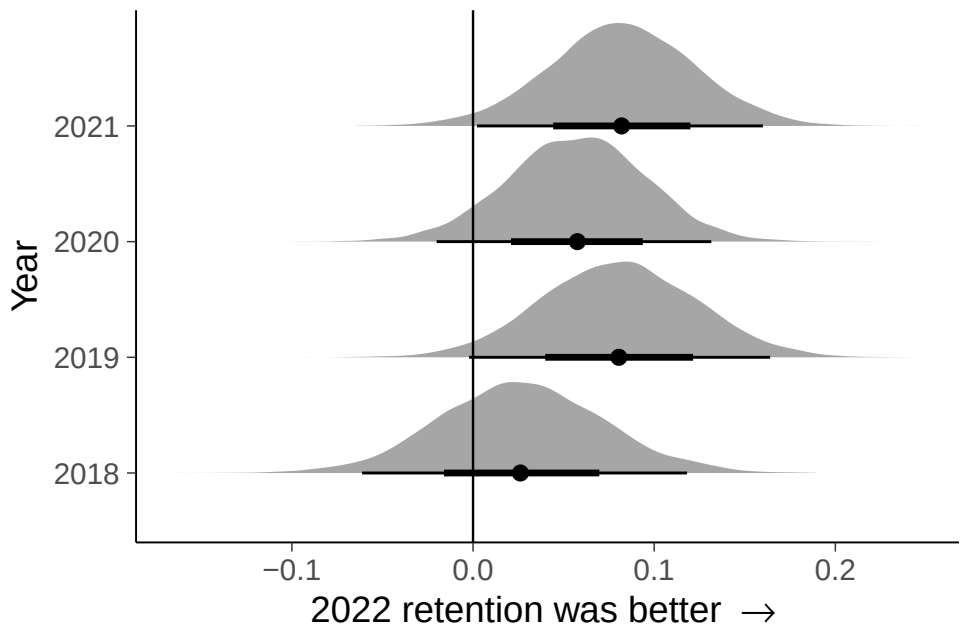


# Student retention separate model

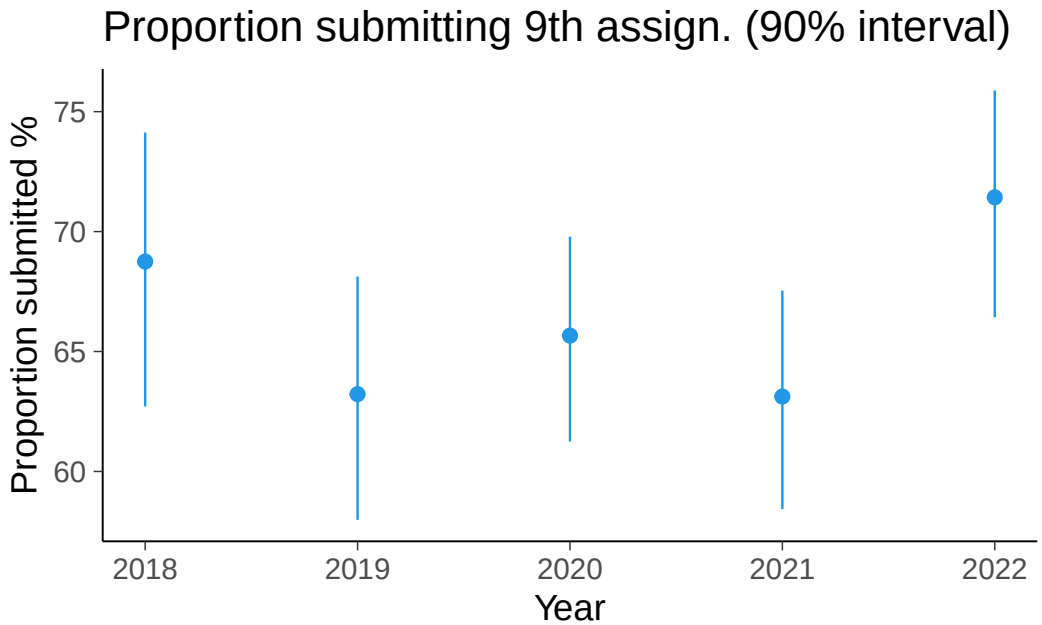


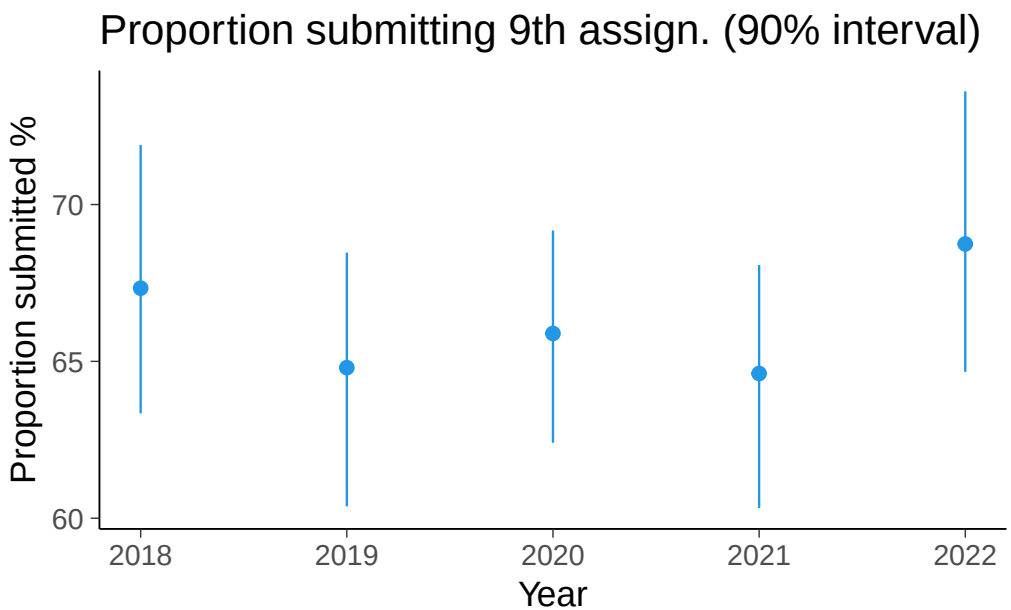
# Student retention separate model



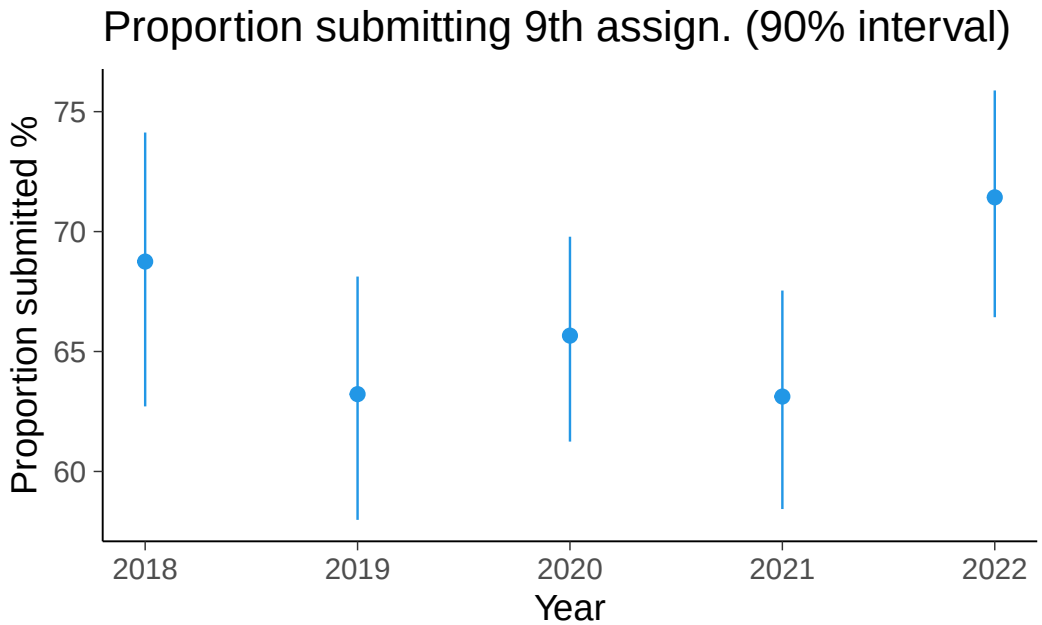



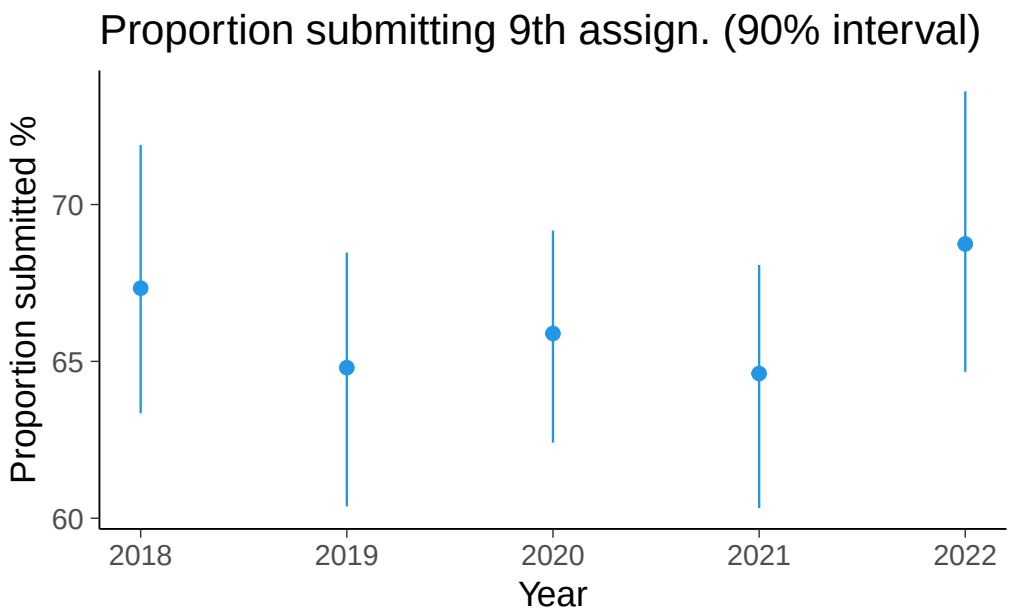
# Student retention separate vs hierarchical model



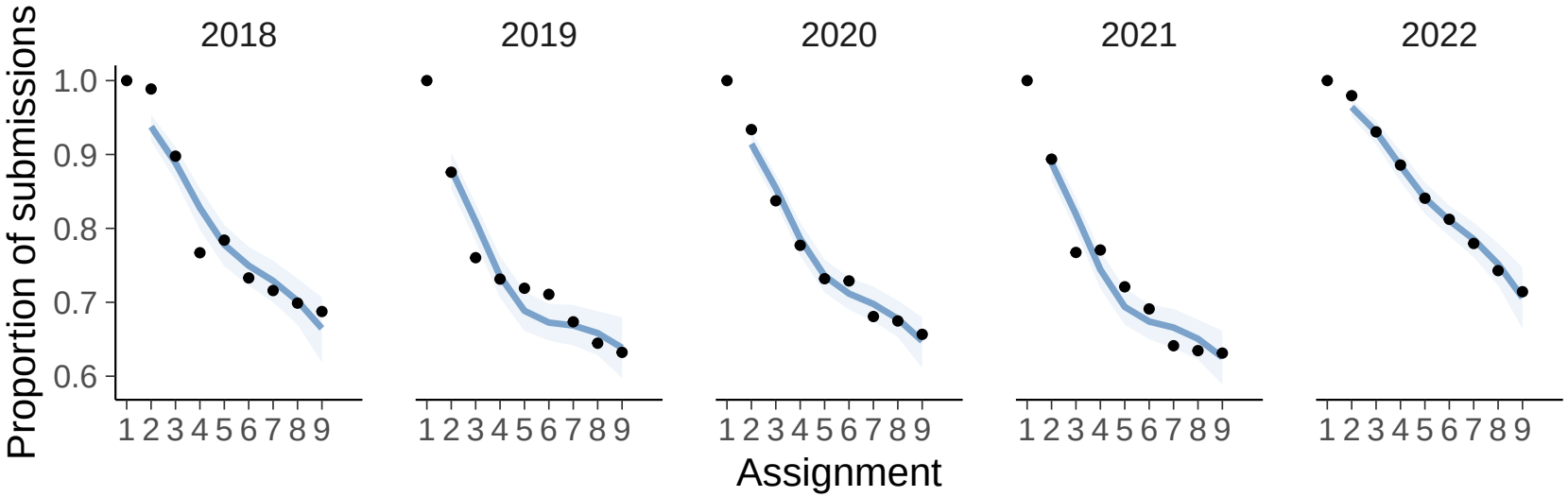
# Student retention separate vs hierarchical model



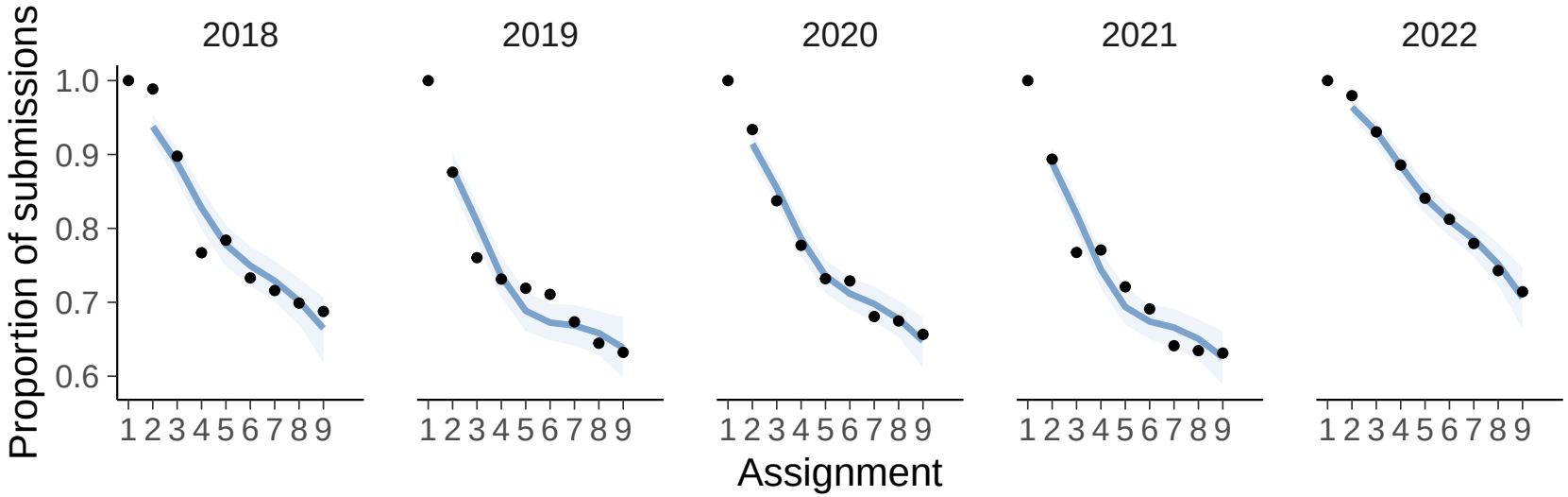

```
nstudents | trials(nstudents1) ~ 1 + (1 | year), family=binomial()
```

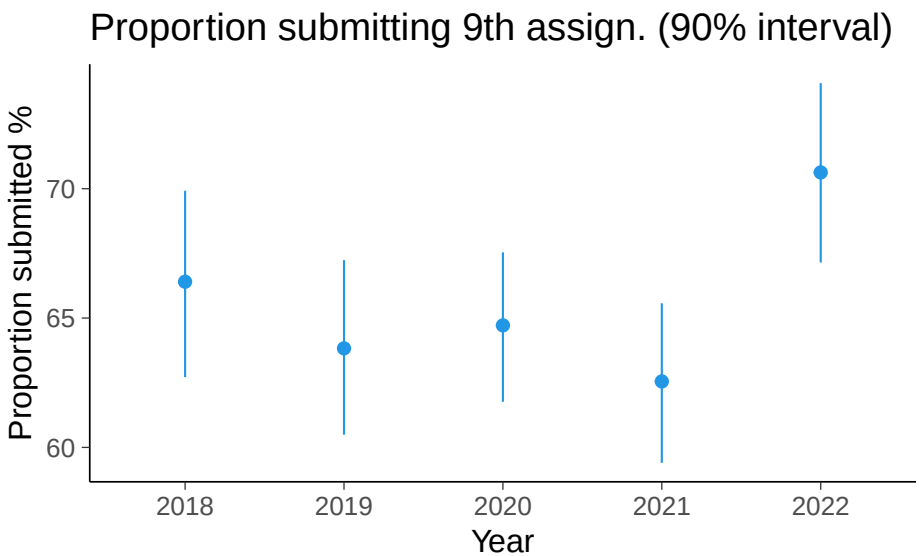
# Student retention latent spline model



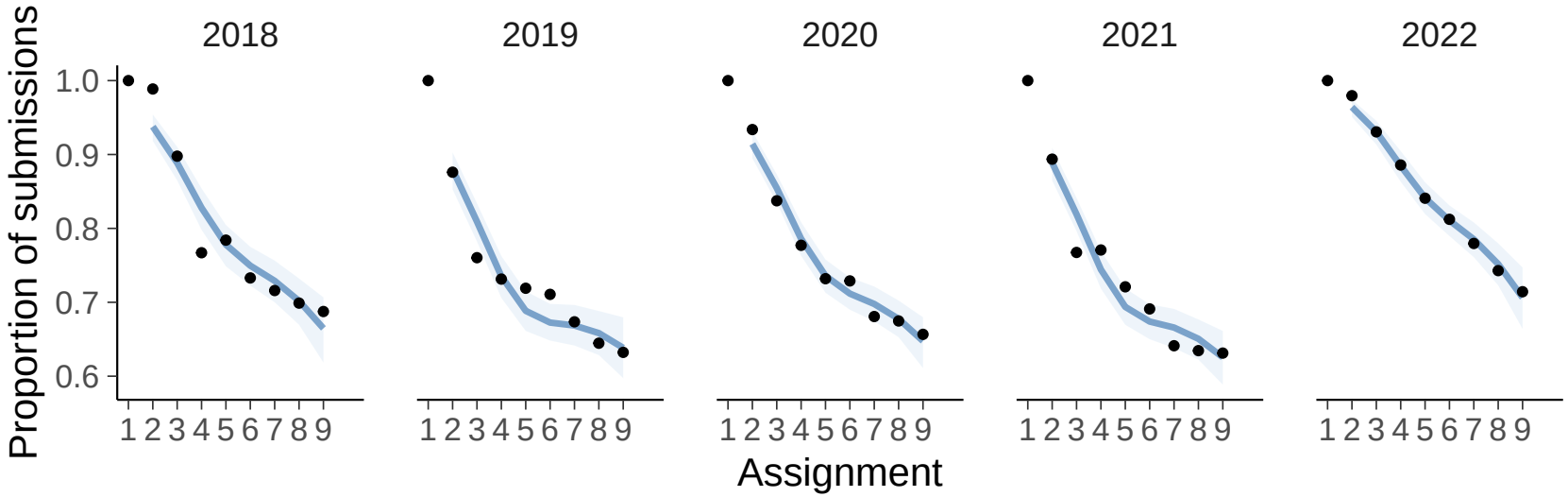

```
nstudents | trials(nstudents1) ~ s(assignment, k=4) + (assignment | year),
family=binomial()
```

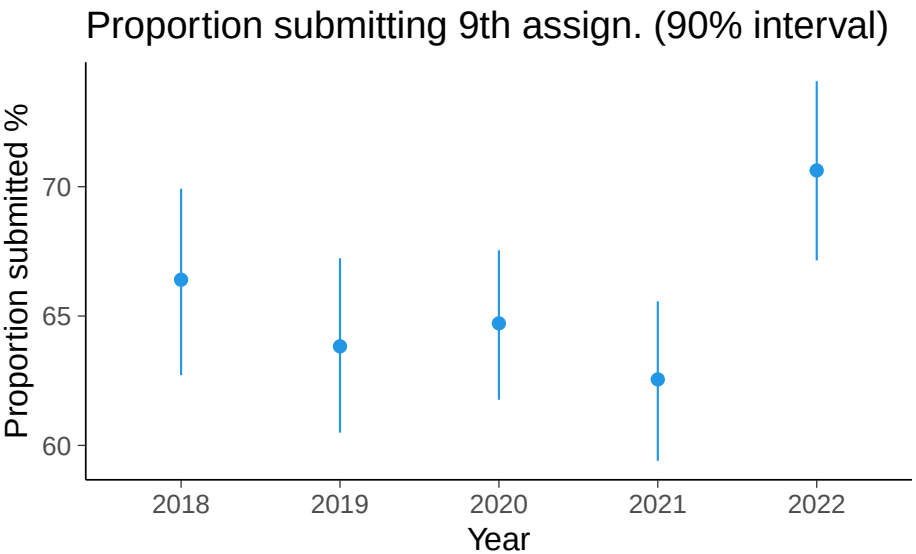
# Student retention latent spline model

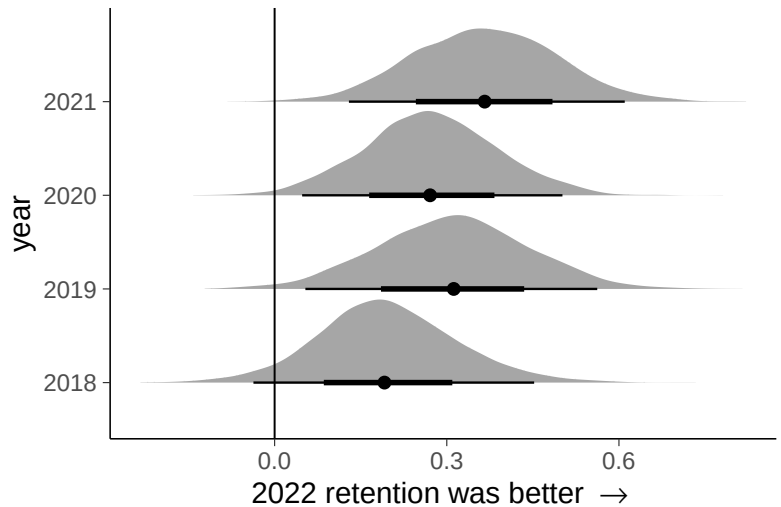


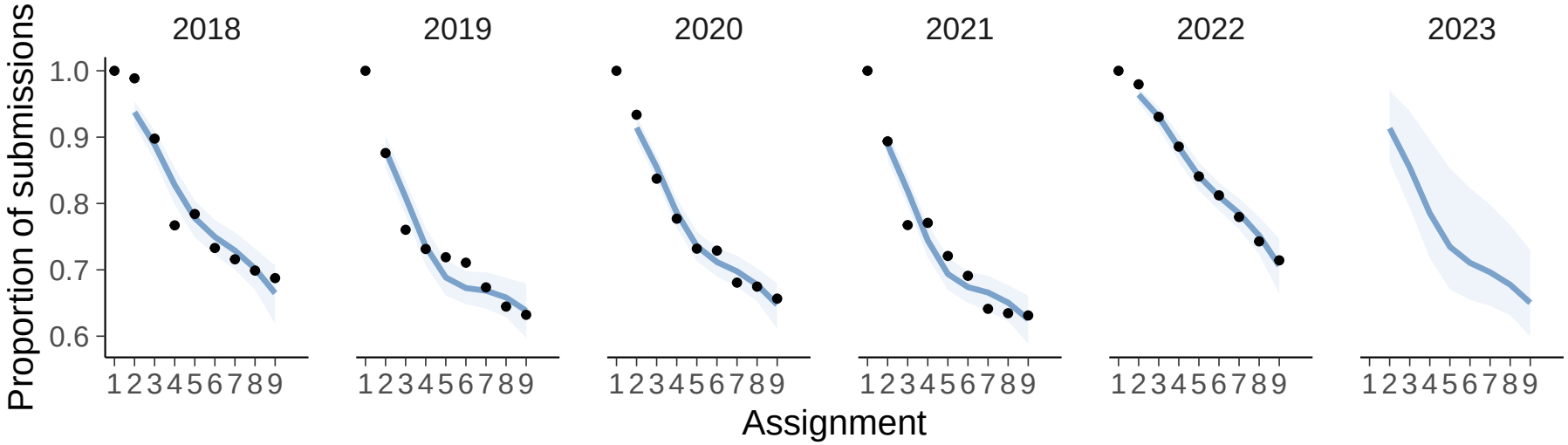
# Student retention latent spline model



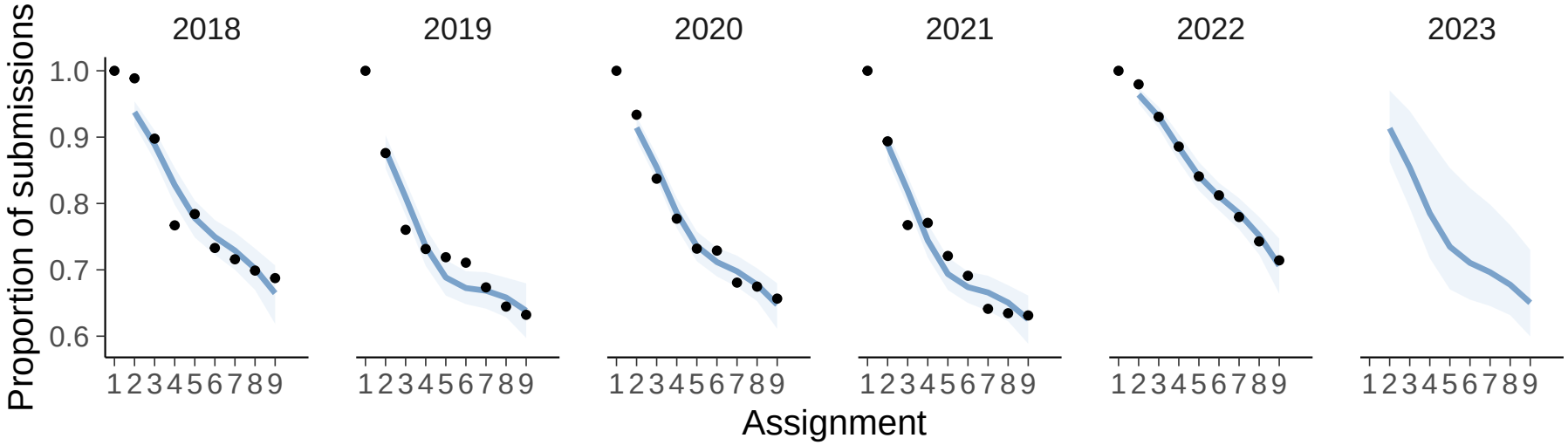



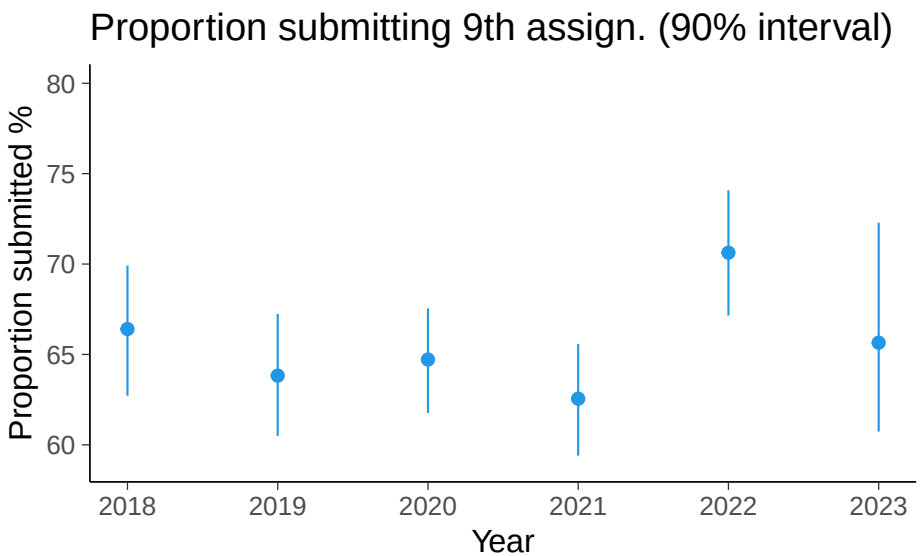



# Student retention latent spline model, year 2023?

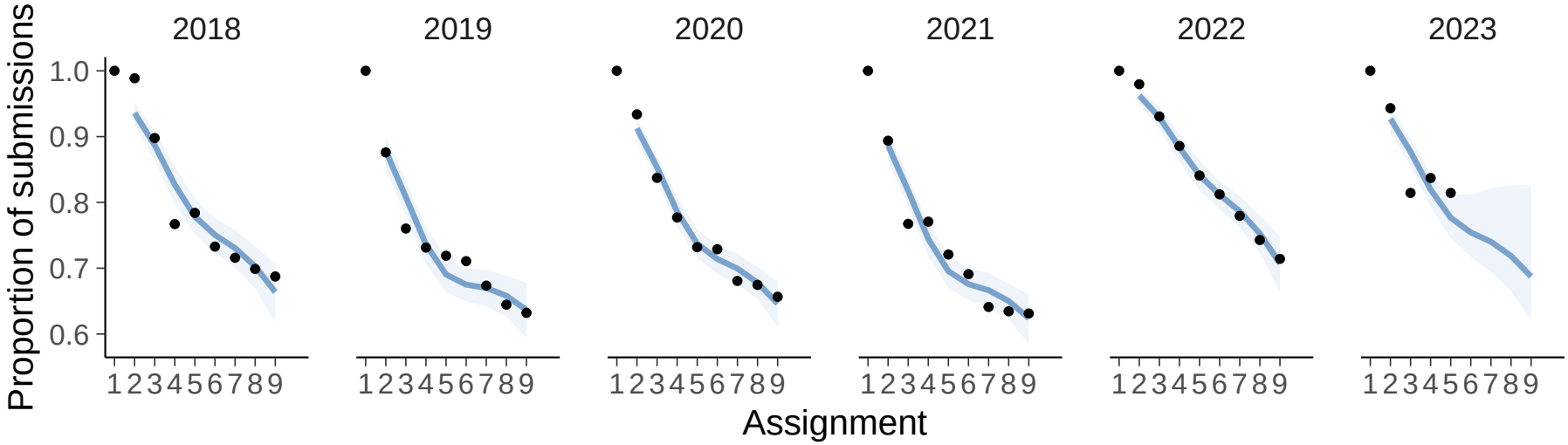


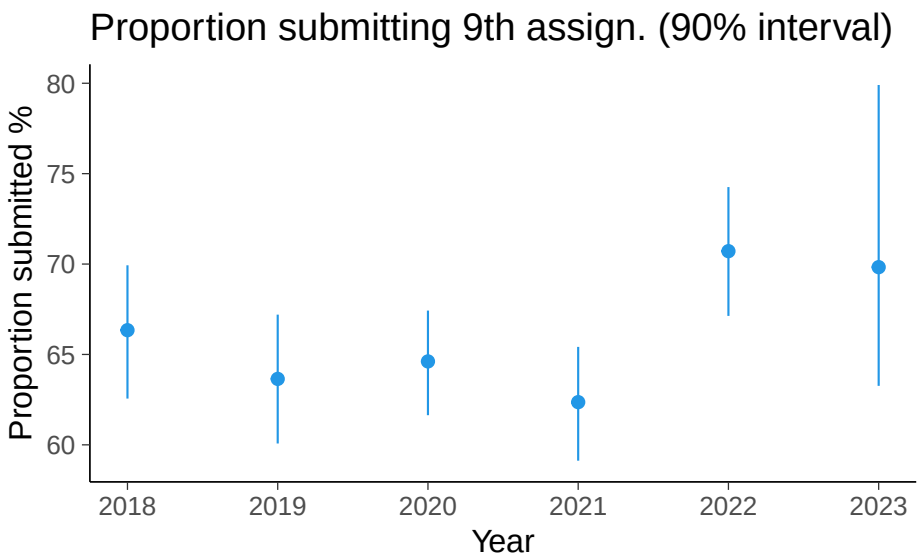
# Student retention latent spline model, year 2023?



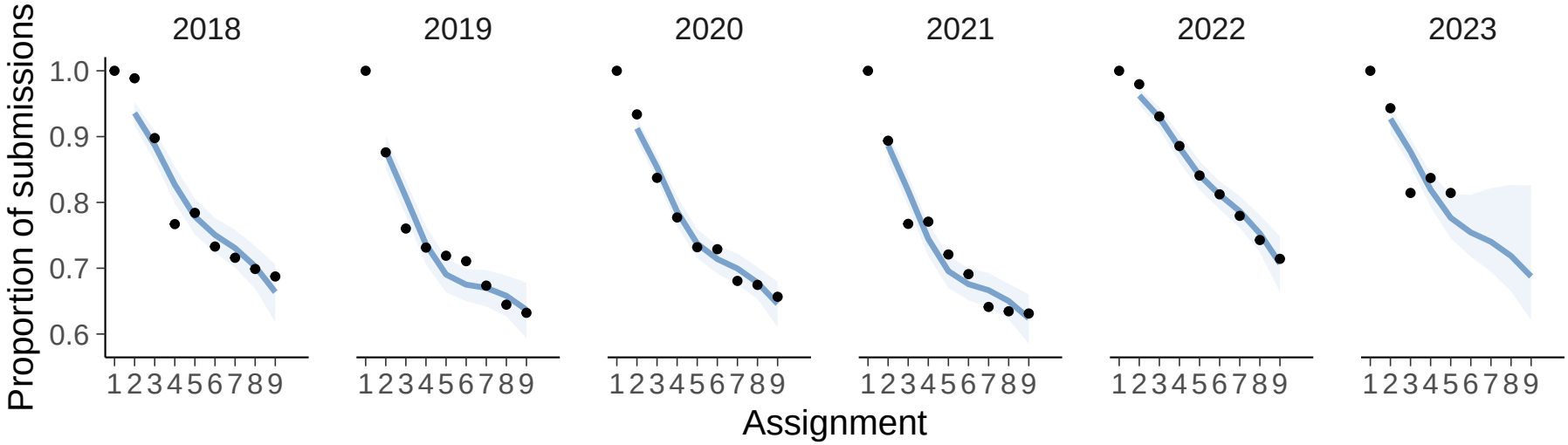



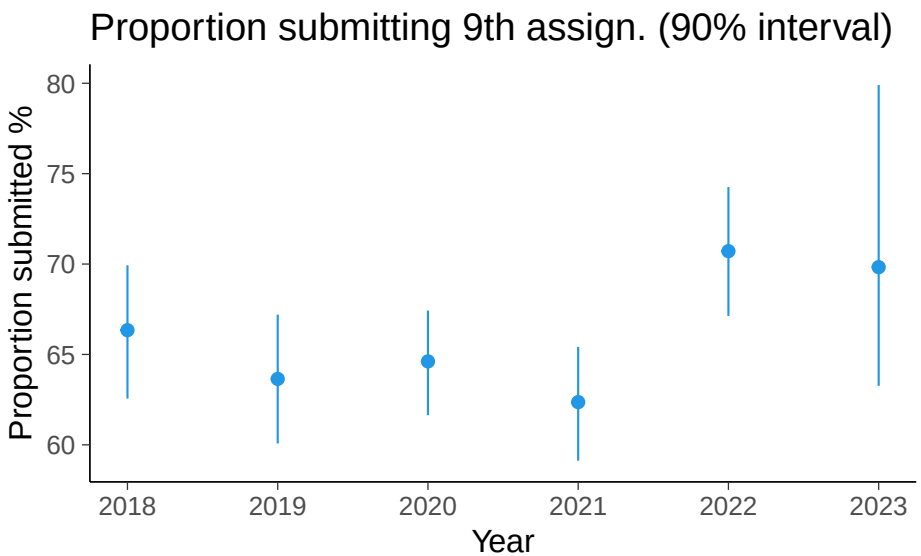
# Student retention latent spline model, year 2023?

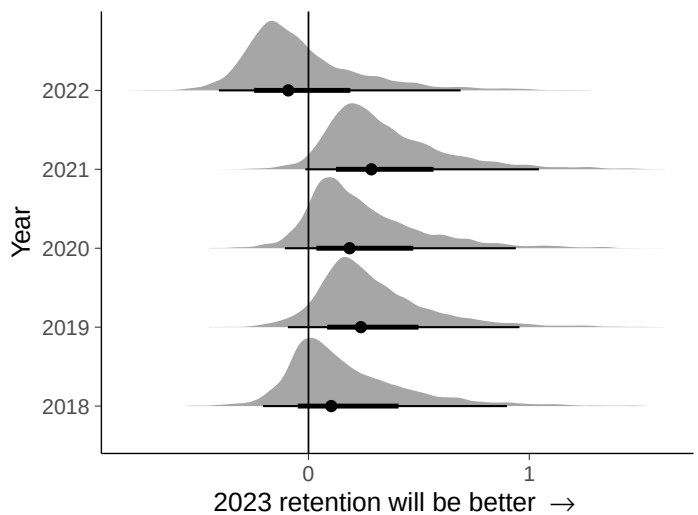


# Student retention latent spline model, year 2023?



# brms summary: one varying coefficient

```
nstudents | trials(nstudents1) ~ 1 + (1 | year), family=binomial()
```

Year specific intercepts have prior $\text{normal}(0, \sigma_{\text{intercept}})$

---

```
 Family: binomial
  Links: mu = logit
Formula: nstudents | trials(nstudents1) ~ 1 + (1 | year)
   Data: filter(tb, assignment == 9) (Number of observations: 5)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~year (Number of levels: 5)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.16      0.13     0.01     0.50 1.00      667      878

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     0.68      0.11     0.45     0.92 1.01      390      242
```

# brms summary: two varying coefficients, nocor

```
nstudents | trials(nstudents1) ~ assignment + (assignment || year), ...
```

Year specific intercepts and slopes have independent priors $\text{normal}(0, \sigma_{\text{intercept}})$ and $\text{normal}(0, \sigma_{\text{assignment}})$

---

```
 Family: binomial
  Links: mu = logit
Formula: nstudents | trials(nstudents1) ~ assignment + (assignment || year)
   Data: filter(tb, assignment > 1) (Number of observations: 40)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000
```

```
Multilevel Hyperparameters:
~year (Number of levels: 5)
               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)      0.79      0.45     0.30     1.95 1.00      972     1470
sd(assignment)     0.07      0.06     0.01     0.23 1.01      592      720
```

```
Regression Coefficients:
           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept      2.37      0.38     1.58     3.14 1.01     1114     1471
assignment    -0.21      0.04    -0.30    -0.14 1.00     1056      803
```

# brms summary: two varying coefficients

```
nstudents | trials(nstudents1) ~ assignment + (assignment | year), ...
```

Year specific intercepts and slopes have a joint prior

$$\mathrm{N}\left[\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}\sigma_{\text{intercept}}\\ \sigma_{\text{slope}}\end{pmatrix}\begin{pmatrix}1 & \rho\\ \rho & 1\end{pmatrix}\begin{pmatrix}\sigma_{\text{intercept}}\\ \sigma_{\text{slope}}\end{pmatrix}^{T}\right]$$

```
 Family: binomial
  Links: mu = logit
Formula: nstudents | trials(nstudents1) ~ assignment + (assignment | year)
   Data: filter(tb, assignment > 1) (Number of observations: 40)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~year (Number of levels: 5)
                        Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)               0.76      0.35     0.34     1.69 1.00     1079     1487
sd(assignment)              0.06      0.03     0.02     0.15 1.00     1211     1979
cor(Intercept,assignment)  -0.85      0.23    -1.00    -0.20 1.00     1730     2265

Regression Coefficients:
           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept      2.36      0.37     1.60     3.10 1.00      860     1179
assignment    -0.21      0.03    -0.27    -0.14 1.00     1073     1464
```

# brms multivariate population prior

```
nstudents | trials(nstudents1) ~ assignment + (assignment | year), ...
```

Year specific intercepts and slopes have a joint prior

$$\mathrm{N}\left[\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}\sigma_{\text{intercept}}\\ \sigma_{\text{slope}}\end{pmatrix}\begin{pmatrix}1 & \rho\\ \rho & 1\end{pmatrix}\begin{pmatrix}\sigma_{\text{intercept}}\\ \sigma_{\text{slope}}\end{pmatrix}^{T}\right]$$

- brms uses LKJ prior for the correlation matrix
- LKJ is a good default weakly informative prior for correlation matrices

# Centered vs non-centered parameterization

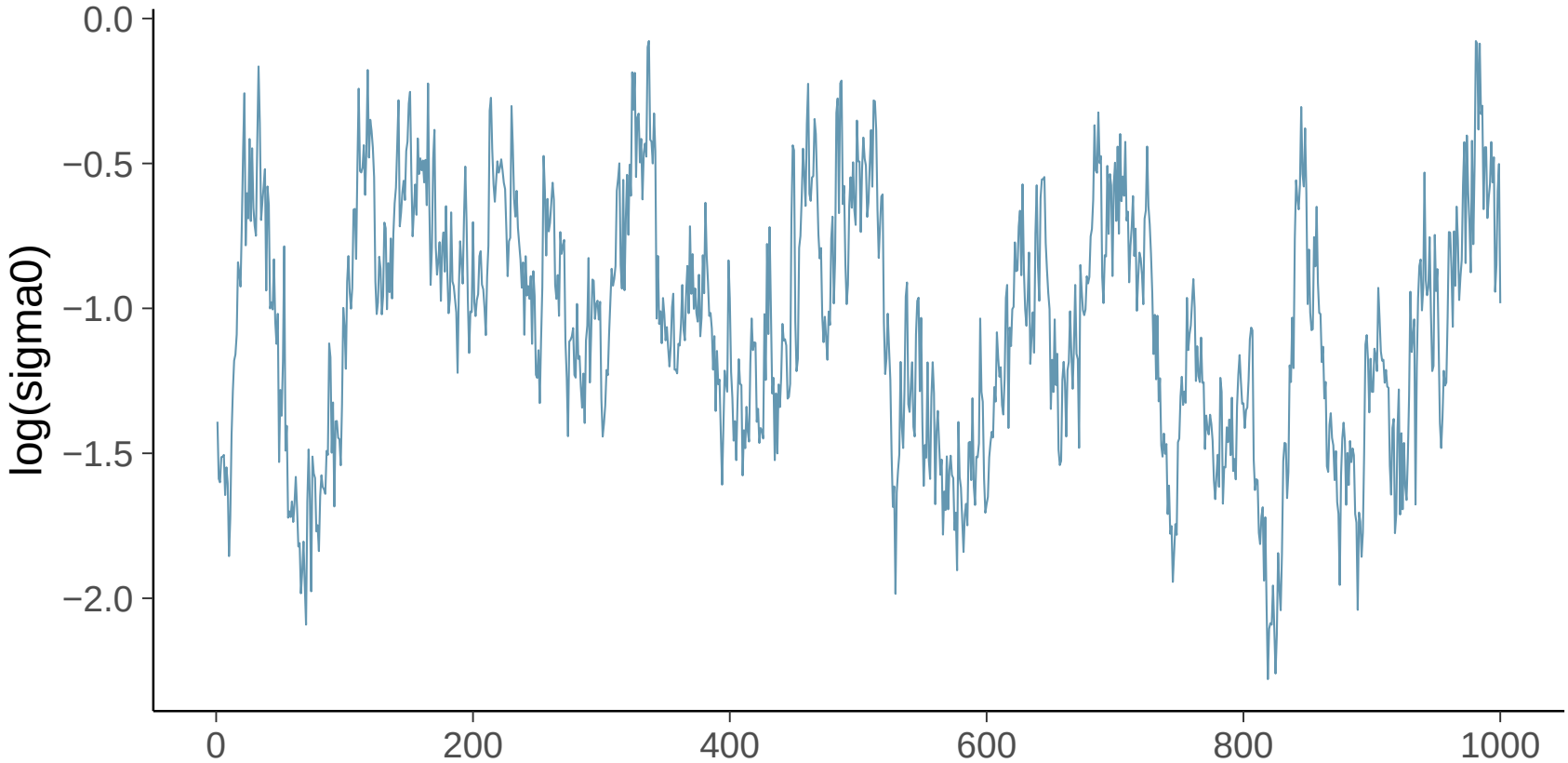
HMC divergences are more likely when using hierarchical models

# Divergences

- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates

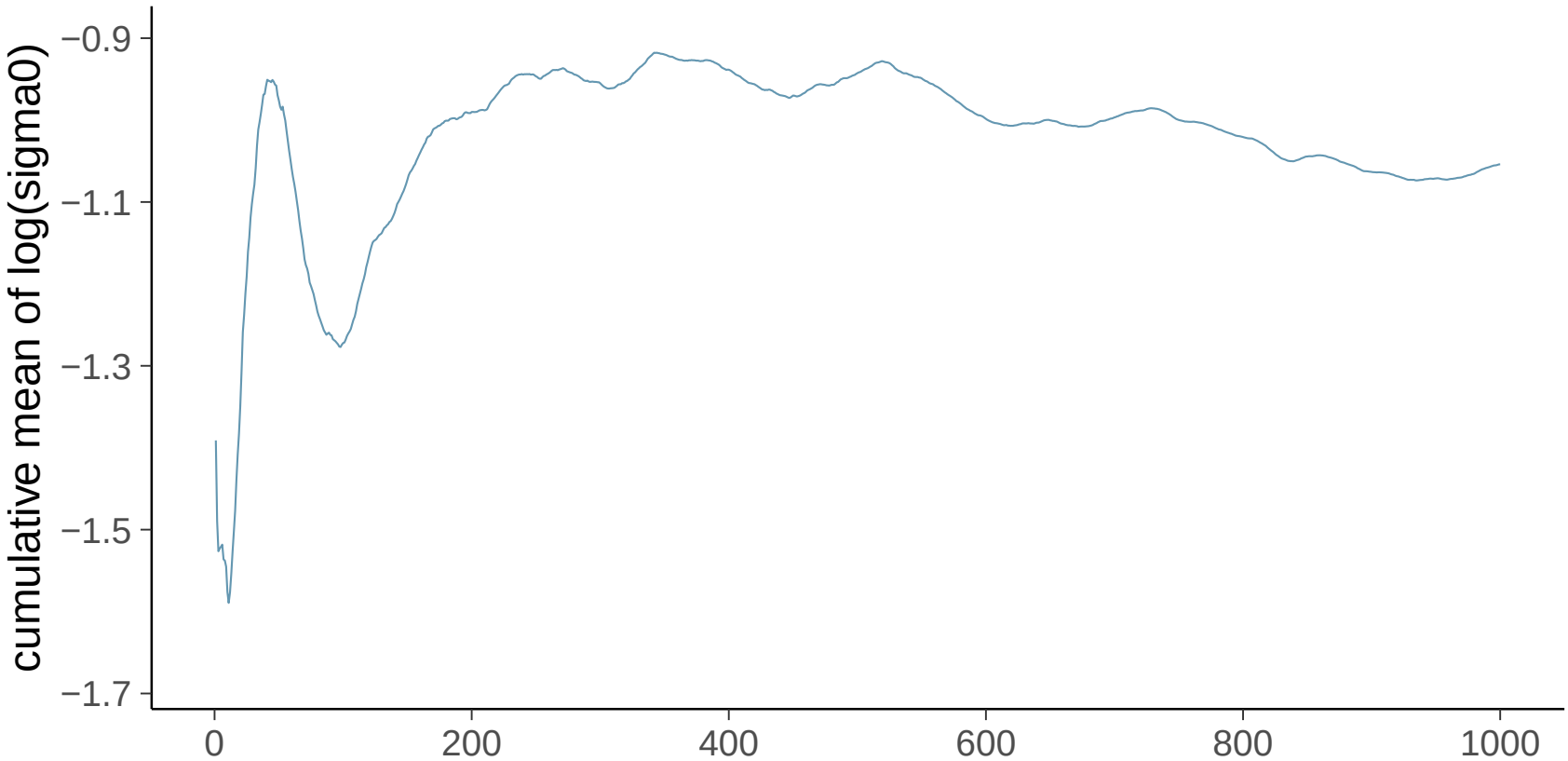
# Divergences

- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
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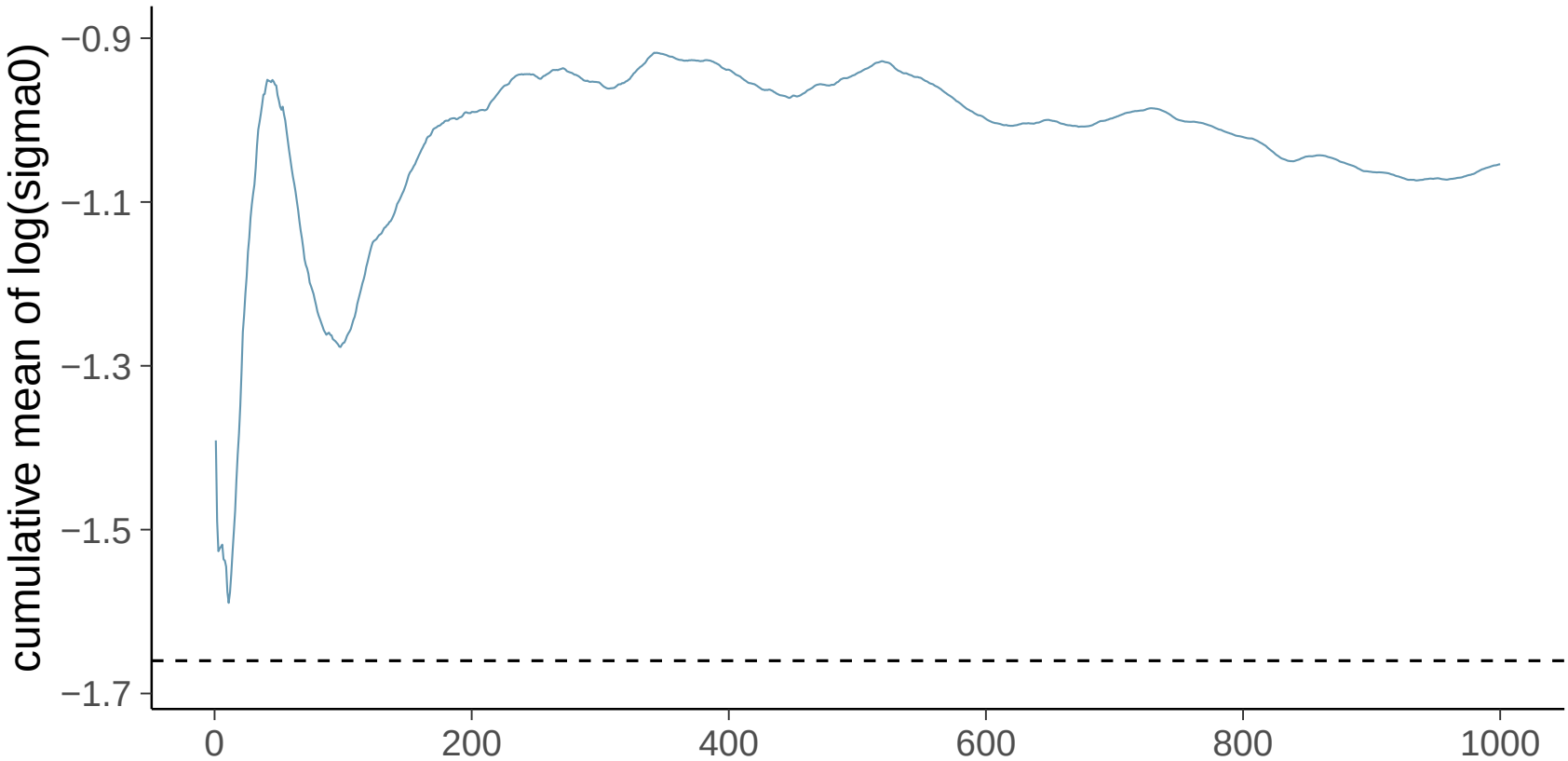
# Divergences

- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates



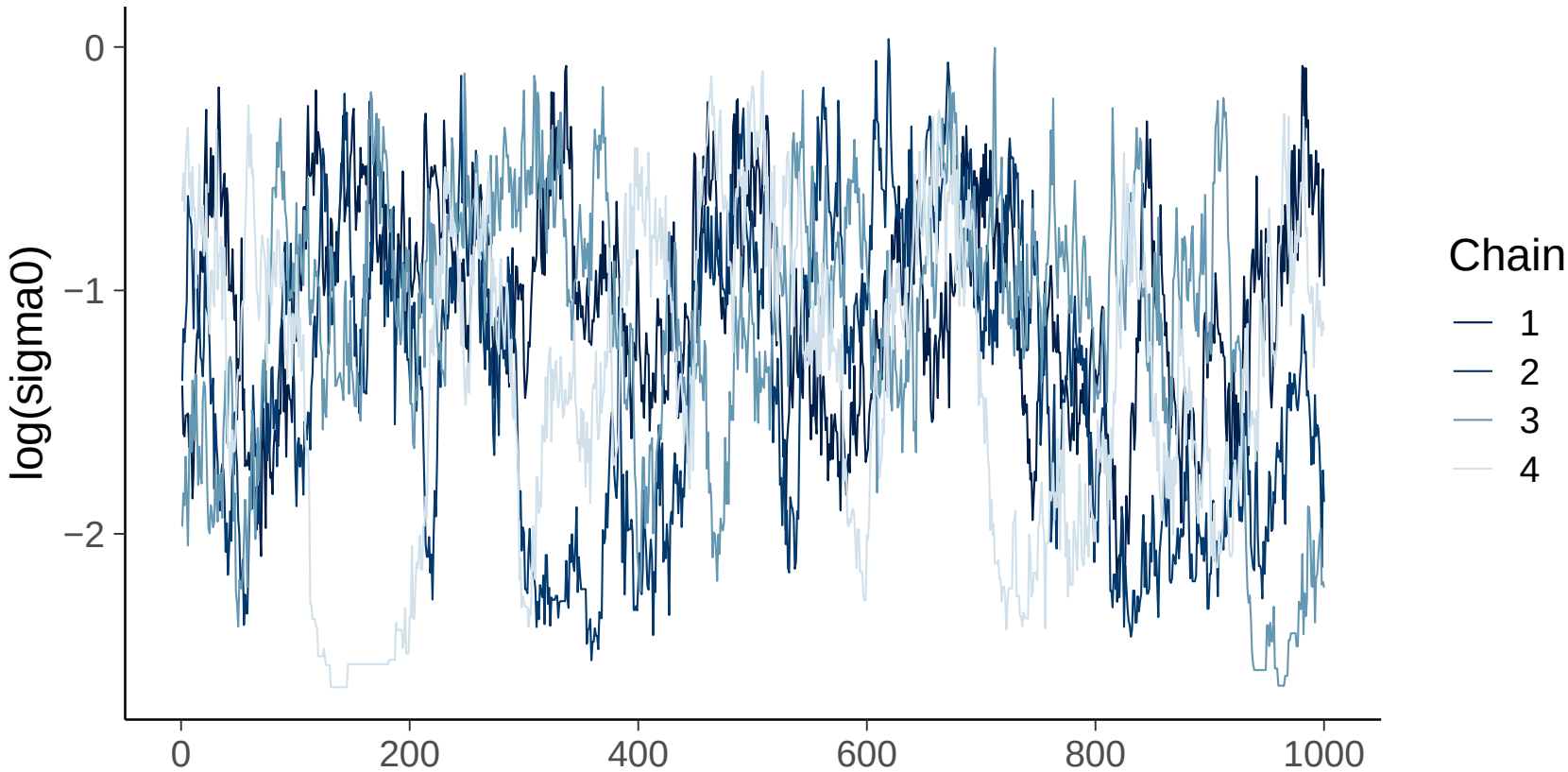
# Divergences

- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
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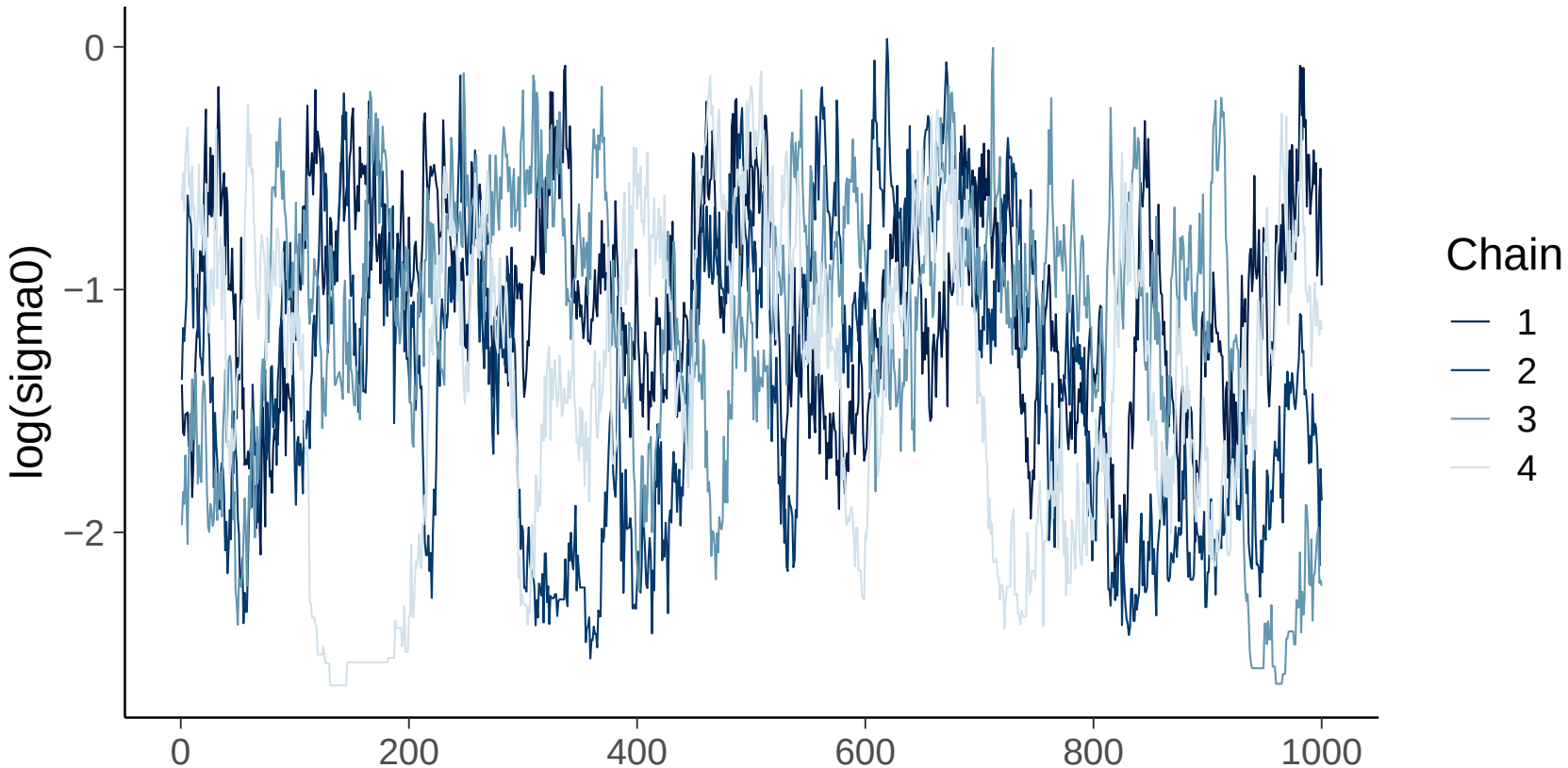
# Divergences

- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates



# Divergences

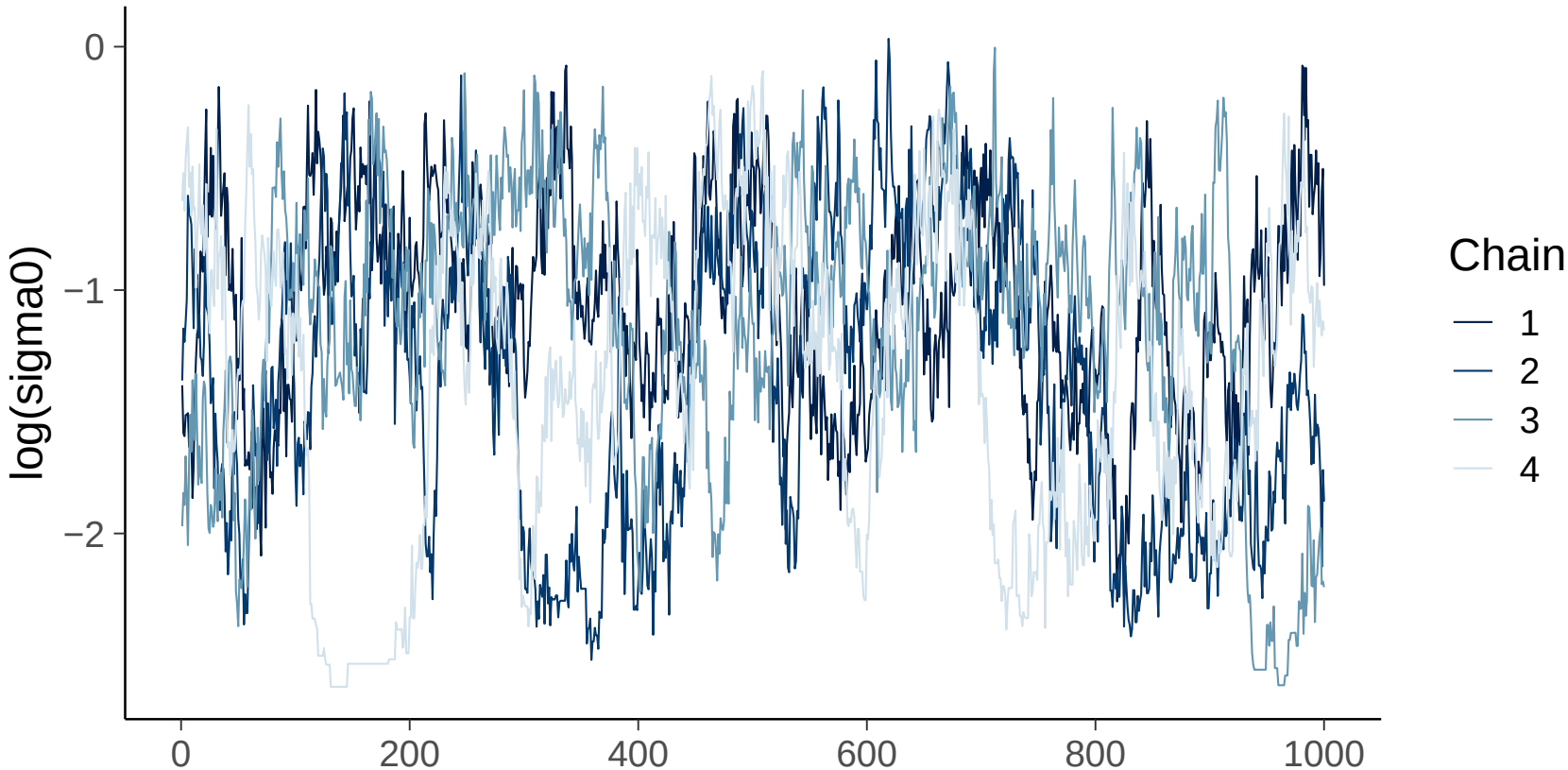
- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates




| variable | mean | median | sd | mad | q5 | q95 | rhat | ess_bulk | ess_tail |
|---|---|---|---|---|---|---|---|---|---|
| lsigma0 | -1.0 | -1.0 | 0.10 | 0.070 | -1.2 | -0.93 | 1.2 | 7.2 | 19. |

# Divergences

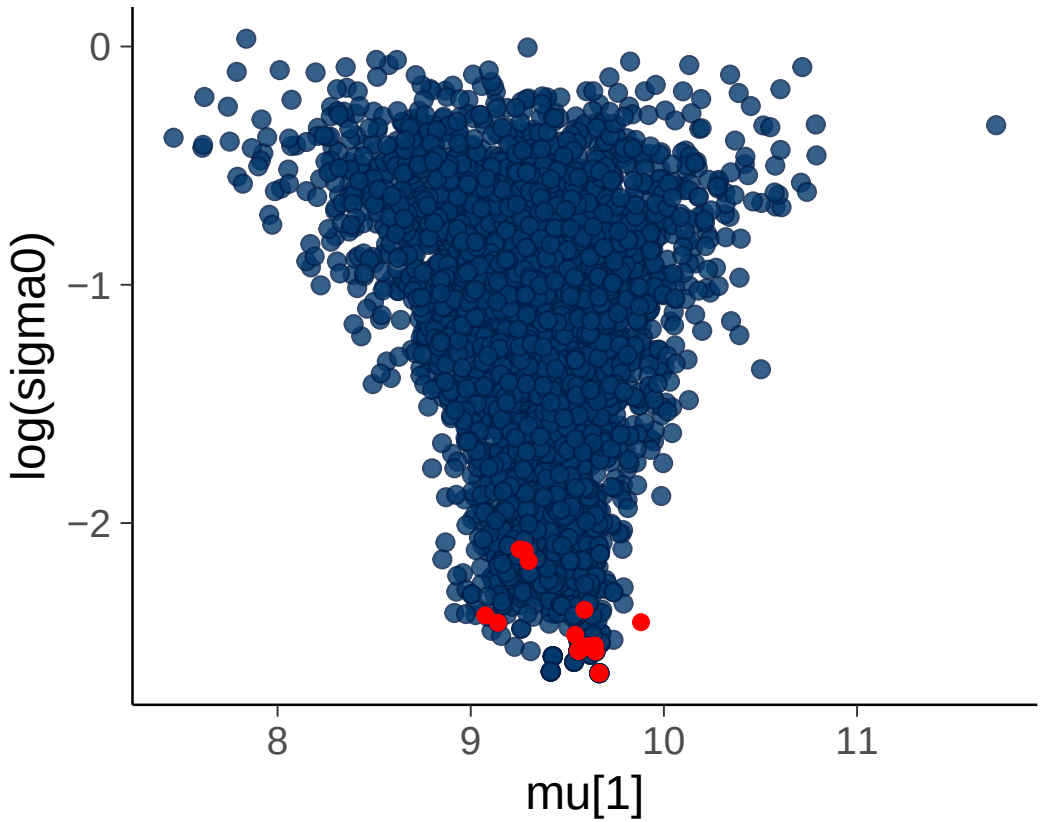
- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates




```
Warning: 29 of 4000 (1.0%) transitions ended with a divergence.
See https://mc-stan.org/misc/warnings for details.
```

# Divergences

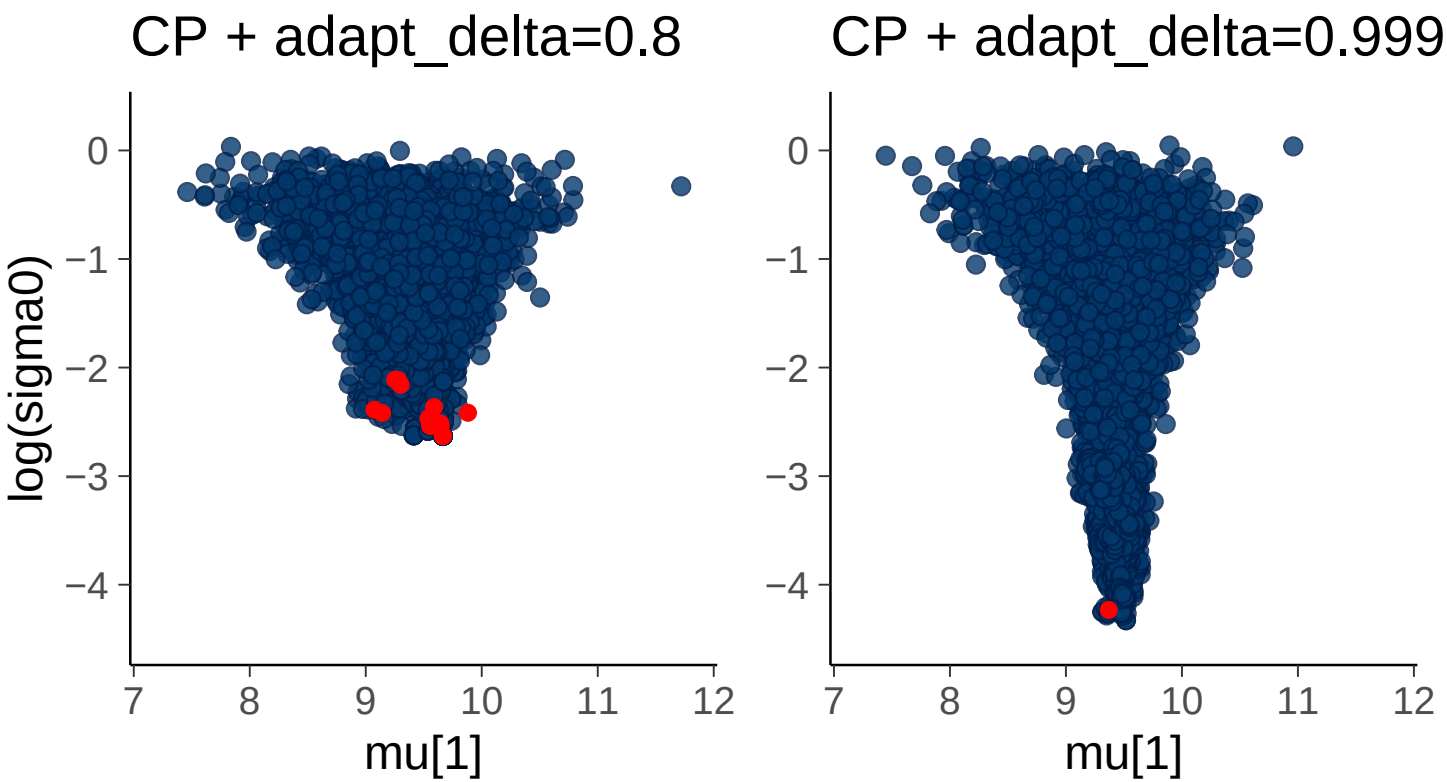
- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
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```
Warning: 29 of 4000 (1.0%) transitions ended with a divergence.
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```

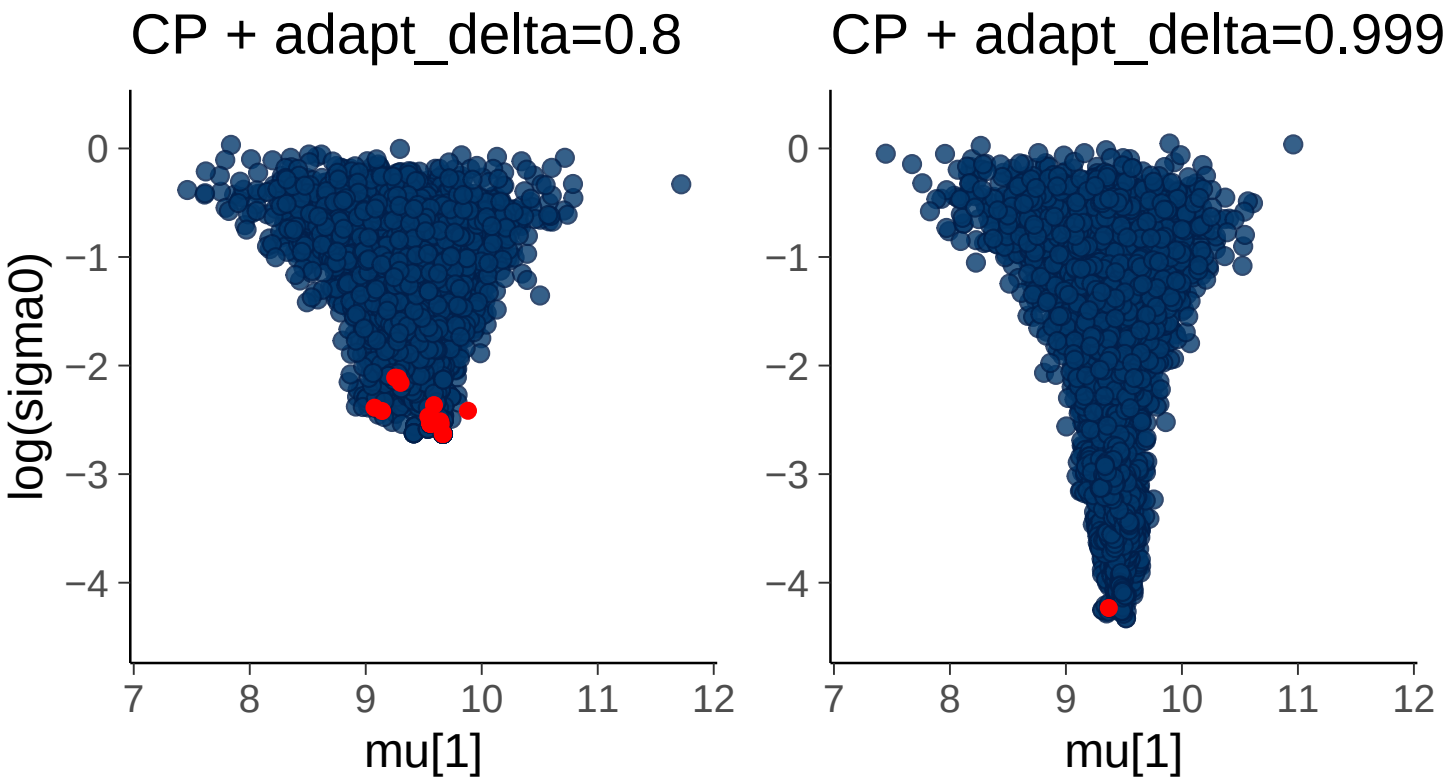
# Divergences

- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates



# Divergences

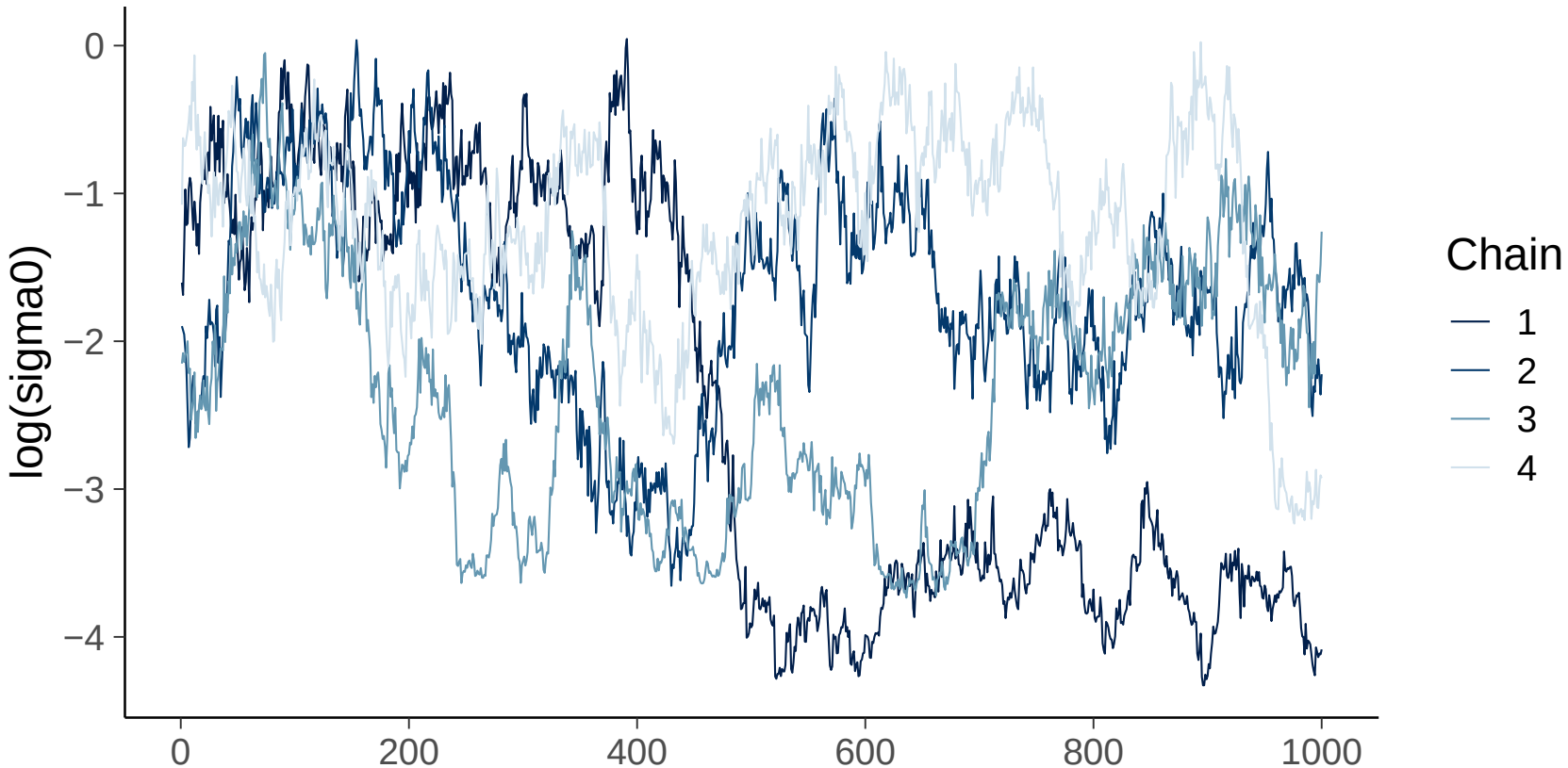
- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates




| variable | mean | median | sd | mad | q5 | q95 | rhat | ess_bulk | ess_tail |
|---|---|---|---|---|---|---|---|---|---|
| lsigma0 | -1.9 | -2.1 | 0.41 | 0.13 | -2.3 | -0.91 | 2.0 | 5.8 | 14. |

# Divergences

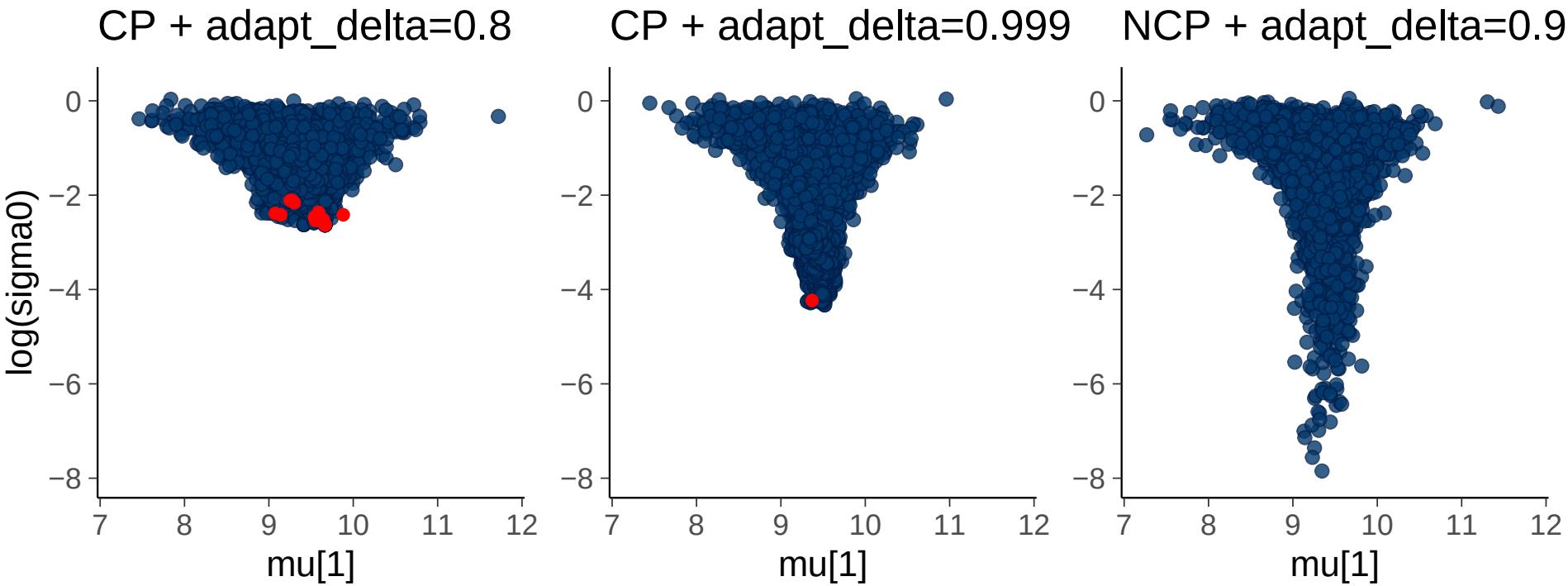
- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates




| variable | mean | median | sd | mad | q5 | q95 | rhat | ess_bulk | ess_tail |
|---|---|---|---|---|---|---|---|---|---|
| lsigma0 | -1.9 | -2.1 | 0.41 | 0.13 | -2.3 | -0.91 | 2.0 | 5.8 | 14. |

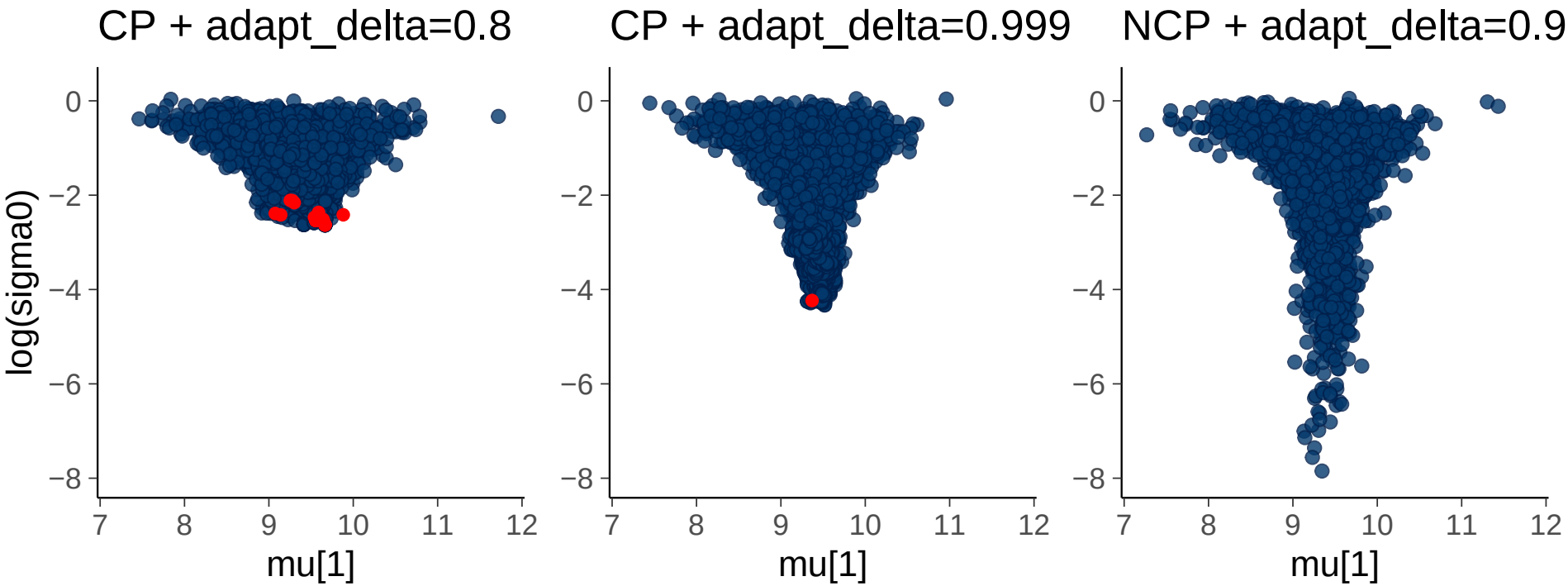
# Divergences

- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates



# Divergences

- HMC specific: indicates that Hamiltonian dynamic simulation has problems with unexpected fast changes in log-density
  - indicates possibility of biased estimates




With NCP parameterization

| variable | mean | median | sd | mad | q5 | q95 | rhat | ess_bulk | ess_tail |
|---|---|---|---|---|---|---|---|---|---|
| sigma0 | 0.27 | 0.24 | 0.19 | 0.20 | 0.023 | 0.64 | 1.0 | 1571. | 1737. |
| lsigma0 | -1.7 | -1.4 | 1.1 | 0.87 | -3.8 | -0.45 | 1.0 | 1571. | 1737. |

# Centered parameterization

Hierarchical model code from the course demos

```
data {
  int<lower=0> N;                     // number of observations
  int<lower=0> K;                     // number of groups
  array[N] int<lower=1, upper=K> x; // discrete group indicators
  vector[N] y;                        // real valued observations
}
```

# Centered parameterization

Hierarchical model code from the course demos

```
data {
  int<lower=0> N;                       // number of observations
  int<lower=0> K;                       // number of groups
  array[N] int<lower=1, upper=K> x; // discrete group indicators
  vector[N] y;                          // real valued observations
}
parameters {
  real mu0;                             // prior mean
  real<lower=0> sigma0;                 // prior std constrained to be pos.
  vector[K] mu;                         // group means
  real<lower=0> sigma;                  // common std constrained to be pos.
}
```

# Centered parameterization

Hierarchical model code from the course demos

```
data {
  int<lower=0> N;                       // number of observations
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  array[N] int<lower=1, upper=K> x; // discrete group indicators
  vector[N] y;                          // real valued observations
}
parameters {
  real mu0;                             // prior mean
  real<lower=0> sigma0;                 // prior std constrained to be pos.
  vector[K] mu;                         // group means
  real<lower=0> sigma;                  // common std constrained to be pos.
}
model {
  mu0 ~ normal(10, 10);                 // weakly informative prior
  sigma0 ~ normal(0, 10);               // weakly informative prior
  mu ~ normal(mu0, sigma0);             // population prior with unknown param.
  sigma ~ lognormal(0, .5);             // weakly informative prior
  y ~ normal(mu[x], sigma);             // observation model
}
```

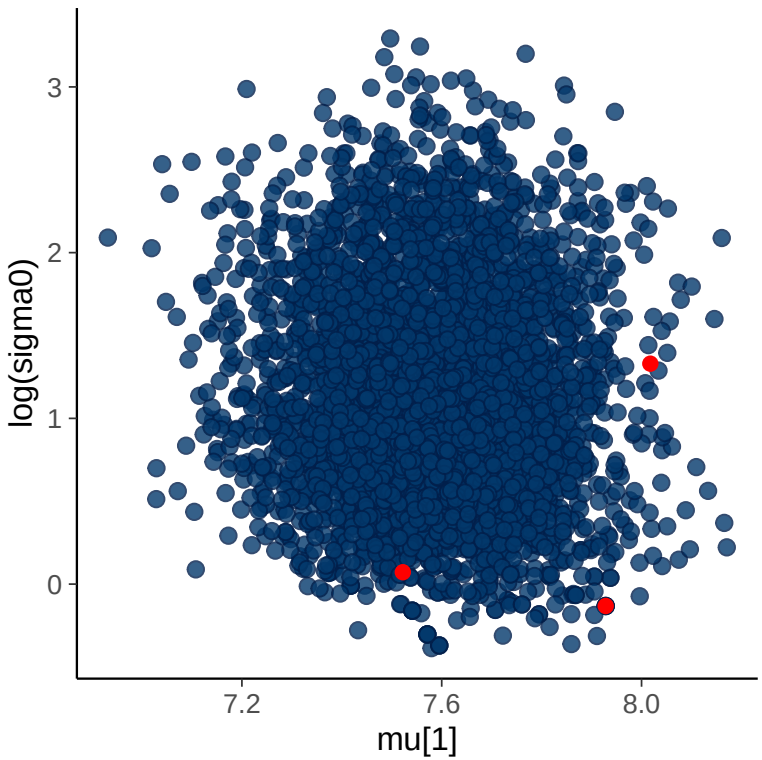
# Centered parameterization

First data with many observations per group: 3 summer months with each having 71 observations.

# Centered parameterization

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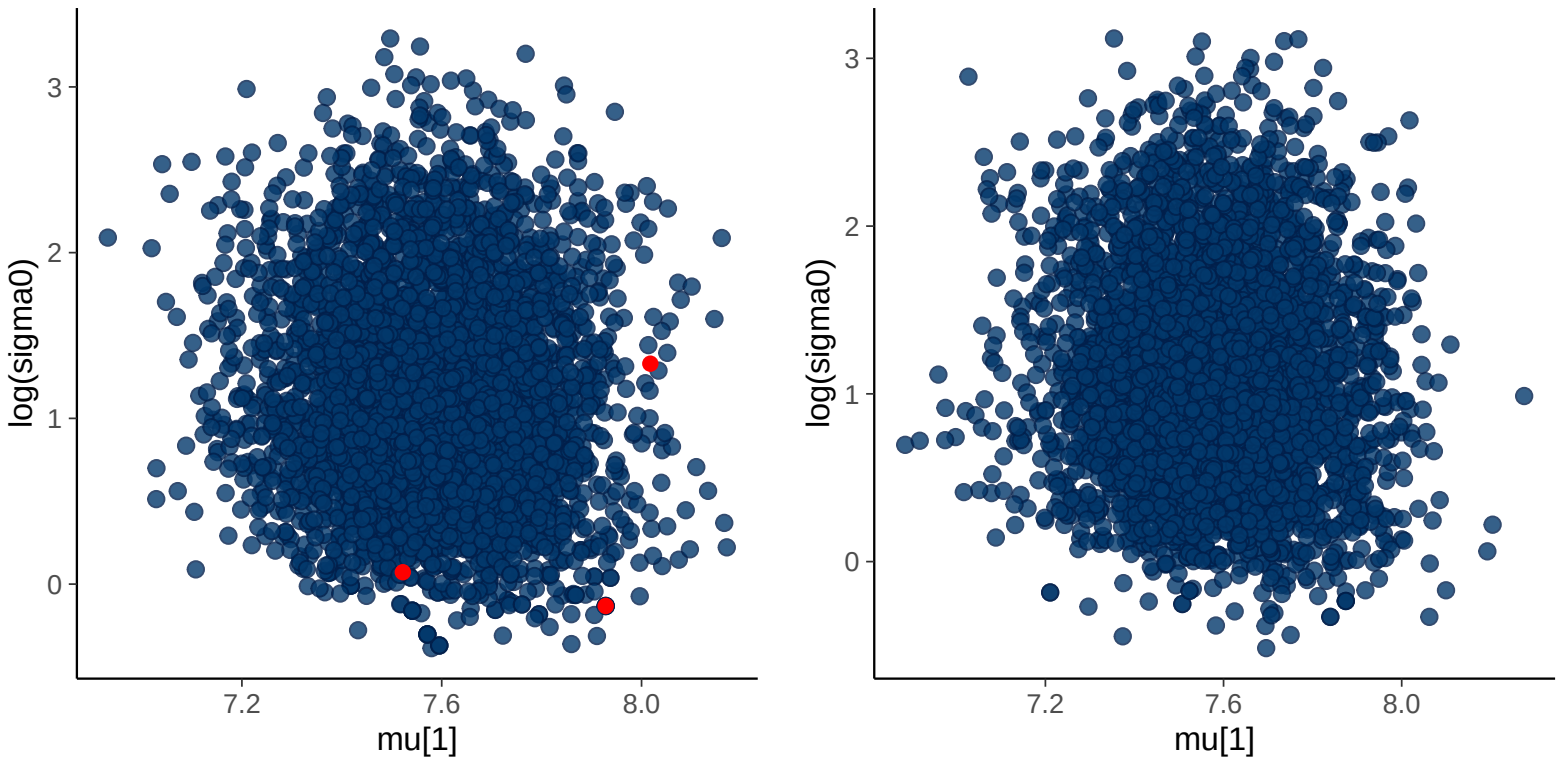
A few divergences that are not clustered.



# Centered parameterization

First data with many observations per group: 3 summer months with each having 71 observations.

And decreasing step size a little helps.



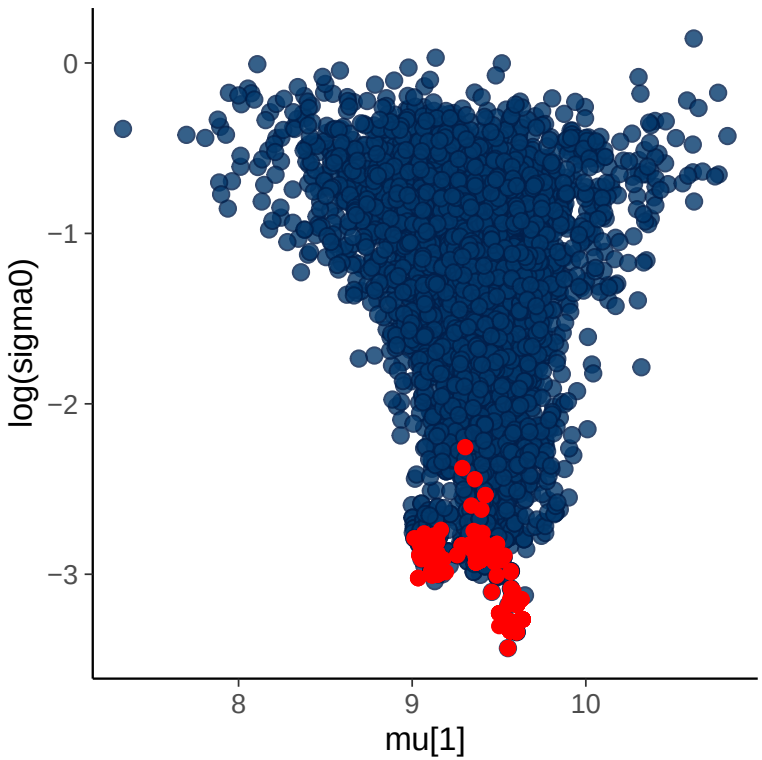
# Centered parameterization

Second data with a few observations per group: 71 years with each having 3 observations.

# Centered parameterization

Second data with a few observations per group: 71 years with each having 3 observations.

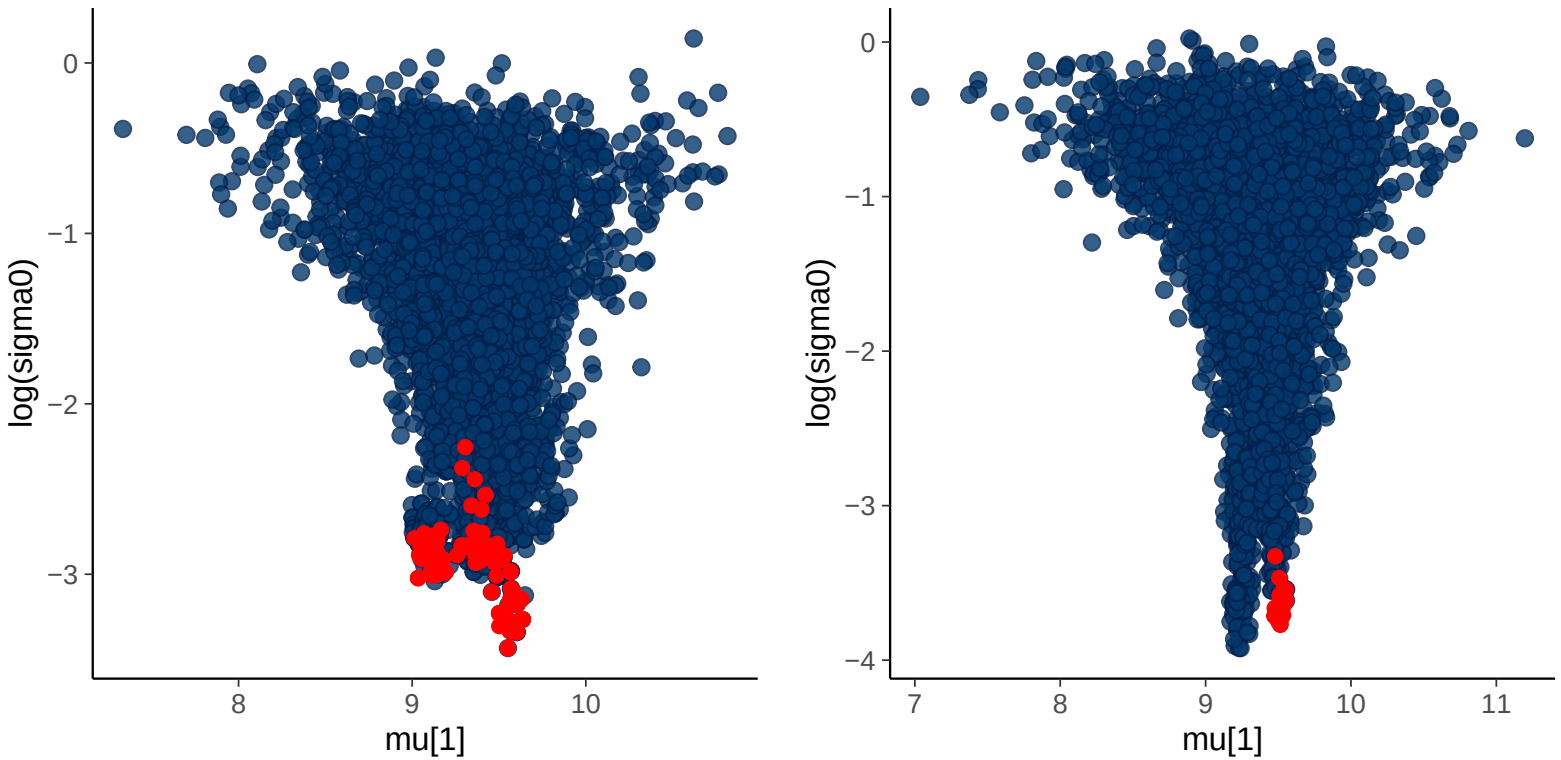
Many divergences that are clustered.



# Centered parameterization

Second data with a few observations per group: 71 years with each having 3 observations.

And decreasing step size doesn't remove the problem.



# Centered parameterization

Hierarchical model code from the course demos

```
parameters {
  real mu0;                       // prior mean
  real<lower=0> sigma0;           // prior std constrained to be pos.
  vector[K] mu;                   // group means
  real<lower=0> sigma;            // common std constrained to be pos.
}
model {
  mu0 ~ normal(10, 10);           // weakly informative prior
  sigma0 ~ normal(0, 10);         // weakly informative prior
  mu ~ normal(mu0, sigma0);       // population prior with unknown param.
  sigma ~ lognormal(0, .5);       // weakly informative prior
  y ~ normal(mu[x], sigma);       // observation model
}
```

# Non-centered parameterization

## Transformation

```
parameters {
  real mu0;                          // prior mean
  real<lower=0> sigma0;              // prior std constrained to be pos.
  vector[K] z;                       // latent variable
  real<lower=0> sigma;               // common std constrained to be pos.
}
transformed parameters {
  vector[K] mu = mu0 + sigma0 * z;  // group means
}
model {
  mu0 ~ normal(10, 10);      // weakly informative prior
  sigma0 ~ normal(0, 10);    // weakly informative prior
  z ~ normal(0, 1);          // unit normal
  sigma ~ lognormal(0, .5); // weakly informative prior
  y ~ normal(mu[x], sigma); // observation model
}
```

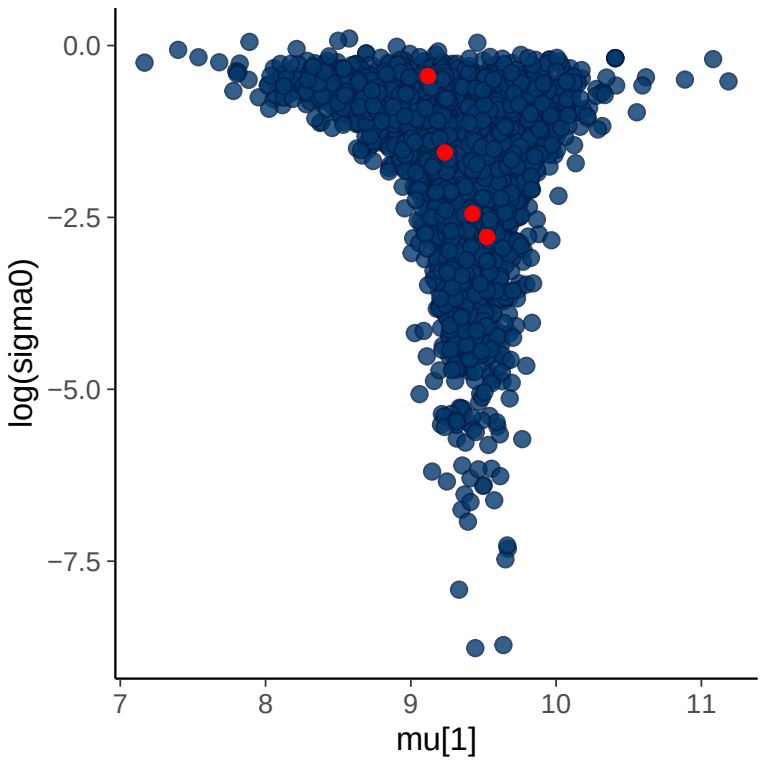
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Second data with a few observations per group: 71 years with each having 3 observations.

# Non-centered parameterization

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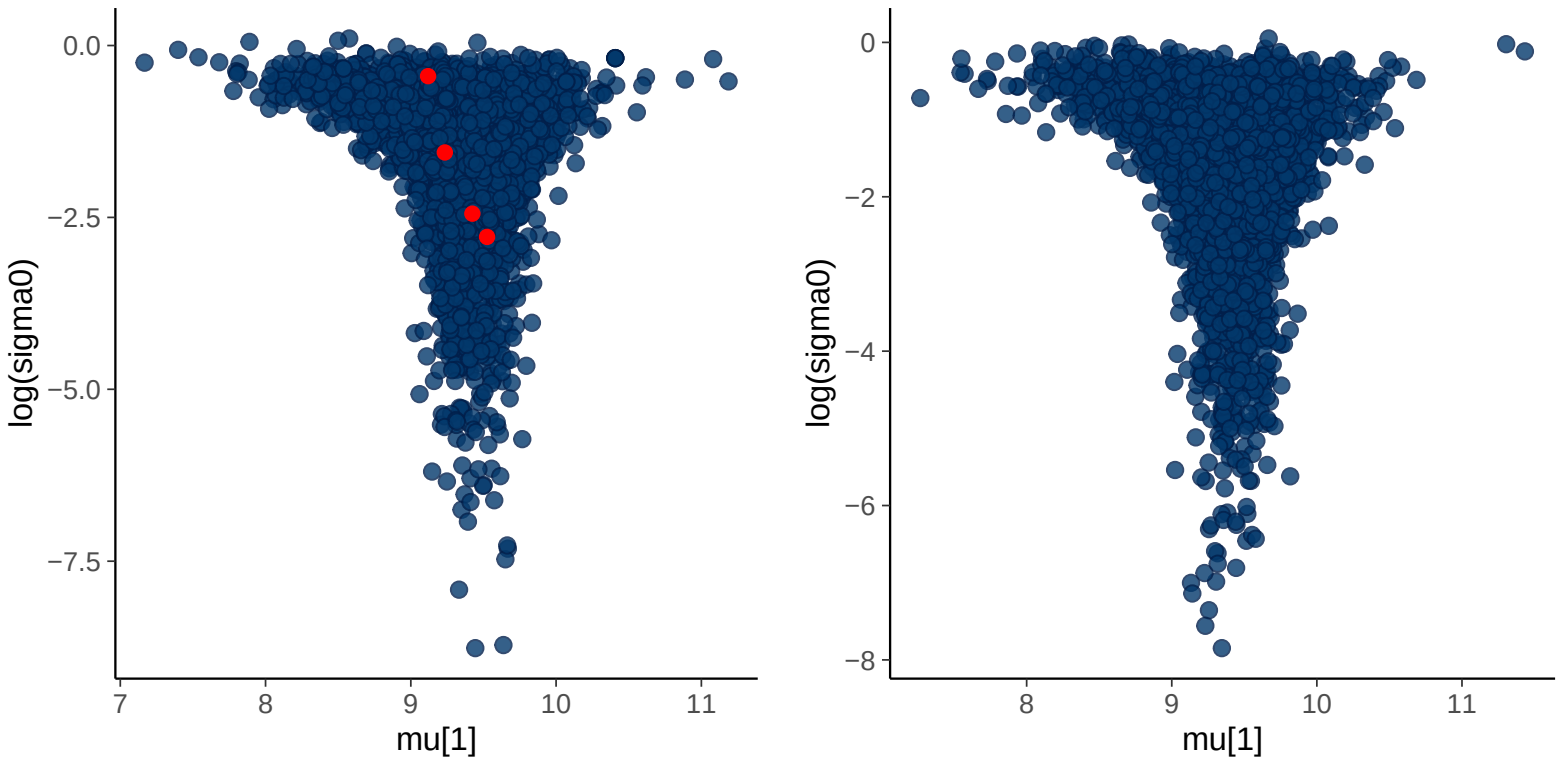
A few divergences that are not clustered.



# Non-centered parameterization

Second data with a few observations per group: 71 years with each having 3 observations.

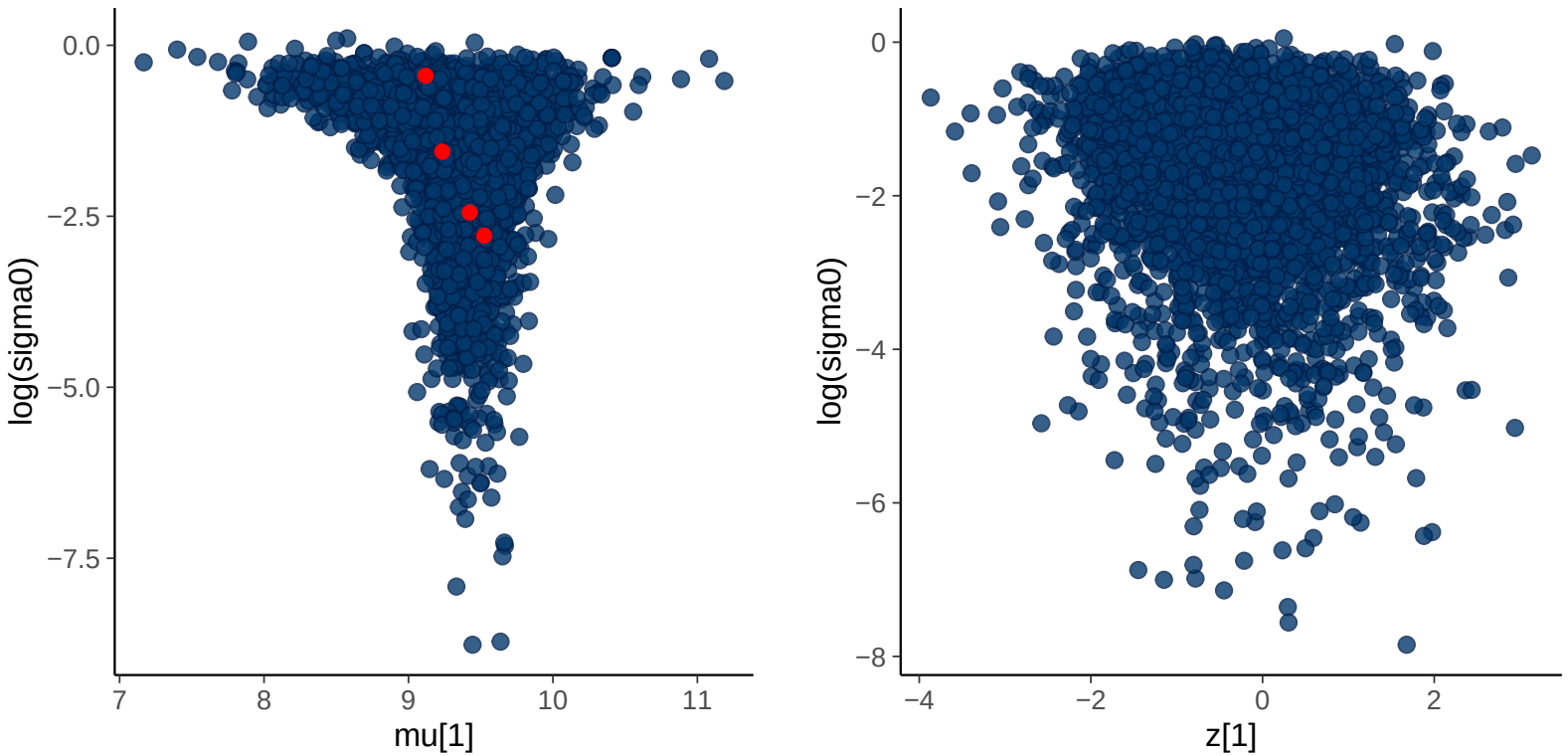
And decreasing step size a little helps.



# Non-centered parameterization

Second data with a few observations per group: 71 years with each having 3 observations.

Because we're actually sampling `z` and not `mu`



# Non-centered parameterization

No free lunch

- non-centered parameterization is good when likelihood is weak
- non-centered parameterization is bad when likelihood is strong

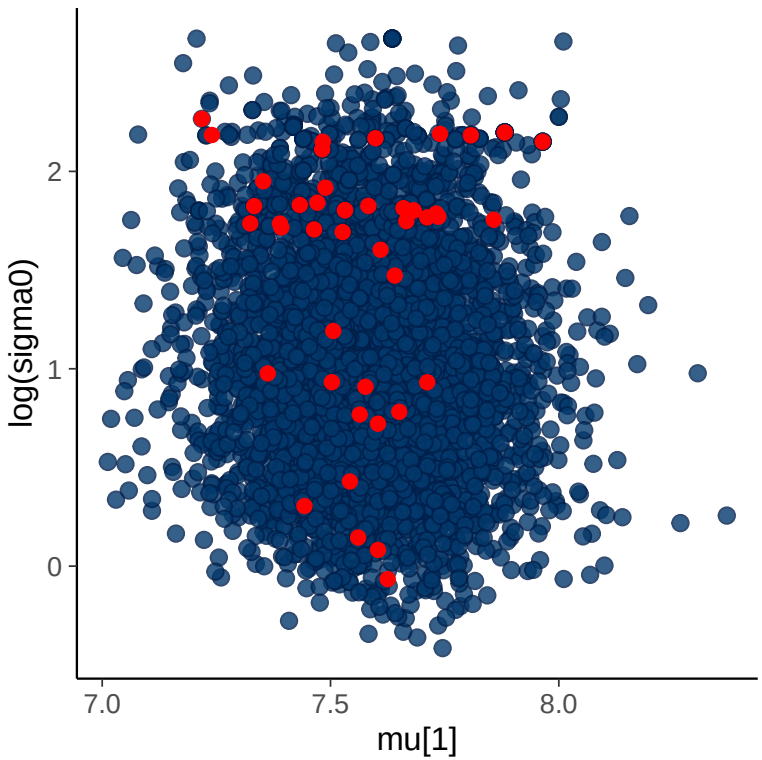
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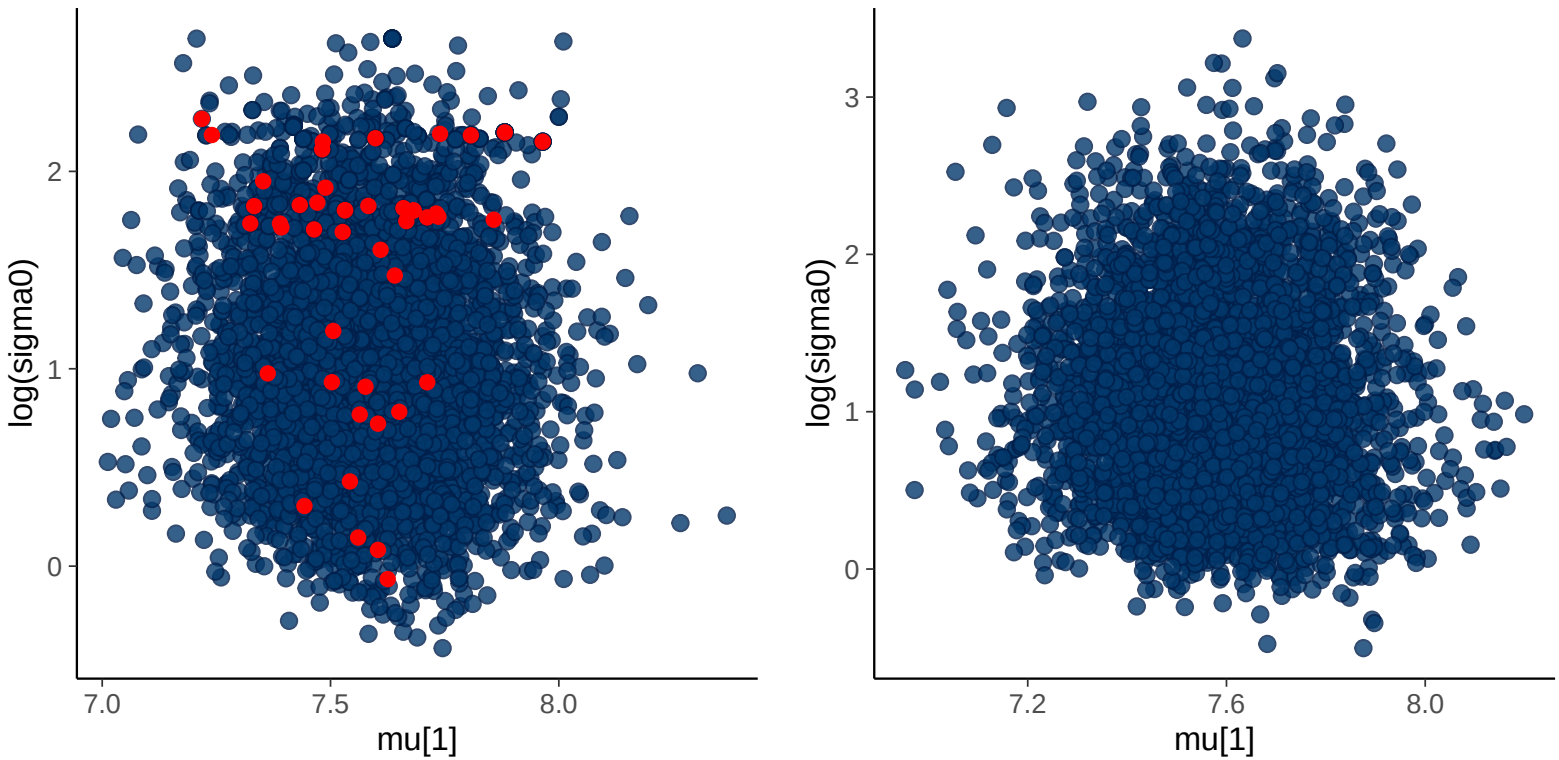
Many divergences that are not clustered.



# Non-centered parameterization

First data with many observations per group: 3 summer months with each having 71 observations.

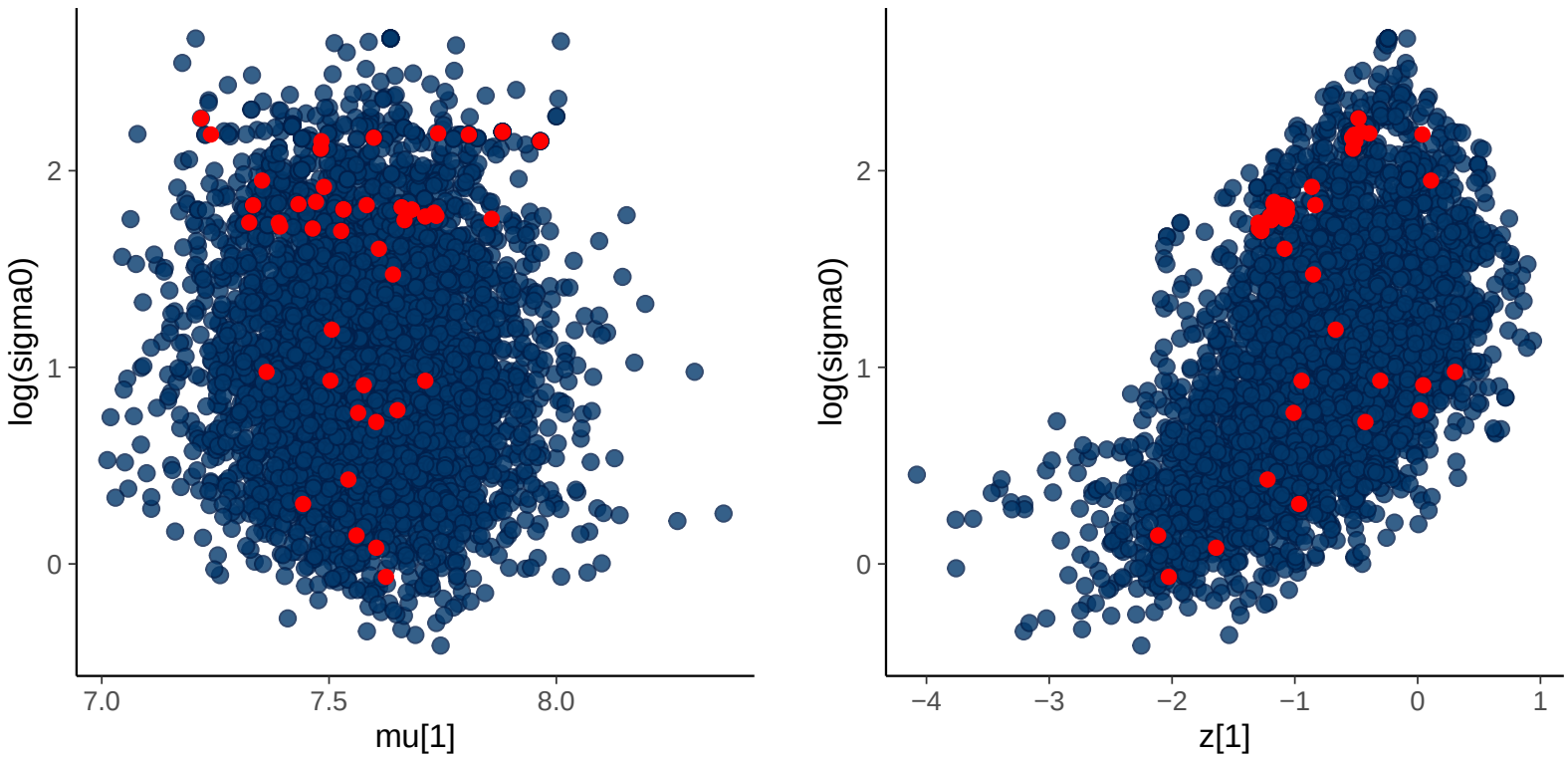
But decreasing step size a lot helps.



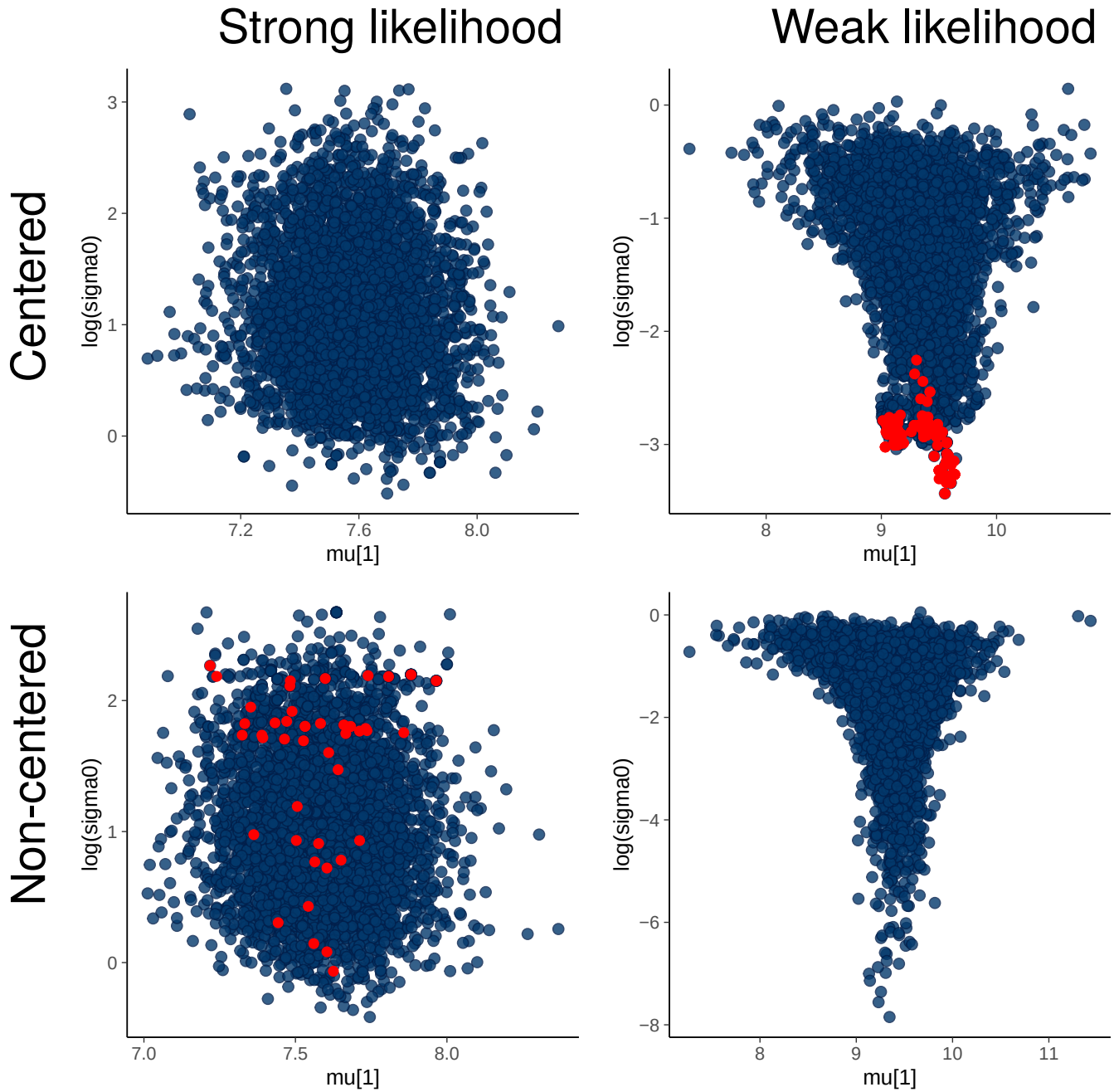
# Non-centered parameterization

First data with many observations per group: 3 summer months with each having 71 observations.

Now the posterior for $z$ is problematic.



# Centered vs. non-centered parameterization



# brms and rstanarm

- brms and rstanarm use non-centered parameterization
  - as hierarchical models and Bayesian inference is most useful when likelihood is weak

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- brms and rstanarm use non-centered parameterization
  - as hierarchical models and Bayesian inference is most useful when likelihood is weak
- If the likelihood is highly informative for all groups, and the non-centered parameterization is causing divergences,
  - use fixed population prior model as the hierarchical part is not that important with strong likelihood. Instead of

    ```
    y ~ 1 + (1 | group)
    ```

    use, for example,

    ```
    y ~ 1 + group,
    prior = prior(normal(0, 10), class="b", coeff="group")
    ```

# brms and rstanarm

- brms and rstanarm use non-centered parameterization
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    y ~ 1 + (1 | group)
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    use, for example,

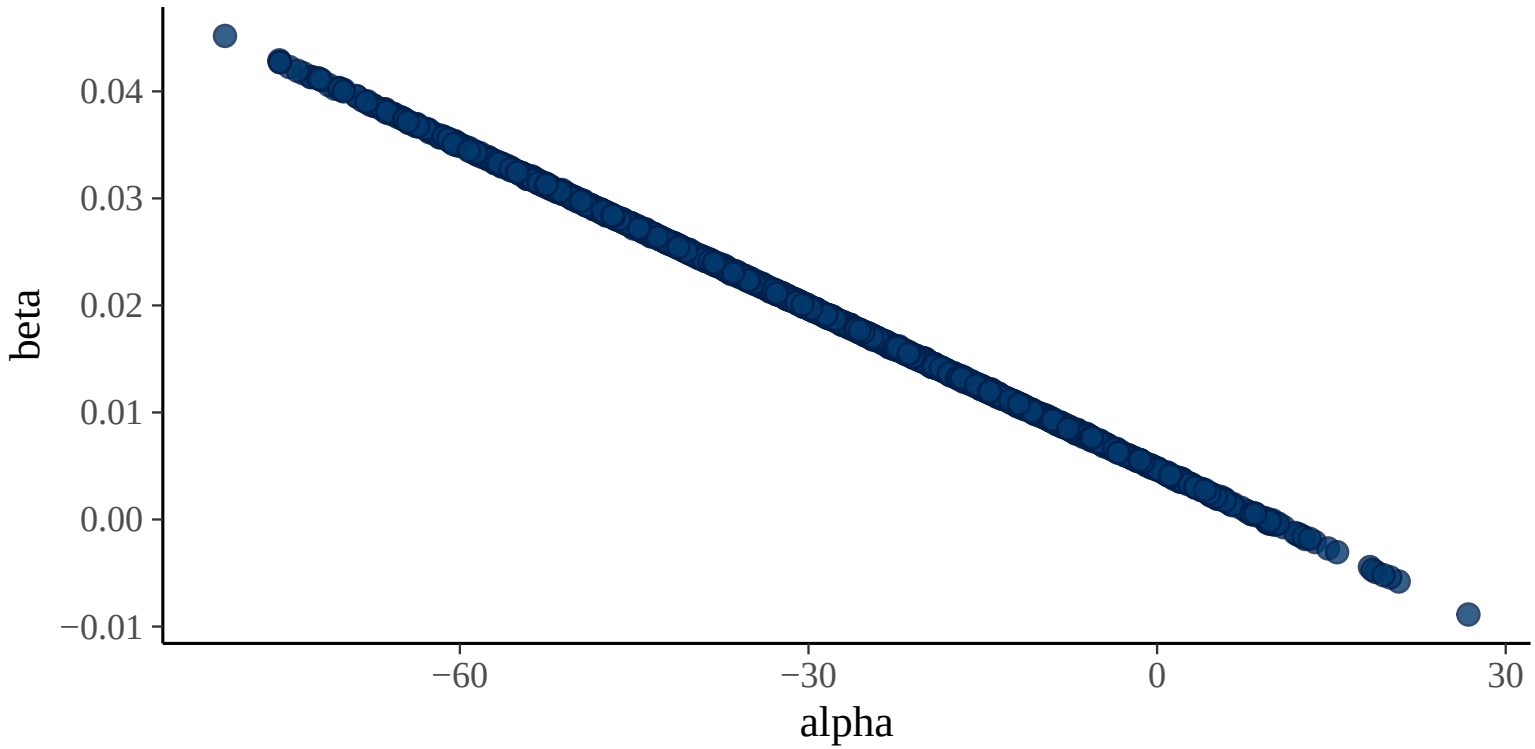
    ```
    y ~ 1 + group,
    prior = prior(normal(0, 10), class="b", coeff="group")
    ```
- There can be need for both centered and non-centered parameterization in the same model
  - automation not easy, but research goes on

# brms non-centered parameterization

```
parameters {
  real Intercept;  // temporary intercept for centered predictors
  real<lower=0> sigma;  // dispersion parameter
  vector<lower=0>[M_1] sd_1;  // group-level standard deviations
  array[M_1] vector[N_1] z_1;  // standardized group-level effects
}
transformed parameters {
  vector[N_1] r_1_1;  // actual group-level effects
  r_1_1 = (sd_1[1] * (z_1[1]));
//...
}
model {
//...
    for (n in 1:N) {
      // add more terms to the linear predictor
      mu[n] += r_1_1[J_1[n]] * Z_1_1[n];
    }
//...
  target += std_normal_lpdf(z_1[1]);
}
```

# Kilpisjärvi summer temperature

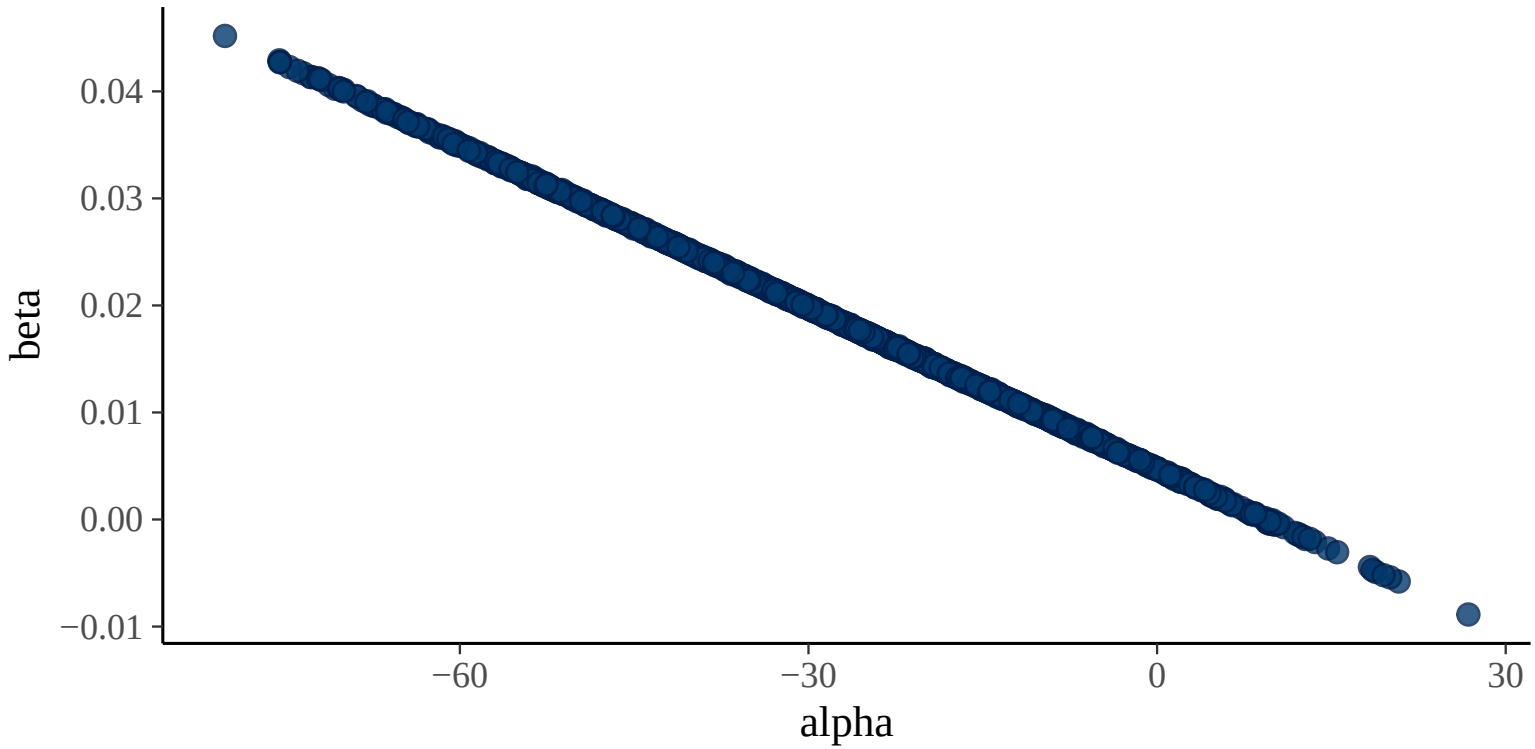
Posterior draws of alpha and beta




```
Warning: 1 of 4000 (0.0%) transitions hit the maximum treedepth limit of 10.
See https://mc-stan.org/misc/warnings for details.
```

# Kilpisjärvi summer temperature

Posterior draws of alpha and beta




```
Warning: 1 of 4000 (0.0%) transitions hit the maximum treedepth limit of 10.
See https://mc-stan.org/misc/warnings for details.
```

Solution was to center the covariate time to have mean 0, so that the intercept is the expected temperature in the middle of the range

# brms covariate centering by default

```
fit_lin <- brm(temp ~ year, data = data_lin)
```

# brms covariate centering by default

```
fit_lin <- brm(temp ~ year, data = data_lin)
```

```
transformed data {
  matrix[N, Kc] Xc;  // centered version of X without an intercept
  vector[Kc] means_X;  // column means of X before centering
  for (i in 2:K) {
    means_X[i - 1] = mean(X[, i]);
    Xc[, i - 1] = X[, i] - means_X[i - 1];
  }
}
parameters {
  vector[Kc] b;  // regression coefficients
  real Intercept;  // temporary intercept for centered predictors
  real<lower=0> sigma;  // dispersion parameter
}
model {
//...
  target += normal_id_glm_lpdf(Y | Xc, Intercept, b, sigma);
//...
}
generated quantities {
  // actual population-level intercept
  real b_Intercept = Intercept - dot_product(means_X, b);
}
```

# brms covariate centering by default

Sometimes we want to turn off the covariate centering, which can be done with additional argument for brms formula function `bf()`

```
fit_lin <- brm(bf(temp ~ year, center=FALSE), data = data_lin)
```

# brms formulas

| model | formula | alternative formula |
|---|---|---|
| intercept only | | |
| $y \sim \mathrm{N}(\alpha, \sigma)$ | y ~ 1 | |

# brms formulas

| model | formula | alternative formula |
|---|---|---|
| intercept only | | |
| $y \sim \mathrm{N}(\alpha, \sigma)$ | y ~ 1 | |
| linear models | | |
| $y \sim \mathrm{N}(\alpha + \beta x, \sigma)$ | y ~ x | y ~ 1 + x |

# brms formulas

| model | formula | alternative formula |
|---|---|---|
| intercept only | | |
| $y \sim \mathrm{N}(\alpha, \sigma)$ | `y ~ 1` | |
| linear models | | |
| $y \sim \mathrm{N}(\alpha + \beta x, \sigma)$ | `y ~ x` | `y ~ 1 + x` |
| $y \sim \mathrm{N}(\beta x, \sigma)$ | `y ~ 0 + x` | `y ~ -1 + x` |

# brms formulas

| model | formula | alternative formula |
|---|---|---|
| intercept only | | |
| $y \sim \mathrm{N}(\alpha, \sigma)$ | `y ~ 1` | |
| linear models | | |
| $y \sim \mathrm{N}(\alpha + \beta x, \sigma)$ | `y ~ x` | `y ~ 1 + x` |
| $y \sim \mathrm{N}(\beta x, \sigma)$ | `y ~ 0 + x` | `y ~ -1 + x` |
| hierarchical models | | |
| $y \sim \mathrm{N}(\alpha_0 + \alpha_g + \beta x, \sigma)$ | `y ~ x + (1 \| g)` | |

# brms formulas

| model | formula | alternative formula |
|---|---|---|
| intercept only | | |
| $y \sim \mathrm{N}(\alpha, \sigma)$ | `y ~ 1` | |
| linear models | | |
| $y \sim \mathrm{N}(\alpha + \beta x, \sigma)$ | `y ~ x` | `y ~ 1 + x` |
| $y \sim \mathrm{N}(\beta x, \sigma)$ | `y ~ 0 + x` | `y ~ -1 + x` |
| hierarchical models | | |
| $y \sim \mathrm{N}(\alpha_0 + \alpha_g + \beta x, \sigma)$ | `y ~ x + (1 \| g)` | |
| $y \sim \mathrm{N}(\alpha_0 + \alpha_g + \beta_0 x + \beta_g x, \sigma)$ | `y ~ x + (x \| g)` | (see above) |

# brms formulas

| model | formula |
|---|---|
| heteroskedastic | |
| $y \sim \mathrm{N}\left(\alpha_\mu + \beta_\mu x,\, \exp(\alpha_\sigma + \beta_\sigma x)\right)$ | `bf(y ~ x, sigma ~ x)` |

# brms families

`family` argument determines the observation model family

| model | brms |
|---|---|
| $y \sim t_\nu(\alpha + \beta x, \sigma)$ | `y ~ x, family = student()` |
| $y \sim \text{Bin}(\text{logit}^{-1}(\alpha + \beta x), N)$ | `y \| trials(N) ~ x, family = binomial()` |
| $y \sim \text{Neg-bin}(\exp(\alpha + \beta x), \phi)$ | `y ~ x, family = negbinomial()` |

# Meta-analysis

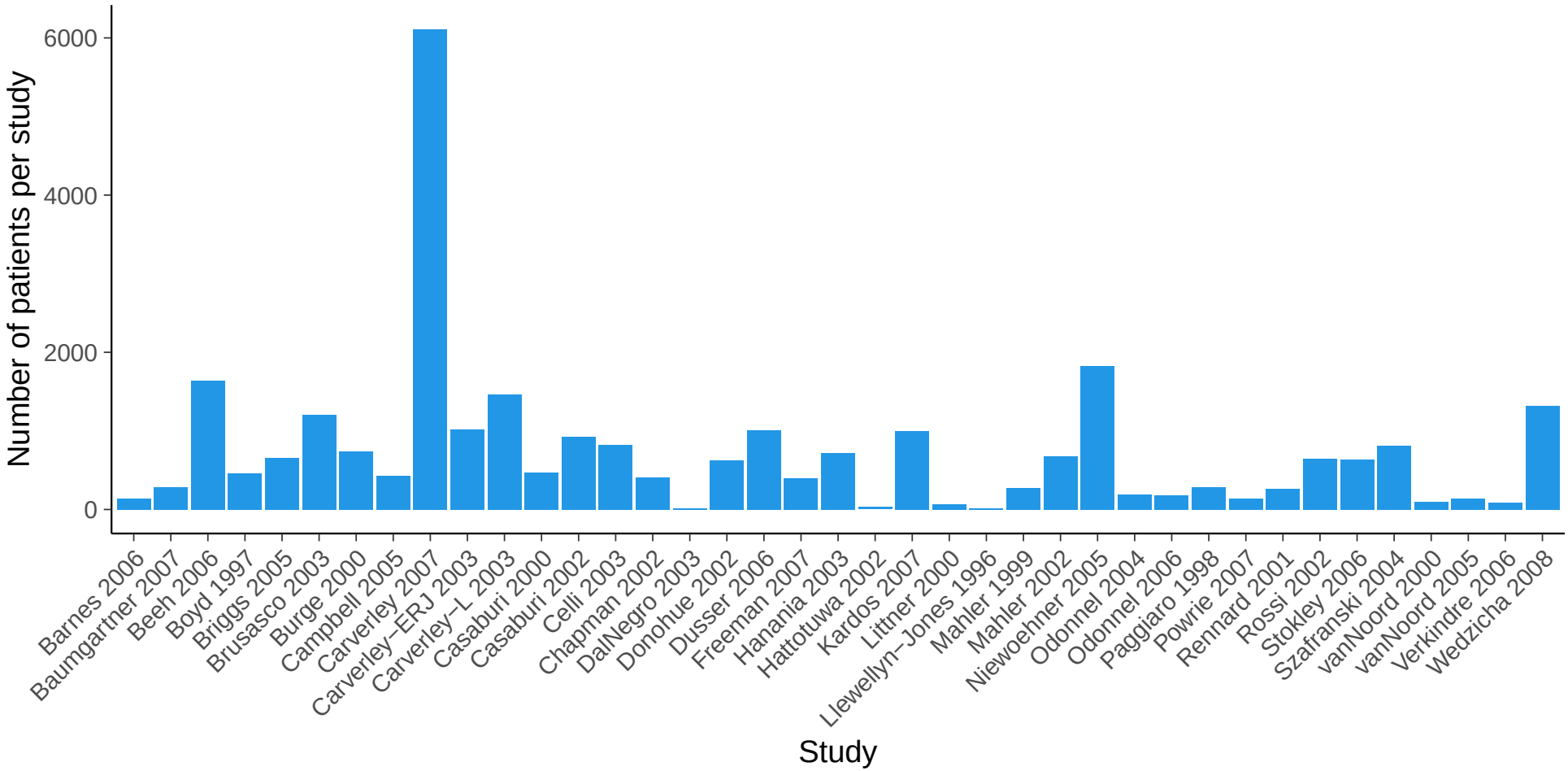
## Pharmacologic treatments for chronic obstructive pulmonary disease

```
> head(dat.baker2009)
                 study year id   treatment exac total
1 Llewellyn-Jones 1996 1996  1 Fluticasone    0     8
2 Llewellyn-Jones 1996 1996  1     Placebo    3     8
3             Boyd 1997 1997  2  Salmeterol   47   229
4             Boyd 1997 1997  2     Placebo   59   227
5         Paggiaro 1998 1998  3 Fluticasone   45   142
6         Paggiaro 1998 1998  3     Placebo   51   139
```

# Meta-analysis

## Pharmacologic treatments for chronic obstructive pulmonary disease

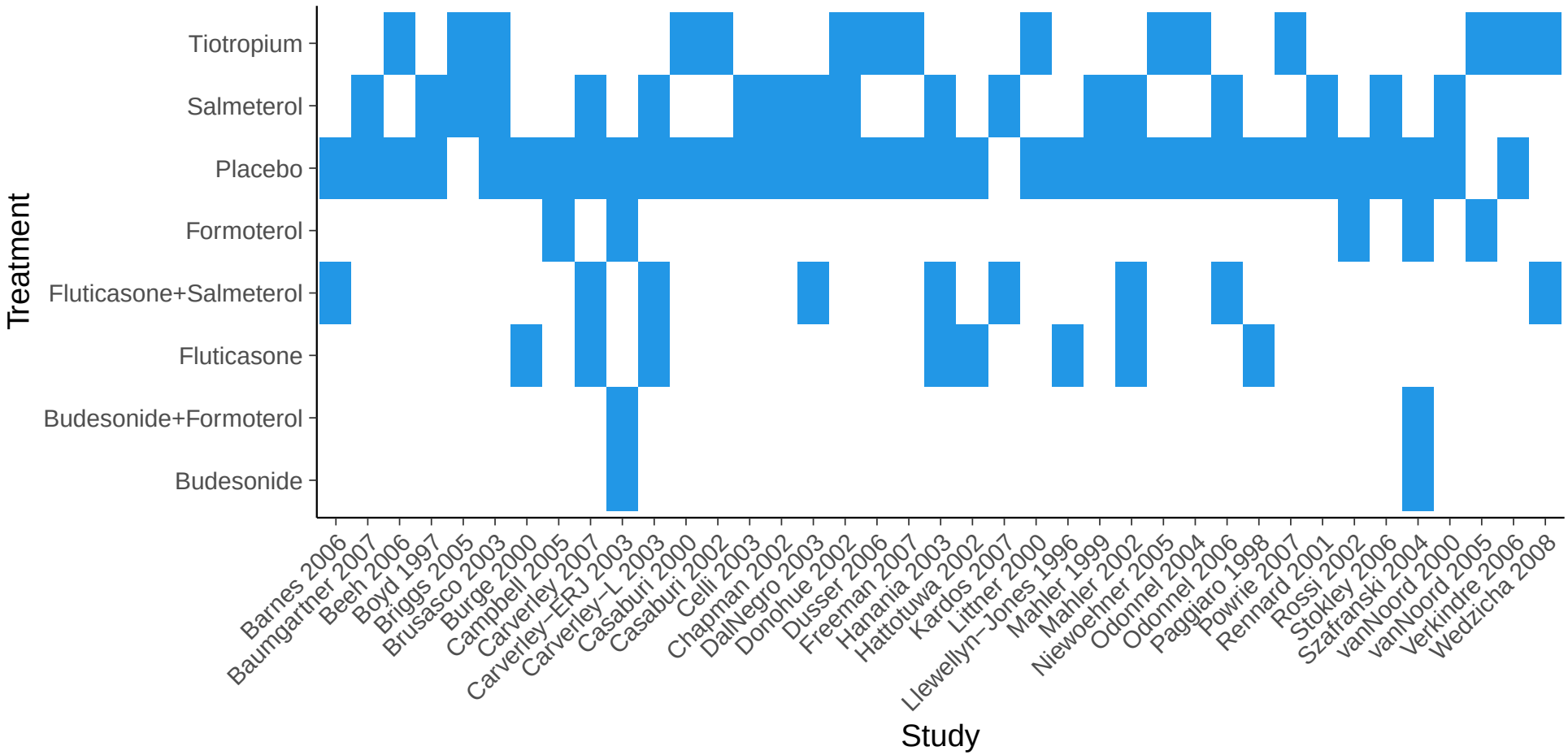
### A big variation in the number of patients per study



# Meta-analysis

## Pharmacologic treatments for chronic obstructive pulmonary disease

2–4 drugs tested per study, and some not including placebo



# Meta-analysis

## Pharmacologic treatments for chronic obstructive pulmonary disease

### Pooled over studies, separate for treatments

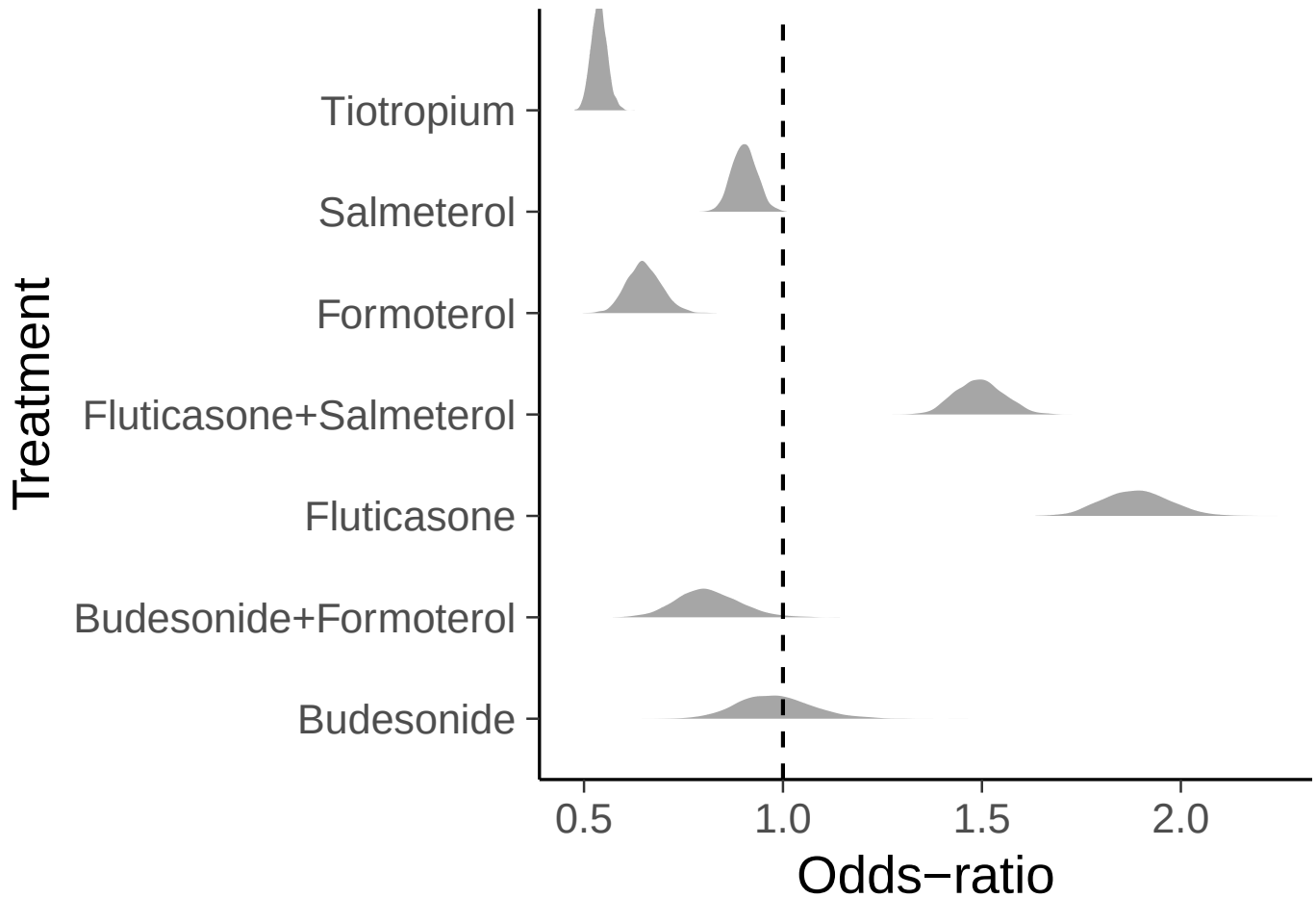
```
fit_pooled <- brm(exac | trials(total) ~ 0 + treatment,
                  prior = prior(student_t(7, 0, 1.5), class='b'),
                  family = binomial(), data = dat.baker2009)
```

# Meta-analysis

## Pharmacologic treatments for chronic obstructive pulmonary disease

### Pooled over studies, separate for treatments

```
fit_pooled <- brm(exac | trials(total) ~ 0 + treatment,
                  prior = prior(student_t(7, 0, 1.5), class='b'),
                  family = binomial(), data = dat.baker2009)
```



# Meta-analysis

## Pharmacologic treatments for chronic obstructive pulmonary disease

### Hirerachical for studies, hierarchical for treatments

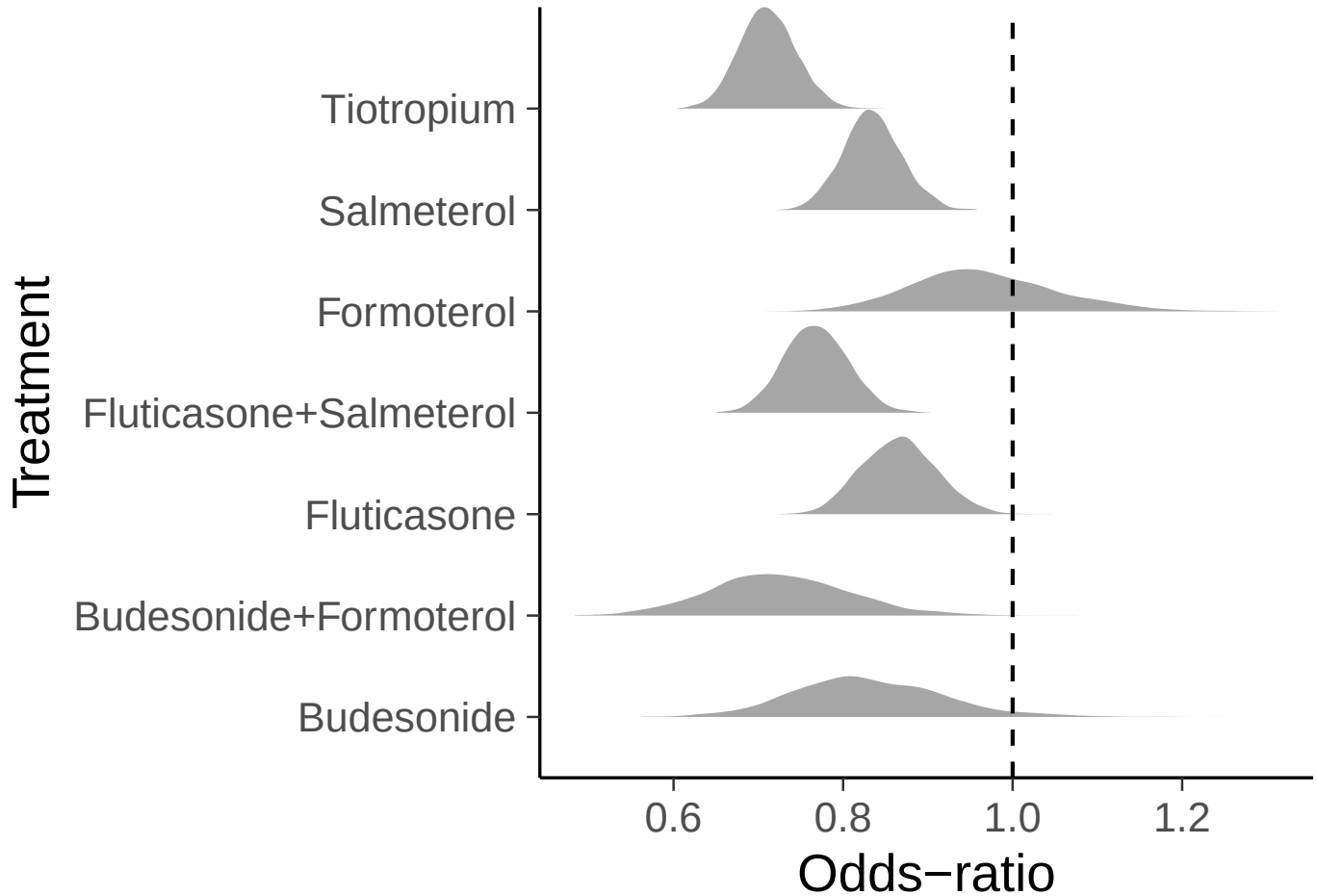
```
fit_hier <- brm(exac | trials(total) ~ (1 | treatment) + (1 | study),
                family = binomial(), data = dat.baker2009)
```

# Meta-analysis

## Pharmacologic treatments for chronic obstructive pulmonary disease

### Hirerachical for studies, hierarchical for treatments

```
fit_hier <- brm(exac | trials(total) ~ (1 | treatment) + (1 | study),
                family = binomial(), data = dat.baker2009)
```

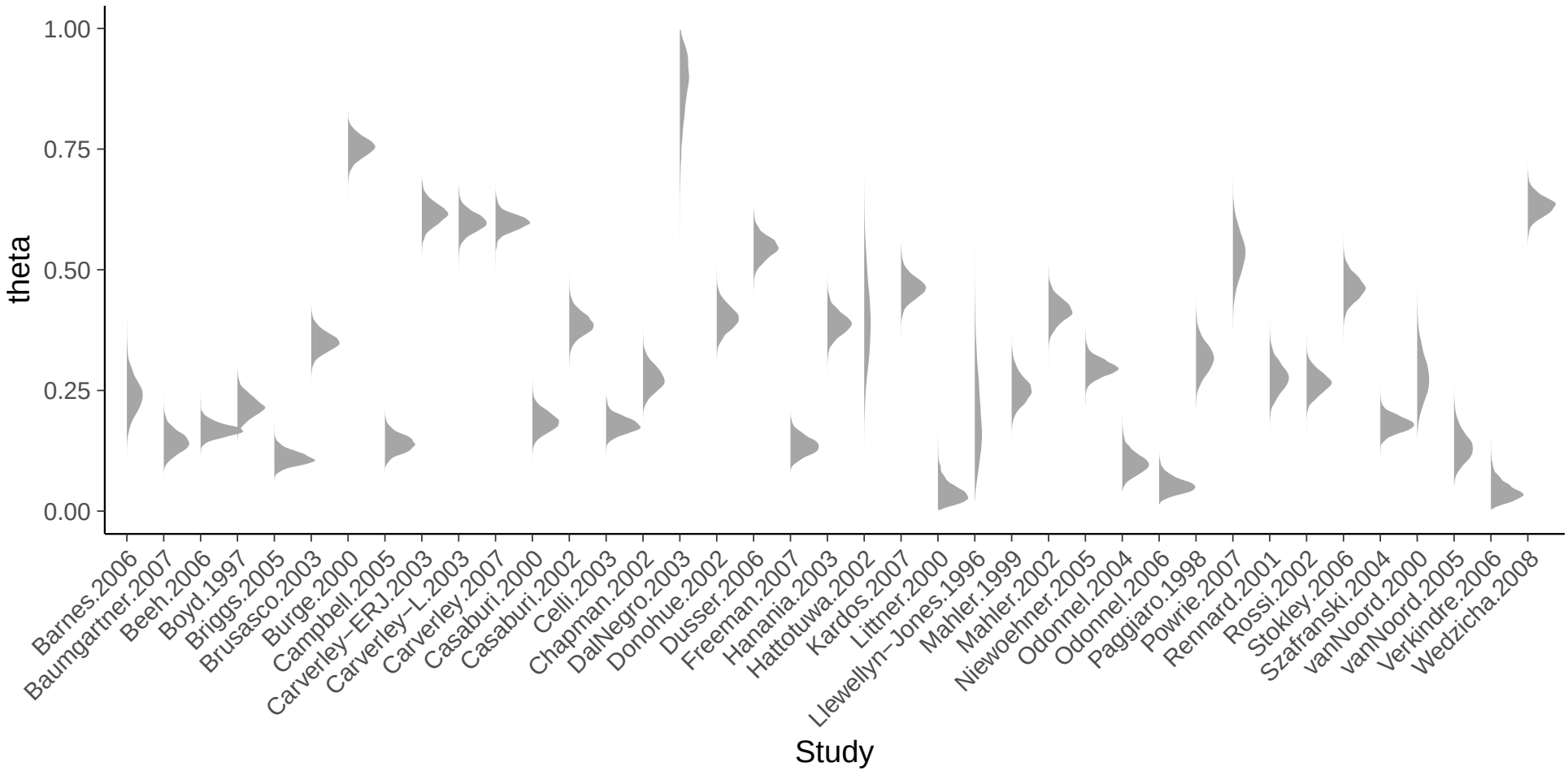


# Meta-analysis

## Pharmacologic treatments for chronic obstructive pulmonary disease

### Hirerachical for studies, hierarchical for treatments

```
fit_hier <- brm(exac | trials(total) ~ (1 | treatment) + (1 | study),
                family = binomial(), data = dat.baker2009)
```



# BDA course demo for brms

Link in the course web site or directly

https://avehtari.github.io/BDA_R_demos/demos_rstan/brms_demo.html

# Exchangeability

- Justifies why we can use
  - a joint model for data
  - a joint prior for a set of parameters
- Less strict than independence

# Exchangeability

- *Exchangeability*: Parameters $\theta_1, \ldots, \theta_J$ (or observations $y_1, \ldots, y_J$) are exchangeable if the joint distribution $p$ is invariant to the permutation of indices $(1, \ldots, J)$
- e.g.

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$

- Exchangeability implies symmetry: If there is no information which can be used *a priori* to separate $\theta_j$ form each other, we can assume exchangeability.
  - "Ignorance implies exchangeability"

# Exchangeability

- Exchangeability does not mean that the results of the experiments could not be different
  - e.g. if we know that the experiments have been in two different laboratories, and we know that the other laboratory has better conditions for the rats, but we do not know which experiments have been made in which laboratory
  - a priori experiments are exchangeable
  - model could have unknown parameter for the laboratory with a conditional prior for rats assumed to come form the same place (clustering model)

# Exchangeability and additional information

- Example: bioassay
  - $y_i$ number of dead animals are not exchangeable alone

# Exchangeability and additional information

- Example: bioassay
  - $y_i$ number of dead animals are not exchangeable alone
  - $x_i$ dose is additional information

# Exchangeability and additional information

- Example: bioassay
  - $y_i$ number of dead animals are not exchangeable alone
  - $x_i$ dose is additional information
  - $(x_i, y_i)$ exchangeable and logistic regression was used

$$p(\alpha, \beta \mid y, n, x) \propto \prod_{i=1}^{n} p(y_i \mid \alpha, \beta, n_i, x_i) p(\alpha, \beta)$$

# Hierarchical exchangeability

- Example: hierarchical rats example
  - all rats not exchangeable

# Hierarchical exchangeability

- Example: hierarchical rats example
  - all rats not exchangeable
  - in a single laboratory rats exchangeable

# Hierarchical exchangeability

- Example: hierarchical rats example
  - all rats not exchangeable
  - in a single laboratory rats exchangeable
  - laboratories exchangeable

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- Example: hierarchical rats example
  - all rats not exchangeable
  - in a single laboratory rats exchangeable
  - laboratories exchangeable
  - $\longrightarrow$ hierarchical model

# Hierarchical exchangeability

- Example: hierarchical rats example
  - all rats not exchangeable
  - in a single laboratory rats exchangeable
  - laboratories exchangeable
  - $\rightarrow$ hierarchical model
- Example: paper helicopters
  - we may assume that the groups had variation in dropping height, dropping technique, timing, and how to define unstable

# Partial or conditional exchangeability

- Conditional exchangeability
  - if $y_i$ is connected to an additional information $x_i$, so that $y_i$ are not exchangeable, but $(y_i, x_i)$ exchangeable use joint model or conditional model $(y_i \mid x_i)$.

# Partial or conditional exchangeability

- Conditional exchangeability
  - if $y_i$ is connected to an additional information $x_i$, so that $y_i$ are not exchangeable, but $(y_i, x_i)$ exchangeable use joint model or conditional model $(y_i \mid x_i)$.
- Partial exchangeability
  - if the observations can be grouped (a priori), then use hierarchical model

# Exchangeability

- The simplest form of the exchangeability (but not the only one) for the parameters $\theta$ conditional independence

$$p(x_1, \ldots, x_J \mid \theta) = \prod_{j=1}^{J} p(x_j \mid \theta)$$

# Exchangeability - Counter example

- A six sided die with probabilities $\theta_1, \ldots, \theta_6$
  - without additional knowledge $\theta_1, \ldots, \theta_6$ exchangeable
  - due to the constraint $\sum_{j=1}^{6} \theta_j$, parameters are not independent and thus joint distribution can not be presented as iid

# Exchangeability

- See more examples in the BDA3 notes - Exchangeability vs. independence